

# A Comparative Study of Modern Inference Techniques for Structured Discrete Energy Minimization Problems : Supplementary Material

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# 1 Evaluation per Model

Table 1: brain-3mm (4 instances)

| algorithm                | runtime     | value       | bound       | mem      | best | opt |
|--------------------------|-------------|-------------|-------------|----------|------|-----|
| $\alpha$ -Exp-pct        | 9.57 sec    | 25164182.75 | $-\infty$   | 2.55 GB  | 1    | 1   |
| $\alpha$ -Exp-VIEW       | 126.94 sec  | 25164077.00 | $-\infty$   | 3.04 GB  | 1    | 0   |
| FastPD                   | 2.02 sec    | 25164518.25 | 22873801.00 | 5.16 GB  | 1    | 1   |
| FastPD-pct               | 7.56 sec    | 25164326.50 | $-\infty$   | 2.55 GB  | 1    | 1   |
| ogm-FastPD-LF1           | 10.51 sec   | 25164518.25 | 22873801.00 | 7.04 GB  | 1    | 1   |
| ogm-FastPD-LF2           | 61.27 sec   | 25163683.50 | 22873801.00 | 7.85 GB  | 1    | 1   |
| ogm-ICM                  | 16.60 sec   | 25425309.50 | $-\infty$   | 2.19 GB  | 1    | 0   |
| ogm-LF-1                 | 9.48 sec    | 25425594.00 | $-\infty$   | 2.87 GB  | 1    | 0   |
| ogm-LF-2                 | 83.39 sec   | 25238328.25 | $-\infty$   | 3.68 GB  | 1    | 0   |
| ogm-TRWS-LF1             | 285.81 sec  | 25162580.25 | 25162493.00 | 3.76 GB  | 1    | 1   |
| ogm-TRWS-LF2             | 343.41 sec  | 25162535.00 | 25162493.00 | 4.56 GB  | 1    | 1   |
| $\alpha\beta$ -Swap-VIEW | 136.43 sec  | 25164671.25 | $-\infty$   | 3.04 GB  | 1    | 0   |
| BPS-TL                   | 644.82 sec  | 25166078.25 | $-\infty$   | 1.80 GB  | 1    | 0   |
| ogm-BPS                  | 2707.16 sec | 25177285.75 | $-\infty$   | 12.18 GB | 1    | 0   |
| ogm-LBP-0.5              | 1930.36 sec | 25166874.25 | $-\infty$   | 12.18 GB | 1    | 0   |
| ogm-LBP-0.95             | 2718.32 sec | 25166927.25 | $-\infty$   | 12.17 GB | 1    | 0   |
| ogm-TRBP-0.5             | 2774.78 sec | 25166817.00 | $-\infty$   | 12.60 GB | 1    | 0   |
| ogm-TRBP-0.95            | 2723.08 sec | 25166836.25 | $-\infty$   | 12.60 GB | 1    | 0   |
| ogm-TRBPS                | 2710.81 sec | 25177148.75 | $-\infty$   | 12.60 GB | 1    | 0   |
| ogm-ADSAL                | 2720.64 sec | 25162633.25 | 25162492.68 | 5.57 GB  | 1    | 1   |
| ogm-BUNDLE-A             | 2708.90 sec | 25166504.00 | 25162118.75 | 9.59 GB  | 1    | 1   |
| ogm-BUNDLE-H             | 2705.77 sec | 25164211.25 | 25162402.99 | 9.60 GB  | 1    | 1   |
| ogm-SG-A                 | 2704.54 sec | 25169609.25 | 25162173.83 | 6.16 GB  | 1    | 1   |
| ogm-BUNDLE-A+            | 2708.06 sec | 25167143.75 | 25161861.94 | 9.60 GB  | 1    | 1   |
| ogm-SG-A+                | 2709.06 sec | 25536961.50 | 24957377.76 | 6.17 GB  | 1    | 1   |
| ogm-BUNDLE-A-            | 2707.39 sec | 25163766.00 | 25162483.90 | 9.58 GB  | 1    | 1   |
| ogm-SG-A-                | 2707.91 sec | 25244347.75 | 25135353.99 | 6.16 GB  | 1    | 1   |
| TRWS-TL                  | 282.73 sec  | 25162700.50 | 25162493.00 | 1.80 GB  | 1    | 1   |
| TRWS-pct                 | 40.16 sec   | 25162648.50 | 25162493.00 | 2.55 GB  | 1    | 1   |
| MCI-pct                  | 27.35 sec   | 25162493.00 | 25162493.00 | 2.77 GB  | 4    | 4   |
| ogm-CombiLP              | 3874.65 sec | 25162580.25 | 25162493.00 | 6.84 GB  | 1    | 1   |

Table 2: brain-5mm (4 instances)

| algorithm                | runtime     | value       | bound       | mem     | best | opt |
|--------------------------|-------------|-------------|-------------|---------|------|-----|
| $\alpha$ -Exp-pct        | 7.02 sec    | 19088999.75 | $-\infty$   | 1.53 GB | 0    | 0   |
| $\alpha$ -Exp-VIEW       | 100.66 sec  | 19089080.00 | $-\infty$   | 1.82 GB | 0    | 0   |
| FastPD                   | 1.32 sec    | 19089484.75 | 17052089.25 | 3.11 GB | 0    | 0   |
| FastPD-pct               | 4.66 sec    | 19089255.25 | $-\infty$   | 1.53 GB | 0    | 0   |
| ogm-FastPD-LF1           | 7.19 sec    | 19089484.75 | 17052089.25 | 4.21 GB | 0    | 0   |
| ogm-FastPD-LF2           | 48.51 sec   | 19088812.00 | 17052089.25 | 4.59 GB | 0    | 0   |
| ogm-ICM                  | 11.30 sec   | 19272820.00 | $-\infty$   | 1.31 GB | 0    | 0   |
| ogm-LF-1                 | 6.49 sec    | 19273079.25 | $-\infty$   | 1.69 GB | 0    | 0   |
| ogm-LF-2                 | 51.81 sec   | 19140692.25 | $-\infty$   | 2.10 GB | 0    | 0   |
| ogm-TRWS-LF1             | 125.92 sec  | 19087665.75 | 19087612.50 | 2.24 GB | 0    | 0   |
| ogm-TRWS-LF2             | 184.22 sec  | 19087628.00 | 19087612.50 | 2.62 GB | 0    | 0   |
| $\alpha\beta$ -Swap-VIEW | 91.84 sec   | 19089768.00 | $-\infty$   | 1.83 GB | 0    | 0   |
| BPS-TL                   | 450.43 sec  | 19090723.25 | $-\infty$   | 1.08 GB | 0    | 0   |
| ogm-BPS                  | 3601.41 sec | 19099086.75 | $-\infty$   | 7.29 GB | 0    | 0   |
| ogm-LBP-0.5              | 1116.27 sec | 19091373.75 | $-\infty$   | 7.29 GB | 0    | 0   |
| ogm-LBP-0.95             | 1574.35 sec | 19091228.75 | $-\infty$   | 7.28 GB | 0    | 0   |
| ogm-TRBP-0.5             | 2948.01 sec | 19091291.75 | $-\infty$   | 7.54 GB | 0    | 0   |
| ogm-TRBP-0.95            | 3463.22 sec | 19091178.00 | $-\infty$   | 7.54 GB | 0    | 0   |
| ogm-TRBPS                | 3602.69 sec | 19099313.50 | $-\infty$   | 7.54 GB | 0    | 0   |
| ogm-ADSAL                | 3610.13 sec | 19087679.25 | 19087612.49 | 3.37 GB | 0    | 0   |
| ogm-BUNDLE-A             | 3605.55 sec | 19088524.25 | 19087585.50 | 5.77 GB | 0    | 0   |
| ogm-BUNDLE-H             | 3603.13 sec | 19088322.25 | 19087572.17 | 5.77 GB | 0    | 0   |
| ogm-SG-A                 | 3602.62 sec | 19090744.50 | 19087462.48 | 3.70 GB | 0    | 0   |
| ogm-BUNDLE-A+            | 3602.10 sec | 19089189.75 | 19087393.52 | 5.77 GB | 0    | 0   |
| ogm-SG-A+                | 3602.34 sec | 19300226.00 | 18969716.96 | 3.69 GB | 0    | 0   |
| ogm-BUNDLE-A-            | 3604.02 sec | 19088258.25 | 19087607.03 | 5.77 GB | 0    | 0   |
| ogm-SG-A-                | 3603.66 sec | 19125604.50 | 19079576.02 | 3.70 GB | 0    | 0   |
| TRWS-TL                  | 120.29 sec  | 19087730.25 | 19087612.50 | 1.08 GB | 0    | 0   |
| TRWS-pct                 | 21.93 sec   | 19087728.50 | 19087612.50 | 1.53 GB | 0    | 0   |
| MCI-pct                  | 25.63 sec   | 19087612.50 | 19087612.50 | 1.99 GB | 4    | 4   |
| ogm-CombiLP              | 2022.87 sec | 19087626.75 | 19087612.50 | 4.84 GB | 3    | 3   |

Table 3: brain-9mm (4 instances)

| algorithm                | runtime     | value      | bound      | mem     | best | opt |
|--------------------------|-------------|------------|------------|---------|------|-----|
| $\alpha$ -Exp-pct        | 3.51 sec    | 9185804.25 | $-\infty$  | 0.85 GB | 1    | 1   |
| $\alpha$ -Exp-VIEW       | 37.19 sec   | 9185865.25 | $-\infty$  | 1.04 GB | 1    | 0   |
| FastPD                   | 0.75 sec    | 9186044.50 | 8260110.75 | 1.71 GB | 1    | 1   |
| FastPD-pct               | 2.63 sec    | 9185930.25 | $-\infty$  | 0.85 GB | 1    | 1   |
| ogm-FastPD-LF1           | 3.71 sec    | 9186044.50 | 8260110.75 | 2.33 GB | 1    | 1   |
| ogm-FastPD-LF2           | 21.41 sec   | 9185787.25 | 8260110.75 | 2.51 GB | 1    | 1   |
| ogm-ICM                  | 5.78 sec    | 9263471.50 | $-\infty$  | 0.73 GB | 1    | 0   |
| ogm-LF-1                 | 3.24 sec    | 9263538.00 | $-\infty$  | 0.93 GB | 1    | 0   |
| ogm-LF-2                 | 29.25 sec   | 9207084.00 | $-\infty$  | 1.13 GB | 1    | 0   |
| ogm-TRWS-LF1             | 80.76 sec   | 9185326.75 | 9185280.75 | 1.25 GB | 1    | 1   |
| ogm-TRWS-LF2             | 93.10 sec   | 9185298.75 | 9185280.75 | 1.42 GB | 1    | 1   |
| $\alpha\beta$ -Swap-VIEW | 43.51 sec   | 9186218.50 | $-\infty$  | 1.01 GB | 1    | 0   |
| BPS-TL                   | 230.93 sec  | 9186485.25 | $-\infty$  | 0.60 GB | 1    | 0   |
| ogm-BPS                  | 1698.94 sec | 9190052.50 | $-\infty$  | 4.03 GB | 1    | 0   |
| ogm-LBP-0.5              | 408.82 sec  | 9186565.25 | $-\infty$  | 4.03 GB | 1    | 0   |
| ogm-LBP-0.95             | 346.08 sec  | 9186506.00 | $-\infty$  | 4.03 GB | 1    | 0   |
| ogm-TRBP-0.5             | 1166.47 sec | 9186545.00 | $-\infty$  | 4.16 GB | 1    | 0   |
| ogm-TRBP-0.95            | 1852.20 sec | 9186507.25 | $-\infty$  | 4.17 GB | 1    | 0   |
| ogm-TRBPS                | 2102.64 sec | 9189796.75 | $-\infty$  | 4.16 GB | 1    | 0   |
| ogm-ADSAL                | 2193.95 sec | 9185310.25 | 9185280.74 | 1.90 GB | 1    | 1   |
| ogm-BUNDLE-A             | 2568.53 sec | 9185566.25 | 9185278.59 | 3.22 GB | 1    | 1   |
| ogm-BUNDLE-H             | 2566.86 sec | 9185559.25 | 9185271.00 | 3.22 GB | 1    | 1   |
| ogm-SG-A                 | 2292.96 sec | 9186984.25 | 9185197.48 | 2.09 GB | 1    | 1   |
| ogm-BUNDLE-A+            | 2568.56 sec | 9185798.75 | 9185240.43 | 3.22 GB | 1    | 1   |
| ogm-SG-A+                | 2468.36 sec | 9266873.50 | 9139457.93 | 2.09 GB | 1    | 1   |
| ogm-BUNDLE-A-            | 2581.48 sec | 9185626.75 | 9185277.57 | 3.22 GB | 1    | 1   |
| ogm-SG-A-                | 2353.54 sec | 9193050.50 | 9184612.03 | 2.09 GB | 1    | 1   |
| TRWS-TL                  | 68.40 sec   | 9185364.75 | 9185280.75 | 0.60 GB | 1    | 1   |
| TRWS-pct                 | 14.13 sec   | 9185347.00 | 9185280.75 | 0.85 GB | 1    | 1   |
| MCI-pct                  | 8.32 sec    | 9185280.75 | 9185280.75 | 1.09 GB | 4    | 4   |
| ogm-CombiLP              | 895.43 sec  | 9185280.75 | 9185280.75 | 2.72 GB | 4    | 4   |

Table 4: cell-tracking (1 instances)

| algorithm     | runtime      | value          | bound      | mem     | best | opt |
|---------------|--------------|----------------|------------|---------|------|-----|
| ogm-ICM       | 0.03 sec     | 40359847.14    | $-\infty$  | 0.03 GB | 0    | 0   |
| ogm-LBP-LF2   | 62.12 sec    | 7515575.61     | $-\infty$  | 0.12 GB | 0    | 0   |
| ogm-LF-1      | 0.04 sec     | 40359847.14    | $-\infty$  | 0.04 GB | 0    | 0   |
| ogm-LF-2      | 0.43 sec     | 14075743.46    | $-\infty$  | 0.06 GB | 0    | 0   |
| ogm-LF-3      | 1.55 sec     | 8461693.24     | $-\infty$  | 0.07 GB | 0    | 0   |
| ogm-BPS       | 60.07 sec    | 207520418.28   | $-\infty$  | 0.08 GB | 0    | 0   |
| ogm-LBP-0.5   | 62.01 sec    | 407516128.01   | $-\infty$  | 0.08 GB | 0    | 0   |
| ogm-LBP-0.95  | 61.97 sec    | 307513873.84   | $-\infty$  | 0.08 GB | 0    | 0   |
| ogm-TRBP-0.5  | 65.94 sec    | 307511908.51   | $-\infty$  | 0.09 GB | 0    | 0   |
| ogm-TRBP-0.95 | 65.97 sec    | 107517017.88   | $-\infty$  | 0.09 GB | 0    | 0   |
| ogm-TRBPS     | 62.92 sec    | 307516517.88   | $-\infty$  | 0.09 GB | 0    | 0   |
| ADDD          | 11.74 sec    | 34008664296.29 | 6206883.23 | 0.08 GB | 0    | 0   |
| MPLP          | 459.93 sec   | 107514359.61   | 7513851.52 | 0.08 GB | 0    | 0   |
| ogm-BUNDLE-A  | 532.38 sec   | 7696631.53     | 7501985.37 | 0.18 GB | 0    | 0   |
| ogm-BUNDLE-H  | 522.32 sec   | 7748583.42     | 7501948.96 | 0.18 GB | 0    | 0   |
| ogm-SG-A      | 522.89 sec   | 927583594.52   | 6899501.32 | 0.15 GB | 0    | 0   |
| ogm-BUNDLE-A+ | 527.17 sec   | 7696631.53     | 7501985.37 | 0.17 GB | 0    | 0   |
| ogm-SG-A+     | 526.20 sec   | 230126875.07   | 6899501.32 | 0.15 GB | 0    | 0   |
| ogm-BUNDLE-A- | 531.65 sec   | 7696631.53     | 7501985.37 | 0.17 GB | 0    | 0   |
| ogm-SG-A-     | 518.55 sec   | 1819216014.16  | 6899501.32 | 0.15 GB | 0    | 0   |
| ogm-LP-LP     | 4.21 sec     | 7516359.61     | 7513851.52 | 0.51 GB | 0    | 0   |
| ADDD-BB       | 38698.77 sec | 7514421.21     | 7411393.72 | 0.10 GB | 1    | 0   |
| ogm-ILP-pct   | 12.36 sec    | 7514421.21     | 7514421.21 | 1.05 GB | 1    | 1   |
| ogm-ILP       | 11.99 sec    | 7514421.21     | 7514421.21 | 1.20 GB | 1    | 1   |

Table 5: color-seg-n4 (9 instances)

| algorithm                    | runtime    | value    | bound     | mem     | best | opt |
|------------------------------|------------|----------|-----------|---------|------|-----|
| $\alpha$ -Exp-pct            | 2.98 sec   | 20034.80 | $-\infty$ | 0.15 GB | 0    | 0   |
| FastPD                       | 0.29 sec   | 20034.80 | 13644.72  | 0.31 GB | 0    | 0   |
| FastPD-pct                   | 1.52 sec   | 20034.85 | $-\infty$ | 0.32 GB | 0    | 0   |
| mrf- $\alpha$ -Exp-TL        | 1.11 sec   | 20033.56 | $-\infty$ | 0.07 GB | 0    | 0   |
| mrf- $\alpha/\beta$ -Swap-TL | 0.64 sec   | 20060.91 | $-\infty$ | 0.07 GB | 0    | 0   |
| ogm-FastPD-LF1               | 1.07 sec   | 20034.80 | 12543.39  | 0.36 GB | 0    | 0   |
| ogm-FastPD-LF2               | 6.15 sec   | 20033.21 | 12543.39  | 0.38 GB | 0    | 0   |
| ogm-ICM                      | 1.07 sec   | 26329.45 | $-\infty$ | 0.11 GB | 0    | 0   |
| ogm-LF-1                     | 0.50 sec   | 26330.66 | $-\infty$ | 0.11 GB | 0    | 0   |
| ogm-LF-2                     | 7.38 sec   | 23775.38 | $-\infty$ | 0.12 GB | 0    | 0   |
| ogm-TRWS-LF1                 | 7.58 sec   | 20012.17 | 20012.14  | 0.15 GB | 7    | 7   |
| ogm-TRWS-LF2                 | 9.08 sec   | 20012.17 | 20012.14  | 0.16 GB | 7    | 7   |
| mrf-LBP-TL                   | 39.87 sec  | 20053.25 | $-\infty$ | 0.09 GB | 0    | 0   |
| mrf-BPS-TL                   | 23.35 sec  | 20094.03 | $-\infty$ | 0.09 GB | 0    | 0   |
| ogm-BPS                      | 419.65 sec | 20377.19 | $-\infty$ | 0.46 GB | 0    | 0   |
| ogm-LBP-0.5                  | 335.15 sec | 20054.27 | $-\infty$ | 0.46 GB | 0    | 0   |
| ogm-LBP-0.95                 | 267.45 sec | 20058.46 | $-\infty$ | 0.46 GB | 0    | 0   |
| ogm-TRBP-0.5                 | 505.13 sec | 20054.07 | $-\infty$ | 0.47 GB | 0    | 0   |
| ogm-TRBP-0.95                | 371.35 sec | 20058.04 | $-\infty$ | 0.47 GB | 0    | 0   |
| ogm-TRBPS                    | 555.27 sec | 20370.77 | $-\infty$ | 0.47 GB | 0    | 0   |
| MCR-TC-MTC                   | 440.57 sec | 20450.12 | 19807.10  | 1.31 GB | 7    | 7   |
| MCR-pct                      | 424.71 sec | 20944.63 | $-\infty$ | 1.68 GB | 3    | 3   |
| mrf-TRWS-TL                  | 23.52 sec  | 20012.18 | 20012.14  | 0.09 GB | 8    | 7   |
| ogm-ADSAL                    | 311.96 sec | 20012.15 | 20012.14  | 0.21 GB | 8    | 7   |
| ogm-BUNDLE-A                 | 224.07 sec | 20024.78 | 20012.01  | 0.38 GB | 7    | 7   |
| ogm-BUNDLE-H                 | 431.91 sec | 20012.44 | 20012.13  | 0.38 GB | 6    | 1   |
| ogm-SG-A                     | 391.10 sec | 20027.98 | 20011.56  | 0.22 GB | 6    | 1   |
| ogm-BUNDLE-A+                | 404.59 sec | 20156.95 | 19979.39  | 0.38 GB | 3    | 3   |
| ogm-SG-A+                    | 404.87 sec | 20198.96 | 19892.88  | 0.22 GB | 1    | 1   |
| ogm-BUNDLE-A-                | 392.39 sec | 20012.19 | 20012.13  | 0.38 GB | 7    | 0   |
| ogm-SG-A-                    | 425.47 sec | 20055.22 | 20011.48  | 0.22 GB | 0    | 0   |
| TRWS-pct                     | 45.62 sec  | 20012.17 | 20012.14  | 0.14 GB | 7    | 7   |
| MCI-TC-MTC-TCI               | 442.89 sec | 20450.11 | 19807.10  | 2.66 GB | 8    | 8   |
| MCI-pct                      | 429.54 sec | 20889.89 | $-\infty$ | 1.67 GB | 8    | 8   |
| ogm-CombiLP                  | 36.68 sec  | 20012.14 | 20012.14  | 0.38 GB | 9    | 9   |

Table 6: color-seg-n8 (9 instances)

| algorithm                | runtime     | value    | bound     | mem     | best | opt |
|--------------------------|-------------|----------|-----------|---------|------|-----|
| $\alpha$ -Exp-pct        | 5.56 sec    | 20011.24 | $-\infty$ | 0.23 GB | 0    | 0   |
| $\alpha$ -Exp-VIEW       | 7.79 sec    | 20011.13 | $-\infty$ | 0.17 GB | 0    | 0   |
| FastPD                   | 0.45 sec    | 20011.14 | 13841.08  | 0.51 GB | 0    | 0   |
| FastPD-pct               | 2.79 sec    | 20011.24 | $-\infty$ | 0.52 GB | 0    | 0   |
| ogm-FastPD-LF1           | 2.07 sec    | 20011.13 | 13121.34  | 0.60 GB | 0    | 0   |
| ogm-FastPD-LF2           | 21.51 sec   | 20010.28 | 13121.34  | 0.64 GB | 0    | 0   |
| ogm-ICM                  | 1.98 sec    | 25302.60 | $-\infty$ | 0.14 GB | 0    | 0   |
| ogm-LF-1                 | 0.93 sec    | 25306.40 | $-\infty$ | 0.16 GB | 0    | 0   |
| ogm-LF-2                 | 25.21 sec   | 21129.07 | $-\infty$ | 0.20 GB | 0    | 0   |
| ogm-TRWS-LF1             | 25.26 sec   | 19991.28 | 19991.16  | 0.23 GB | 3    | 1   |
| ogm-TRWS-LF2             | 43.80 sec   | 19991.27 | 19991.16  | 0.27 GB | 4    | 1   |
| $\alpha\beta$ -Swap-VIEW | 9.75 sec    | 20038.26 | $-\infty$ | 0.17 GB | 0    | 0   |
| BPS-TL                   | 62.69 sec   | 20120.79 | $-\infty$ | 0.13 GB | 0    | 0   |
| ogm-BPS                  | 878.41 sec  | 20080.07 | $-\infty$ | 0.79 GB | 0    | 0   |
| ogm-LBP-0.5              | 850.32 sec  | 20087.13 | $-\infty$ | 0.79 GB | 0    | 0   |
| ogm-LBP-0.95             | 786.09 sec  | 20081.55 | $-\infty$ | 0.79 GB | 0    | 0   |
| ogm-TRBP-0.5             | 1157.33 sec | 20086.99 | $-\infty$ | 0.81 GB | 0    | 0   |
| ogm-TRBP-0.95            | 1078.86 sec | 20081.26 | $-\infty$ | 0.81 GB | 0    | 0   |
| ogm-TRBPS                | 1146.51 sec | 20079.65 | $-\infty$ | 0.81 GB | 0    | 0   |
| MCR-TC-MTC               | 539.88 sec  | 20671.38 | 19713.14  | 1.73 GB | 2    | 1   |
| MCR-pct                  | 486.38 sec  | 20988.67 | $-\infty$ | 2.20 GB | 3    | 2   |
| ogm-ADSAL                | 1320.13 sec | 19991.25 | 19991.16  | 0.35 GB | 5    | 2   |
| ogm-BUNDLE-A             | 935.94 sec  | 20020.56 | 19990.56  | 0.80 GB | 4    | 1   |
| ogm-BUNDLE-H             | 978.44 sec  | 19991.94 | 19991.11  | 0.80 GB | 3    | 0   |
| ogm-SG-A                 | 854.29 sec  | 20360.63 | 19979.94  | 0.38 GB | 1    | 1   |
| ogm-BUNDLE-A+            | 973.20 sec  | 20248.63 | 19857.07  | 0.80 GB | 1    | 1   |
| ogm-SG-A+                | 896.80 sec  | 21203.60 | 19476.10  | 0.39 GB | 0    | 0   |
| ogm-BUNDLE-A-            | 986.69 sec  | 19991.51 | 19991.12  | 0.80 GB | 1    | 0   |
| ogm-SG-A-                | 913.14 sec  | 20128.58 | 19988.55  | 0.38 GB | 0    | 0   |
| TRWS-TL                  | 24.80 sec   | 19991.33 | 19991.16  | 0.13 GB | 2    | 1   |
| TRWS-pct                 | 91.86 sec   | 19991.33 | 19991.16  | 0.21 GB | 1    | 1   |
| MCI-TC-MTC-TCI           | 836.33 sec  | 20670.74 | 19713.17  | 4.52 GB | 8    | 8   |
| MCI-pct                  | 534.88 sec  | 20893.77 | $-\infty$ | 2.55 GB | 7    | 8   |
| ogm-CombiLP              | 118.53 sec  | 19991.21 | 19991.21  | 0.69 GB | 9    | 9   |

Table 7: color-seg (3 instances)

| algorithm                | runtime     | value        | bound        | mem     | best | opt |
|--------------------------|-------------|--------------|--------------|---------|------|-----|
| $\alpha$ -Exp-pct        | 0.82 sec    | 308472274.33 | $-\infty$    | 0.56 GB | 3    | 0   |
| $\alpha$ -Exp-VIEW       | 5.98 sec    | 308472275.67 | $-\infty$    | 0.70 GB | 2    | 0   |
| FastPD                   | 0.31 sec    | 308472275.00 | 308420090.33 | 1.00 GB | 2    | 0   |
| FastPD-pct               | 0.80 sec    | 308472274.67 | $-\infty$    | 0.56 GB | 2    | 0   |
| ogm-FastPD-LF1           | 1.25 sec    | 308472275.00 | 308420090.33 | 1.42 GB | 2    | 0   |
| ogm-FastPD-LF2           | 8.58 sec    | 308472275.00 | 308420090.33 | 1.51 GB | 2    | 0   |
| ogm-ICM                  | 2.38 sec    | 311333421.33 | $-\infty$    | 0.47 GB | 0    | 0   |
| ogm-LF-1                 | 1.94 sec    | 311333771.00 | $-\infty$    | 0.63 GB | 0    | 0   |
| ogm-LF-2                 | 11.82 sec   | 309850181.00 | $-\infty$    | 0.74 GB | 0    | 0   |
| ogm-TRWS-LF1             | 104.99 sec  | 308472294.33 | 308472270.43 | 0.83 GB | 2    | 1   |
| ogm-TRWS-LF2             | 102.44 sec  | 308472294.33 | 308472270.43 | 0.92 GB | 2    | 1   |
| $\alpha\beta$ -Swap-VIEW | 6.25 sec    | 308472292.33 | $-\infty$    | 0.70 GB | 2    | 0   |
| BPS-TL                   | 68.24 sec   | 308733349.67 | $-\infty$    | 0.40 GB | 0    | 0   |
| ogm-BPS                  | 106.57 sec  | 308494459.00 | $-\infty$    | 2.76 GB | 0    | 0   |
| ogm-LBP-0.5              | 362.43 sec  | 308492950.67 | $-\infty$    | 2.76 GB | 0    | 0   |
| ogm-LBP-0.95             | 117.09 sec  | 308494213.33 | $-\infty$    | 2.76 GB | 0    | 0   |
| ogm-TRBP-0.5             | 1245.66 sec | 308492909.67 | $-\infty$    | 2.85 GB | 0    | 0   |
| ogm-TRBP-0.95            | 1237.92 sec | 308494370.67 | $-\infty$    | 2.85 GB | 0    | 0   |
| ogm-TRBPS                | 1009.29 sec | 308494514.33 | $-\infty$    | 2.85 GB | 0    | 0   |
| MCR-TC-MTC               | 89.46 sec   | 308472274.33 | 308472274.33 | 3.76 GB | 3    | 3   |
| MCR-pct                  | 0.82 sec    | 308472274.33 | 308472274.33 | 0.56 GB | 3    | 3   |
| ogm-ADSAL                | 2156.82 sec | 308472289.00 | 308472273.99 | 1.19 GB | 2    | 2   |
| ogm-BUNDLE-A             | 1661.42 sec | 308472329.33 | 308472259.92 | 2.25 GB | 1    | 0   |
| ogm-BUNDLE-H             | 1585.35 sec | 308472407.67 | 308472264.89 | 2.25 GB | 1    | 0   |
| ogm-SG-A                 | 1399.36 sec | 308485243.33 | 308471868.96 | 1.47 GB | 0    | 0   |
| ogm-BUNDLE-A+            | 1625.25 sec | 308472378.67 | 308472268.33 | 2.26 GB | 1    | 1   |
| ogm-SG-A+                | 1325.76 sec | 308736418.67 | 308454764.27 | 1.47 GB | 0    | 0   |
| ogm-BUNDLE-A-            | 1522.39 sec | 308472520.67 | 308472231.05 | 2.26 GB | 0    | 0   |
| ogm-SG-A-                | 1389.03 sec | 308481954.33 | 308472044.84 | 1.46 GB | 0    | 0   |
| TRWS-TL                  | 90.76 sec   | 308472310.67 | 308472270.43 | 0.40 GB | 2    | 1   |
| TRWS-pct                 | 1.07 sec    | 308472290.67 | 308472274.33 | 0.56 GB | 2    | 2   |
| MCI-TC-MTC-TCI           | 80.19 sec   | 308472274.33 | 308472274.33 | 3.76 GB | 3    | 3   |
| MCI-pct                  | 0.98 sec    | 308472274.33 | 308472274.33 | 0.56 GB | 3    | 3   |
| ogm-CombiLP              | 483.64 sec  | 308472274.33 | 308472274.33 | 1.84 GB | 3    | 3   |

Table 8: correlation-clustering (715 instances)

| algorithm       | runtime  | value   | bound     | mem     | best | opt | VI     | RI     |
|-----------------|----------|---------|-----------|---------|------|-----|--------|--------|
| ogm-ICM         | 1.28 sec | -585.60 | $-\infty$ | 0.01 GB | 0    | 0   | 2.6245 | 0.5154 |
| ogm-LF-1        | 0.63 sec | -585.60 | $-\infty$ | 0.02 GB | 0    | 0   | 2.6245 | 0.5154 |
| MCR-CC          | 0.12 sec | -626.76 | -628.89   | 0.04 GB | 166  | 98  | 2.0463 | 0.8362 |
| MCR-CCFDB       | 0.08 sec | -626.75 | -628.90   | 0.03 GB | 164  | 98  | 2.0463 | 0.8362 |
| MCR-CCFDB-OWC   | 0.08 sec | -626.77 | -628.89   | 0.03 GB | 166  | 100 | 2.0460 | 0.8362 |
| MCI-CCFDB-CCIFD | 0.78 sec | -628.16 | -628.16   | 0.08 GB | 715  | 713 | 2.0406 | 0.8350 |
| MCI-CCI         | 1.28 sec | -628.16 | -628.17   | 0.08 GB | 715  | 707 | 2.0406 | 0.8350 |
| MCI-CCIFD       | 1.25 sec | -628.16 | -628.16   | 0.07 GB | 715  | 713 | 2.0406 | 0.8350 |



Table 9: dtf-chinesechar (100 instances)

| algorithm     | runtime        | value     | bound     | mem           | best | opt | PA     |
|---------------|----------------|-----------|-----------|---------------|------|-----|--------|
| LSA-TR (euc.) | 0.05 sec       | -49548.10 | $-\infty$ | 0.10 GB       | 30   | 0   | 0.6712 |
| LSA-TR (ham.) | 0.06 sec       | -49536.76 | $-\infty$ | 0.10 GB       | 1    | 0   | 0.6433 |
| ogm-ICM       | 0.21 sec       | -49515.98 | $-\infty$ | 0.12 GB       | 1    | 0   | 0.5723 |
| ogm-LBP-LF2   | 481.32 sec     | -49520.26 | $-\infty$ | 0.91 GB       | 7    | 0   | 0.6393 |
| ogm-LF-1      | 0.23 sec       | -49516.08 | $-\infty$ | 0.16 GB       | 1    | 0   | 0.5725 |
| ogm-LF-2      | 7.34 sec       | -49531.11 | $-\infty$ | 0.25 GB       | 7    | 0   | 0.6003 |
| ogm-LF-3      | 637.92 sec     | -49535.37 | $-\infty$ | 2.29 GB       | 16   | 0   | 0.6119 |
| ogm-TRWS-LF1  | 84.41 sec      | -49505.84 | -50119.41 | 0.23 GB       | 4    | 0   | 0.5747 |
| ogm-TRWS-LF2  | 83.78 sec      | -49519.42 | -50119.41 | 0.31 GB       | 10   | 0   | 0.5945 |
| BPS-TAB       | 62.69 sec      | -49537.08 | $-\infty$ | 0.12 GB       | 30   | 0   | 0.6715 |
| ogm-BPS       | 456.79 sec     | -49497.39 | $-\infty$ | 0.72 GB       | 11   | 0   | 0.6336 |
| ogm-LBP-0.5   | 482.00 sec     | -49509.81 | $-\infty$ | 0.72 GB       | 12   | 0   | 0.6406 |
| ogm-LBP-0.95  | 479.42 sec     | -49506.02 | $-\infty$ | 0.72 GB       | 11   | 0   | 0.6388 |
| ogm-TRBP-0.5  | 561.98 sec     | -49515.07 | $-\infty$ | 0.75 GB       | 12   | 0   | 0.6535 |
| ogm-TRBP-0.95 | 540.99 sec     | -49512.19 | $-\infty$ | 0.75 GB       | 15   | 0   | 0.6514 |
| ogm-TRBPS     | 466.66 sec     | -49497.04 | $-\infty$ | 0.75 GB       | 9    | 0   | 0.6545 |
| ADDD          | 9.74 sec       | -48656.71 | -50119.38 | 0.77 GB       | 0    | 0   | 0.5079 |
| MPLP          | 516.36 sec     | -49040.57 | -50119.46 | 0.94 GB       | 0    | 0   | 0.6064 |
| ogm-ADSAL     | 730.78 sec     | -49524.30 | -50119.39 | 0.31 GB       | 1    | 0   | 0.6445 |
| ogm-BUNDLE-A  | 445.46 sec     | -49471.84 | -50125.37 | 0.72 GB       | 0    | 0   | 0.5806 |
| ogm-BUNDLE-H  | 454.52 sec     | -49489.73 | -50120.92 | 0.72 GB       | 0    | 0   | 0.5890 |
| ogm-SG-A      | 420.01 sec     | -49476.28 | -50152.59 | 0.48 GB       | 0    | 0   | 0.5745 |
| ogm-BUNDLE-A+ | 443.99 sec     | -49469.63 | -50211.67 | 0.72 GB       | 0    | 0   | 0.5867 |
| ogm-SG-A+     | 417.48 sec     | -49457.71 | -50666.34 | 0.48 GB       | 0    | 0   | 0.5795 |
| ogm-BUNDLE-A- | 447.56 sec     | -49468.24 | -50120.76 | 0.72 GB       | 0    | 0   | 0.5729 |
| ogm-SG-A-     | 416.33 sec     | -49478.46 | -50138.29 | 0.51 GB       | 0    | 0   | 0.5836 |
| ogm-LP-LP     | 1335.58 sec    | -48913.03 | -50134.52 | 3.29 GB       | 0    | 0   | 0.5272 |
| QPBO          | 0.17 sec       | -49501.95 | -50119.38 | 0.13 GB       | 0    | 0   | 0.5520 |
| TRWS-TAB      | 78.84 sec      | -49497.01 | -50119.41 | 0.13 GB       | 3    | 0   | 0.5649 |
| TRWS-pct      | 4.43 sec       | -49496.76 | -50119.38 | 0.13 GB       | 2    | 0   | 0.5636 |
| ogm-ILP-pct   | 3553.71 sec    | -49547.41 | -50061.15 | 1.06 GB       | 63   | 0   | 0.6556 |
| MCBC-pct      | 2053.89 sec    | -49550.10 | -49612.38 | <i>NaN</i> GB | 80   | 56  | 0.6624 |
| ogm-ILP       | 3569.52 sec    | -49536.00 | -50092.16 | 7.53 GB       | 8    | 0   | 0.6444 |
| SA            | <i>NaN</i> sec | -49533.02 | $-\infty$ | <i>NaN</i> GB | 13   | 0   | 0.6541 |

Table 10: geo-surf-3 (300 instances)

| algorithm        | runtime   | value  | bound     | mem     | best | opt | PA     |
|------------------|-----------|--------|-----------|---------|------|-----|--------|
| $\alpha$ -Fusion | 0.00 sec  | 132.14 | $-\infty$ | 0.01 GB | 294  | 0   | 0.8255 |
| ogm-ICM          | 0.00 sec  | 132.77 | $-\infty$ | 0.01 GB | 133  | 0   | 0.8194 |
| ogm-LBP-LF1      | 0.07 sec  | 132.17 | $-\infty$ | 0.01 GB | 272  | 0   | 0.8266 |
| ogm-LBP-LF2      | 0.09 sec  | 132.17 | $-\infty$ | 0.01 GB | 274  | 0   | 0.8266 |
| ogm-LF-1         | 0.00 sec  | 132.77 | $-\infty$ | 0.01 GB | 133  | 0   | 0.8194 |
| ogm-LF-2         | 0.02 sec  | 132.36 | $-\infty$ | 0.01 GB | 207  | 0   | 0.8213 |
| ogm-LF-3         | 0.30 sec  | 132.21 | $-\infty$ | 0.01 GB | 263  | 0   | 0.8258 |
| ogm-BPS          | 0.02 sec  | 132.17 | $-\infty$ | 0.01 GB | 272  | 0   | 0.8266 |
| ogm-LBP-0.5      | 0.07 sec  | 132.17 | $-\infty$ | 0.01 GB | 272  | 0   | 0.8266 |
| ogm-LBP-0.95     | 0.03 sec  | 132.17 | $-\infty$ | 0.01 GB | 272  | 0   | 0.8266 |
| ogm-TRBP-0.5     | 1.14 sec  | 132.16 | $-\infty$ | 0.01 GB | 283  | 0   | 0.8264 |
| ogm-TRBP-0.95    | 0.60 sec  | 132.17 | $-\infty$ | 0.01 GB | 284  | 0   | 0.8264 |
| ogm-TRBPS        | 0.49 sec  | 132.17 | $-\infty$ | 0.01 GB | 286  | 0   | 0.8265 |
| ADDD             | 0.12 sec  | 132.14 | 132.14    | 0.01 GB | 294  | 288 | 0.8260 |
| MPLP             | 0.15 sec  | 132.16 | 132.14    | 0.01 GB | 294  | 189 | 0.8265 |
| MPLP-C           | 0.16 sec  | 132.15 | 132.14    | 0.01 GB | 296  | 194 | 0.8245 |
| ogm-BUNDLE-A     | 21.88 sec | 132.14 | 132.14    | 0.05 GB | 300  | 200 | 0.8259 |
| ogm-BUNDLE-H     | 9.57 sec  | 132.14 | 132.14    | 0.04 GB | 300  | 277 | 0.8259 |
| ogm-SG-A         | 22.37 sec | 132.14 | 132.14    | 0.03 GB | 300  | 185 | 0.8259 |
| ogm-BUNDLE-A+    | 2.78 sec  | 132.14 | 132.14    | 0.05 GB | 300  | 300 | 0.8259 |
| ogm-SG-A+        | 1.80 sec  | 132.14 | 132.14    | 0.03 GB | 300  | 300 | 0.8259 |
| ogm-BUNDLE-A-    | 31.74 sec | 132.14 | 132.09    | 0.05 GB | 296  | 9   | 0.8260 |
| ogm-SG-A-        | 31.06 sec | 132.14 | 131.99    | 0.03 GB | 269  | 6   | 0.8261 |
| ogm-LP-LP        | 0.55 sec  | 132.14 | 132.14    | 0.08 GB | 300  | 300 | 0.8259 |
| BRAOBB-1         | 0.66 sec  | 132.14 | $-\infty$ | 0.14 GB | 298  | 0   | 0.8260 |
| BRAOBB-2         | 6.67 sec  | 132.14 | $-\infty$ | 0.14 GB | 299  | 0   | 0.8259 |
| BRAOBB-3         | 17.99 sec | 132.14 | $-\infty$ | 2.10 GB | 300  | 0   | 0.8259 |
| ogm-ILP          | 0.13 sec  | 132.14 | 132.14    | 0.12 GB | 300  | 300 | 0.8259 |

Table 11: geo-surf-7 (300 instances)

| algorithm        | runtime    | value  | bound     | mem      | best | opt | PA     |
|------------------|------------|--------|-----------|----------|------|-----|--------|
| $\alpha$ -Fusion | 0.02 sec   | 477.83 | $-\infty$ | 0.01 GB  | 257  | 0   | 0.6474 |
| ogm-ICM          | 0.01 sec   | 839.28 | $-\infty$ | 0.01 GB  | 8    | 0   | 0.5093 |
| ogm-LBP-LF1      | 0.60 sec   | 498.45 | $-\infty$ | 0.01 GB  | 66   | 0   | 0.6988 |
| ogm-LBP-LF2      | 0.65 sec   | 498.44 | $-\infty$ | 0.01 GB  | 66   | 0   | 0.6988 |
| ogm-LF-1         | 0.01 sec   | 838.32 | $-\infty$ | 0.01 GB  | 8    | 0   | 0.5086 |
| ogm-LF-2         | 0.12 sec   | 692.68 | $-\infty$ | 0.01 GB  | 14   | 0   | 0.5696 |
| ogm-LF-3         | 9.45 sec   | 620.56 | $-\infty$ | 0.01 GB  | 27   | 0   | 0.6080 |
| ogm-BPS          | 0.37 sec   | 498.34 | $-\infty$ | 0.01 GB  | 69   | 0   | 0.7035 |
| ogm-LBP-0.5      | 0.60 sec   | 498.45 | $-\infty$ | 0.01 GB  | 67   | 0   | 0.6988 |
| ogm-LBP-0.95     | 0.33 sec   | 498.66 | $-\infty$ | 0.01 GB  | 66   | 0   | 0.7039 |
| ogm-TRBP-0.5     | 8.07 sec   | 486.42 | $-\infty$ | 0.01 GB  | 128  | 0   | 0.6768 |
| ogm-TRBP-0.95    | 7.42 sec   | 495.05 | $-\infty$ | 0.01 GB  | 120  | 0   | 0.6601 |
| ogm-TRBPS        | 6.85 sec   | 502.98 | $-\infty$ | 0.01 GB  | 114  | 0   | 0.6605 |
| ADDD             | 0.55 sec   | 476.95 | 476.94    | 0.03 GB  | 296  | 293 | 0.6531 |
| MPLP             | 1.31 sec   | 477.56 | 476.94    | 0.02 GB  | 278  | 195 | 0.6529 |
| MPLP-C           | 1.43 sec   | 477.34 | 476.95    | 0.05 GB  | 282  | 198 | 0.6529 |
| ogm-BUNDLE-A     | 53.41 sec  | 477.08 | 476.10    | 0.08 GB  | 261  | 22  | 0.6545 |
| ogm-BUNDLE-H     | 41.45 sec  | 476.95 | 476.86    | 0.07 GB  | 299  | 180 | 0.6529 |
| ogm-SG-A         | 52.38 sec  | 479.02 | 473.05    | 0.03 GB  | 164  | 14  | 0.6548 |
| ogm-BUNDLE-A+    | 32.29 sec  | 476.95 | 476.91    | 0.06 GB  | 298  | 238 | 0.6529 |
| ogm-SG-A+        | 29.96 sec  | 477.28 | 476.76    | 0.03 GB  | 294  | 246 | 0.6532 |
| ogm-BUNDLE-A-    | 53.63 sec  | 479.37 | 470.50    | 0.08 GB  | 112  | 2   | 0.6547 |
| ogm-SG-A-        | 52.14 sec  | 488.41 | 452.65    | 0.03 GB  | 52   | 1   | 0.6623 |
| ogm-LP-LP        | 2.74 sec   | 476.95 | 476.94    | 0.42 GB  | 299  | 299 | 0.6530 |
| BRAOBB-1         | 973.62 sec | 479.82 | $-\infty$ | 17.72 GB | 214  | 0   | 0.6602 |
| BRAOBB-2         | 957.11 sec | 477.10 | $-\infty$ | 17.28 GB | 267  | 0   | 0.6532 |
| BRAOBB-3         | 685.84 sec | 477.11 | $-\infty$ | 5.69 GB  | 269  | 0   | 0.6531 |
| ogm-ILP          | 0.95 sec   | 476.95 | 476.95    | 0.74 GB  | 300  | 300 | 0.6529 |

Table 12: image-seg-3rdorder (100 instances)

| algorithm       | runtime   | value   | bound     | mem     | best | opt | VI     | RI     |
|-----------------|-----------|---------|-----------|---------|------|-----|--------|--------|
| ogm-ICM         | 6.30 sec  | 6030.49 | $-\infty$ | 0.01 GB | 0    | 0   | 2.7089 | 0.5031 |
| ogm-LF-1        | 2.23 sec  | 6030.29 | $-\infty$ | 0.01 GB | 0    | 0   | 2.7095 | 0.5033 |
| MCR-CC          | 32.56 sec | 5822.31 | 5465.15   | 0.28 GB | 0    | 0   | 2.7722 | 0.7512 |
| MCR-CCFDB       | 20.85 sec | 5823.09 | 5465.15   | 0.27 GB | 0    | 0   | 2.7705 | 0.7511 |
| MCR-CCFDB-OWC   | 21.63 sec | 5823.59 | 5465.29   | 0.27 GB | 0    | 0   | 2.7705 | 0.7511 |
| MCI-CCFDB-CCIFD | 46.68 sec | 5627.52 | 5627.52   | 0.68 GB | 100  | 100 | 2.6586 | 0.7727 |
| MCI-CCI         | 70.54 sec | 5628.39 | 5627.49   | 0.73 GB | 99   | 98  | 2.6589 | 0.7721 |
| MCI-CCIFD       | 50.78 sec | 5627.52 | 5627.52   | 0.72 GB | 99   | 100 | 2.6586 | 0.7727 |

Table 13: image-seg (100 instances)

| algorithm       | runtime  | value   | bound     | mem     | best | opt | VI     | RI     |
|-----------------|----------|---------|-----------|---------|------|-----|--------|--------|
| ogm-ICM         | 3.98 sec | 4705.07 | $-\infty$ | 0.01 GB | 0    | 0   | 2.8580 | 0.5954 |
| ogm-KL          | 1.46 sec | 4608.49 | $-\infty$ | 0.01 GB | 0    | 0   | 2.6432 | 0.6400 |
| ogm-LF-1        | 1.35 sec | 4705.01 | $-\infty$ | 0.01 GB | 0    | 0   | 2.8583 | 0.5953 |
| MCR-CC          | 8.54 sec | 4447.14 | 4442.34   | 0.15 GB | 35   | 35  | 2.5471 | 0.7822 |
| MCR-CCFDB       | 4.34 sec | 4447.14 | 4442.34   | 0.06 GB | 35   | 35  | 2.5469 | 0.7822 |
| MCR-CCFDB-OWC   | 4.34 sec | 4447.09 | 4442.34   | 0.06 GB | 35   | 35  | 2.5468 | 0.7822 |
| MCI-CCFDB-CCIFD | 4.89 sec | 4442.64 | 4442.64   | 0.17 GB | 100  | 100 | 2.5365 | 0.7821 |
| MCI-CCI         | 2.43 sec | 4442.64 | 4442.64   | 0.14 GB | 100  | 100 | 2.5365 | 0.7821 |
| MCI-CCIFD       | 2.24 sec | 4442.64 | 4442.64   | 0.09 GB | 100  | 100 | 2.5367 | 0.7821 |

Table 14: inclusion (10 instances)

| algorithm        | runtime     | value      | bound     | mem     | best | opt | PA     |
|------------------|-------------|------------|-----------|---------|------|-----|--------|
| $\alpha$ -Fusion | 0.04 sec    | 1587.13    | $-\infty$ | 0.01 GB | 0    | 0   | 0.6771 |
| ogm-ICM          | 0.01 sec    | 1556.20    | $-\infty$ | 0.01 GB | 0    | 0   | 0.6206 |
| ogm-LBP-LF1      | 19.33 sec   | 1400.66    | $-\infty$ | 0.01 GB | 4    | 0   | 0.9490 |
| ogm-LBP-LF2      | 19.37 sec   | 1400.61    | $-\infty$ | 0.02 GB | 7    | 0   | 0.9495 |
| ogm-LF-1         | 0.01 sec    | 1556.20    | $-\infty$ | 0.01 GB | 0    | 0   | 0.6206 |
| ogm-LF-2         | 0.09 sec    | 1476.39    | $-\infty$ | 0.01 GB | 0    | 0   | 0.7630 |
| ogm-LF-3         | 1.14 sec    | 1461.23    | $-\infty$ | 0.01 GB | 0    | 0   | 0.8011 |
| ogm-BPS          | 21.42 sec   | 2200.68    | $-\infty$ | 0.01 GB | 6    | 0   | 0.9489 |
| ogm-LBP-0.5      | 19.77 sec   | 2100.61    | $-\infty$ | 0.01 GB | 7    | 0   | 0.9487 |
| ogm-LBP-0.95     | 19.82 sec   | 2700.74    | $-\infty$ | 0.01 GB | 8    | 0   | 0.9492 |
| ogm-TRBP-0.5     | 21.42 sec   | 1900.84    | $-\infty$ | 0.01 GB | 5    | 0   | 0.9491 |
| ogm-TRBP-0.95    | 20.65 sec   | 2600.73    | $-\infty$ | 0.01 GB | 8    | 0   | 0.9481 |
| ogm-TRBPS        | 20.68 sec   | 2300.80    | $-\infty$ | 0.01 GB | 7    | 0   | 0.9491 |
| ADDD             | 6.23 sec    | 3400.81    | 1400.31   | 0.03 GB | 1    | 1   | 0.9479 |
| MPLP             | 5.94 sec    | 4000.44    | 1400.30   | 0.02 GB | 2    | 1   | 0.9479 |
| MPLP-C           | 3579.25 sec | 4200.37    | 1400.35   | 0.08 GB | 2    | 1   | 0.9470 |
| ogm-BUNDLE-A     | 77.24 sec   | 4007.73    | 1392.01   | 0.02 GB | 2    | 0   | 0.9391 |
| ogm-BUNDLE-H     | 73.24 sec   | 1400.76    | 1400.32   | 0.02 GB | 3    | 1   | 0.9496 |
| ogm-SG-A         | 78.38 sec   | 26797.36   | 1309.23   | 0.01 GB | 0    | 0   | 0.8518 |
| ogm-BUNDLE-A+    | 77.44 sec   | 3909.68    | 1389.99   | 0.02 GB | 0    | 0   | 0.9336 |
| ogm-SG-A+        | 77.57 sec   | 26797.36   | 1309.23   | 0.01 GB | 0    | 0   | 0.8518 |
| ogm-BUNDLE-A-    | 76.92 sec   | 1400.69    | 1400.29   | 0.02 GB | 3    | 0   | 0.9499 |
| ogm-SG-A-        | 75.57 sec   | 11920.10   | 1375.81   | 0.01 GB | 2    | 0   | 0.9210 |
| ogm-LP-LP        | 18.27 sec   | 4100.60    | 1400.33   | 0.30 GB | 1    | 1   | 0.9482 |
| MCI-TC-MTC-TCI   | 61.46 sec   | 1400.57    | 1400.57   | 0.56 GB | 10   | 10  | 0.9496 |
| BRAOBB-1         | 3600.01 sec | <i>Inf</i> | $-\infty$ | 0.90 GB | 0    | 0   | 0.3692 |
| BRAOBB-2         | 3600.01 sec | 1401.73    | $-\infty$ | 0.80 GB | 0    | 0   | 0.9461 |
| BRAOBB-3         | 3600.01 sec | 1401.64    | $-\infty$ | 1.79 GB | 0    | 0   | 0.9467 |
| ogm-ILP          | 6.21 sec    | 1400.57    | 1400.57   | 0.74 GB | 10   | 10  | 0.9496 |

Table 15: inpainting-n4 (2 instances)

| algorithm                    | runtime     | value     | bound     | mem     | best | opt |
|------------------------------|-------------|-----------|-----------|---------|------|-----|
| $\alpha$ -Exp-pct            | 0.11 sec    | 454.75    | $-\infty$ | 0.02 GB | 1    | 0   |
| FastPD                       | 0.01 sec    | 454.75    | 294.89    | 0.03 GB | 1    | 0   |
| FastPD-pct                   | 0.07 sec    | 454.75    | $-\infty$ | 0.03 GB | 1    | 0   |
| mrf- $\alpha$ -Exp-TL        | 0.01 sec    | 454.75    | $-\infty$ | 0.01 GB | 1    | 0   |
| mrf- $\alpha/\beta$ -Swap-TL | 0.01 sec    | 454.35    | $-\infty$ | 0.01 GB | 2    | 0   |
| ogm-FastPD-LF1               | 0.05 sec    | 454.75    | 294.89    | 0.04 GB | 1    | 0   |
| ogm-FastPD-LF2               | 0.20 sec    | 454.75    | 294.89    | 0.04 GB | 1    | 0   |
| ogm-ICM                      | 0.06 sec    | 3453.29   | $-\infty$ | 0.02 GB | 0    | 0   |
| ogm-LF-1                     | 0.04 sec    | 3453.29   | $-\infty$ | 0.02 GB | 0    | 0   |
| ogm-LF-2                     | 0.18 sec    | 3453.29   | $-\infty$ | 0.03 GB | 0    | 0   |
| ogm-TRWS-LF1                 | 1.32 sec    | 489.70    | 448.09    | 0.02 GB | 1    | 1   |
| ogm-TRWS-LF2                 | 1.45 sec    | 489.30    | 448.09    | 0.03 GB | 1    | 1   |
| mrf-LBP-TL                   | 4.25 sec    | 475.56    | $-\infty$ | 0.01 GB | 1    | 0   |
| mrf-BPS-TL                   | 1.69 sec    | 454.35    | $-\infty$ | 0.01 GB | 2    | 0   |
| ogm-BPS                      | 11.99 sec   | 2042.43   | $-\infty$ | 0.06 GB | 1    | 0   |
| ogm-LBP-0.5                  | 17.10 sec   | 480.27    | $-\infty$ | 0.06 GB | 1    | 0   |
| ogm-LBP-0.95                 | 14.70 sec   | 483.41    | $-\infty$ | 0.06 GB | 1    | 0   |
| ogm-TRBP-0.5                 | 20.63 sec   | 480.27    | $-\infty$ | 0.06 GB | 1    | 0   |
| ogm-TRBP-0.95                | 17.29 sec   | 485.38    | $-\infty$ | 0.06 GB | 1    | 0   |
| ogm-TRBPS                    | 13.66 sec   | 2042.43   | $-\infty$ | 0.06 GB | 1    | 0   |
| MCR-TC-MTC                   | 1386.81 sec | 645.89    | 448.27    | 0.20 GB | 1    | 1   |
| MCR-pct                      | 1248.88 sec | 1179.00   | 448.27    | 0.18 GB | 0    | 0   |
| mrf-TRWS-TL                  | 0.97 sec    | 490.48    | 448.09    | 0.01 GB | 1    | 1   |
| ogm-ADSAL                    | 59.91 sec   | 454.75    | 448.27    | 0.03 GB | 1    | 1   |
| ogm-BUNDLE-A                 | 39.98 sec   | 455.25    | 448.23    | 0.04 GB | 1    | 0   |
| ogm-BUNDLE-H                 | 19.13 sec   | 455.25    | 448.22    | 0.04 GB | 1    | 1   |
| ogm-SG-A                     | 21.38 sec   | 455.25    | 447.76    | 0.03 GB | 1    | 1   |
| ogm-BUNDLE-A+                | 19.49 sec   | 455.25    | 447.84    | 0.04 GB | 1    | 1   |
| ogm-SG-A+                    | 20.25 sec   | 455.25    | 441.36    | 0.03 GB | 1    | 1   |
| ogm-BUNDLE-A-                | 39.31 sec   | 455.25    | 448.08    | 0.04 GB | 1    | 0   |
| ogm-SG-A-                    | 39.48 sec   | 454.35    | 447.89    | 0.03 GB | 2    | 0   |
| TRWS-pct                     | 2.77 sec    | 489.30    | 448.10    | 0.02 GB | 1    | 1   |
| MCI-TC-MTC-TCI               | 1812.16 sec | 462.60    | 448.86    | 0.44 GB | 1    | 1   |
| MCI-pct                      | 1807.10 sec | 270479.80 | $-\infty$ | 0.41 GB | 1    | 1   |
| ogm-CombiLP                  | 129.04 sec  | 461.81    | 446.66    | 0.49 GB | 1    | 1   |

Table 16: inpainting-n8 (2 instances)

| algorithm                | runtime     | value     | bound     | mem     | best | opt |
|--------------------------|-------------|-----------|-----------|---------|------|-----|
| $\alpha$ -Exp-pct        | 0.24 sec    | 464.76    | $-\infty$ | 0.04 GB | 2    | 0   |
| $\alpha$ -Exp-VIEW       | 0.43 sec    | 465.02    | $-\infty$ | 0.03 GB | 1    | 0   |
| FastPD                   | 0.09 sec    | 465.02    | 136.28    | 0.04 GB | 1    | 0   |
| FastPD-pct               | 0.17 sec    | 464.76    | $-\infty$ | 0.04 GB | 2    | 0   |
| ogm-FastPD-LF1           | 0.17 sec    | 465.02    | -67.89    | 0.05 GB | 1    | 0   |
| ogm-FastPD-LF2           | 0.67 sec    | 465.02    | -67.89    | 0.07 GB | 1    | 0   |
| ogm-ICM                  | 0.09 sec    | 3451.36   | $-\infty$ | 0.02 GB | 0    | 0   |
| ogm-LF-1                 | 0.06 sec    | 3451.36   | $-\infty$ | 0.03 GB | 0    | 0   |
| ogm-LF-2                 | 0.56 sec    | 3451.36   | $-\infty$ | 0.04 GB | 0    | 0   |
| ogm-TRWS-LF1             | 11.25 sec   | 499.36    | 453.96    | 0.03 GB | 1    | 0   |
| ogm-TRWS-LF2             | 10.24 sec   | 499.30    | 453.96    | 0.05 GB | 1    | 0   |
| $\alpha\beta$ -Swap-VIEW | 0.38 sec    | 465.02    | $-\infty$ | 0.03 GB | 1    | 0   |
| BPS-TL                   | 7.76 sec    | 468.21    | $-\infty$ | 0.02 GB | 1    | 0   |
| ogm-BPS                  | 9.64 sec    | 493.96    | $-\infty$ | 0.10 GB | 1    | 0   |
| ogm-LBP-0.5              | 41.77 sec   | 495.52    | $-\infty$ | 0.10 GB | 1    | 0   |
| ogm-LBP-0.95             | 11.67 sec   | 493.96    | $-\infty$ | 0.10 GB | 1    | 0   |
| ogm-TRBP-0.5             | 49.98 sec   | 495.80    | $-\infty$ | 0.10 GB | 1    | 0   |
| ogm-TRBP-0.95            | 13.36 sec   | 493.96    | $-\infty$ | 0.10 GB | 1    | 0   |
| ogm-TRBPS                | 10.58 sec   | 493.96    | $-\infty$ | 0.10 GB | 1    | 0   |
| MCR-TC-MTC               | 2073.02 sec | 679.26    | 445.02    | 0.17 GB | 0    | 0   |
| MCR-pct                  | 1871.11 sec | 270610.25 | $-\infty$ | 0.26 GB | 0    | 0   |
| ogm-ADSAL                | 297.48 sec  | 467.20    | 455.48    | 0.05 GB | 1    | 0   |
| ogm-BUNDLE-A             | 96.60 sec   | 465.26    | 455.43    | 0.10 GB | 1    | 0   |
| ogm-BUNDLE-H             | 97.86 sec   | 465.34    | 455.43    | 0.11 GB | 0    | 0   |
| ogm-SG-A                 | 93.27 sec   | 466.81    | 454.78    | 0.06 GB | 0    | 0   |
| ogm-BUNDLE-A+            | 102.22 sec  | 465.26    | 454.31    | 0.11 GB | 1    | 0   |
| ogm-SG-A+                | 100.14 sec  | 465.76    | 446.05    | 0.06 GB | 0    | 0   |
| ogm-BUNDLE-A-            | 99.87 sec   | 465.26    | 455.23    | 0.11 GB | 1    | 0   |
| ogm-SG-A-                | 86.02 sec   | 466.81    | 453.50    | 0.06 GB | 0    | 0   |
| TRWS-TL                  | 10.15 sec   | 500.09    | 453.96    | 0.02 GB | 1    | 0   |
| TRWS-pct                 | 19.74 sec   | 467.69    | 453.97    | 0.03 GB | 1    | 0   |
| MCI-TC-MTC-TCI           | 2166.25 sec | 534.07    | 445.02    | 0.40 GB | 1    | 1   |
| MCI-pct                  | 1879.20 sec | 270485.10 | $-\infty$ | 0.26 GB | 1    | 1   |
| ogm-CombiLP              | 2117.77 sec | 465.35    | 450.06    | 1.05 GB | 1    | 1   |

Table 17: knott-3d-150 (8 instances)

| algorithm       | runtime   | value    | bound     | mem     | best | opt | VI     | RI     |
|-----------------|-----------|----------|-----------|---------|------|-----|--------|--------|
| ogm-ICM         | 2.05 sec  | -4179.30 | $-\infty$ | 0.01 GB | 0    | 0   | 2.3947 | 0.7811 |
| ogm-KL          | 0.15 sec  | -4431.67 | $-\infty$ | 0.01 GB | 0    | 0   | 2.0648 | 0.8085 |
| ogm-LF-1        | 0.72 sec  | -4179.30 | $-\infty$ | 0.01 GB | 0    | 0   | 2.3947 | 0.7811 |
| MCR-CC          | 17.72 sec | -4568.90 | -4572.10  | 0.08 GB | 3    | 3   | 0.9178 | 0.9232 |
| MCR-CCFDB       | 1.87 sec  | -4568.90 | -4572.10  | 0.03 GB | 3    | 3   | 0.9178 | 0.9232 |
| MCR-CCFDB-OWC   | 2.00 sec  | -4570.87 | -4571.86  | 0.03 GB | 7    | 7   | 0.9076 | 0.9236 |
| MCI-CCFDB-CCIFD | 1.96 sec  | -4571.69 | -4571.69  | 0.03 GB | 8    | 8   | 0.9063 | 0.9236 |
| MCI-CCI         | 0.57 sec  | -4571.69 | -4571.69  | 0.04 GB | 8    | 8   | 0.9063 | 0.9236 |
| MCI-CCIFD       | 0.58 sec  | -4571.69 | -4571.69  | 0.03 GB | 8    | 8   | 0.9063 | 0.9236 |

Table 18: knott-3d-300 (8 instances)

| algorithm       | runtime     | value     | bound     | mem     | best | opt | VI     | RI     |
|-----------------|-------------|-----------|-----------|---------|------|-----|--------|--------|
| ogm-ICM         | 84.37 sec   | -25196.51 | $-\infty$ | 0.01 GB | 0    | 0   | 4.1365 | 0.7235 |
| ogm-KL          | 13.16 sec   | -25556.93 | $-\infty$ | 0.01 GB | 0    | 0   | 4.1318 | 0.6858 |
| ogm-LF-1        | 29.08 sec   | -25243.76 | $-\infty$ | 0.02 GB | 0    | 0   | 4.1297 | 0.7241 |
| MCR-CC          | 3423.65 sec | -26161.81 | -27434.30 | 0.57 GB | 1    | 1   | 1.7995 | 0.8405 |
| MCR-CCFDB       | 1338.99 sec | -27276.12 | -27307.22 | 0.15 GB | 1    | 1   | 1.6336 | 0.8850 |
| MCR-CCFDB-OWC   | 1367.03 sec | -27287.23 | -27309.62 | 0.15 GB | 6    | 6   | 1.6342 | 0.8849 |
| MCI-CCFDB-CCIFD | 1261.99 sec | -26826.57 | -27308.19 | 0.37 GB | 6    | 6   | 1.7010 | 0.8840 |
| MCI-CCI         | 220.30 sec  | -27302.78 | -27305.02 | 0.28 GB | 8    | 7   | 1.6352 | 0.8849 |
| MCI-CCIFD       | 104.55 sec  | -27302.78 | -27302.78 | 0.16 GB | 8    | 8   | 1.6352 | 0.8849 |

Table 19: knott-3d-450 (8 instances)

| algorithm       | runtime     | value     | bound     | mem     | best | opt | VI     | RI     |
|-----------------|-------------|-----------|-----------|---------|------|-----|--------|--------|
| ogm-ICM         | 883.63 sec  | -72464.54 | $-\infty$ | 0.03 GB | 0    | 0   | 4.9342 | 0.6783 |
| ogm-KL          | 186.89 sec  | -73188.82 | $-\infty$ | 0.03 GB | 0    | 0   | 4.9270 | 0.6409 |
| ogm-LF-1        | 298.07 sec  | -72479.60 | $-\infty$ | 0.04 GB | 0    | 0   | 4.9325 | 0.6783 |
| MCR-CC          | 9814.45 sec | -4892.36  | -83272.85 | 0.39 GB | 0    | 0   | 4.6616 | 0.0998 |
| MCR-CCFDB       | 6404.34 sec | -4892.36  | -83272.85 | 0.20 GB | 0    | 0   | 4.6616 | 0.0998 |
| MCR-CCFDB-OWC   | 6455.21 sec | -4892.36  | -83272.85 | 0.19 GB | 0    | 0   | 4.6616 | 0.0998 |
| MCI-CCFDB-CCIFD | 6404.14 sec | -4892.36  | -83272.85 | 0.19 GB | 0    | 0   | 4.6616 | 0.0998 |
| MCI-CCI         | 1196.62 sec | -78135.34 | -78518.55 | 0.83 GB | 7    | 6   | 2.0027 | 0.8653 |
| MCI-CCIFD       | 1379.90 sec | -78180.20 | -78507.25 | 0.54 GB | 7    | 6   | 2.0096 | 0.8643 |

Table 20: matching (4 instances)

| algorithm        | runtime     | value           | bound     | mem     | best | opt | MPE    |
|------------------|-------------|-----------------|-----------|---------|------|-----|--------|
| $\alpha$ -Fusion | 0.00 sec    | 146500000000.00 | $-\infty$ | 0.01 GB | 0    | 0   | 7.2531 |
| ogm-ICM          | 0.00 sec    | 95.73           | $-\infty$ | 0.01 GB | 0    | 0   | 6.3151 |
| ogm-LBP-LF1      | 0.00 sec    | 85.87           | $-\infty$ | 0.01 GB | 0    | 0   | 6.0827 |
| ogm-LBP-LF2      | 0.19 sec    | 38.07           | $-\infty$ | 0.01 GB | 1    | 0   | 5.4469 |
| ogm-LF-1         | 0.00 sec    | 95.73           | $-\infty$ | 0.01 GB | 0    | 0   | 6.3151 |
| ogm-LF-2         | 0.29 sec    | 40.79           | $-\infty$ | 0.01 GB | 0    | 0   | 5.7689 |
| ogm-LF-3         | 12.35 sec   | 39.81           | $-\infty$ | 0.01 GB | 0    | 0   | 5.6346 |
| ogm-TRWS-LF1     | 0.03 sec    | 61.46           | 15.22     | 0.01 GB | 0    | 0   | 3.7921 |
| ogm-TRWS-LF2     | 0.32 sec    | 33.31           | 15.22     | 0.01 GB | 0    | 0   | 3.1763 |
| BPS-TAB          | 0.11 sec    | 40.26           | $-\infty$ | 0.01 GB | 1    | 0   | 4.9692 |
| ogm-BPS          | 0.00 sec    | 105000000082.06 | $-\infty$ | 0.01 GB | 0    | 0   | 6.0228 |
| ogm-LBP-0.5      | 0.00 sec    | 105000000082.06 | $-\infty$ | 0.01 GB | 0    | 0   | 6.0228 |
| ogm-LBP-0.95     | 0.00 sec    | 105000000082.06 | $-\infty$ | 0.01 GB | 0    | 0   | 6.0228 |
| ogm-TRBP-0.5     | 0.00 sec    | 105000000082.06 | $-\infty$ | 0.01 GB | 0    | 0   | 6.0228 |
| ogm-TRBP-0.95    | 0.00 sec    | 105000000082.06 | $-\infty$ | 0.01 GB | 0    | 0   | 6.0228 |
| ogm-TRBPS        | 0.00 sec    | 105000000082.06 | $-\infty$ | 0.01 GB | 0    | 0   | 6.0228 |
| ADDD             | 1.38 sec    | 105000000039.23 | 16.35     | 0.01 GB | 0    | 0   | 3.2429 |
| MPLP             | 0.26 sec    | 65000000041.76  | 15.16     | 0.01 GB | 0    | 0   | 3.1630 |
| MPLP-C           | 3.51 sec    | 21.22           | 21.22     | 0.03 GB | 4    | 4   | 0.0907 |
| ogm-ADSAL        | 1380.55 sec | 32.47           | 15.62     | 0.01 GB | 0    | 0   | 2.9236 |
| ogm-BUNDLE-A     | 1.56 sec    | 10000000041.82  | 14.47     | 0.01 GB | 0    | 0   | 3.0245 |
| ogm-BUNDLE-H     | 1.57 sec    | 7500000053.86   | 15.91     | 0.01 GB | 0    | 0   | 3.5518 |
| ogm-SG-A         | 1.38 sec    | 7500000099.44   | 9.27      | 0.01 GB | 0    | 0   | 4.9390 |
| ogm-BUNDLE-A+    | 1.56 sec    | 10000000048.72  | 14.25     | 0.01 GB | 0    | 0   | 3.4616 |
| ogm-SG-A+        | 1.37 sec    | 7500000101.08   | 9.27      | 0.01 GB | 0    | 0   | 5.1003 |
| ogm-BUNDLE-A-    | 1.57 sec    | 10000000048.72  | 14.49     | 0.01 GB | 0    | 0   | 3.4616 |
| ogm-SG-A-        | 1.32 sec    | 7500000096.42   | 9.27      | 0.01 GB | 0    | 0   | 5.9589 |
| ogm-LP-LP        | 21.38 sec   | 102500000036.76 | 16.35     | 0.07 GB | 0    | 0   | 3.3482 |
| TRWS-TAB         | 0.03 sec    | 64.19           | 15.22     | 0.01 GB | 0    | 0   | 3.8159 |
| BRAOBB-1         | 2.05 sec    | 21.22           | $-\infty$ | 0.06 GB | 4    | 0   | 0.0907 |
| BRAOBB-2         | 8.09 sec    | 21.22           | $-\infty$ | 0.06 GB | 4    | 0   | 0.0907 |
| BRAOBB-3         | 41.69 sec   | 21.22           | $-\infty$ | 0.81 GB | 4    | 0   | 0.0907 |
| ADDD-BB          | 1192.02 sec | 37500000029.63  | $-\infty$ | 0.02 GB | 3    | 2   | 1.4615 |
| ogm-ASTAR        | 0.80 sec    | 21.22           | 21.22     | 0.05 GB | 4    | 4   | 0.0907 |
| ogm-CombiLP      | 314.52 sec  | 21.22           | 21.22     | 0.17 GB | 4    | 4   | 0.0907 |
| ogm-ILP          | 402.09 sec  | 21.22           | 21.22     | 0.17 GB | 4    | 4   | 0.0907 |

Table 21: modularity-clustering (6 instances)

| algorithm       | runtime     | value   | bound     | mem     | best | opt |
|-----------------|-------------|---------|-----------|---------|------|-----|
| ogm-ICM         | 0.09 sec    | 0.0000  | $-\infty$ | 0.01 GB | 0    | 0   |
| ogm-KL          | 0.01 sec    | -0.4860 | $-\infty$ | 0.01 GB | 3    | 0   |
| ogm-LF-1        | 0.03 sec    | 0.0000  | $-\infty$ | 0.01 GB | 0    | 0   |
| MCR-CC          | 100.37 sec  | -0.4543 | -0.5094   | 0.14 GB | 2    | 1   |
| MCR-CCFDB       | 2.15 sec    | -0.4543 | -0.5094   | 0.03 GB | 1    | 1   |
| MCR-CCFDB-OWC   | 602.75 sec  | -0.4652 | -0.4962   | 0.03 GB | 5    | 5   |
| MCI-CCFDB-CCIFD | 601.38 sec  | -0.4400 | -0.5021   | 1.58 GB | 5    | 5   |
| MCI-CCI         | 1207.07 sec | -0.4312 | -0.5158   | 2.69 GB | 4    | 4   |
| MCI-CCIFD       | 1204.03 sec | -0.4399 | -0.5176   | 3.02 GB | 4    | 4   |



Table 22: mrf-inpainting (2 instances)

| algorithm                   | runtime      | value       | bound         | mem     | best | opt | CE      |
|-----------------------------|--------------|-------------|---------------|---------|------|-----|---------|
| FastPD                      | 7.72 sec     | 32939430.00 | 0.00          | 4.10 GB | 0    | 0   | 14.7039 |
| mrf- $\alpha$ -Exp-TL       | 42.07 sec    | 27266168.50 | $-\infty$     | 0.30 GB | 0    | 0   | 11.5700 |
| mrf- $\alpha\beta$ -Swap-TL | 92.06 sec    | 27055552.00 | $-\infty$     | 0.33 GB | 0    | 0   | 11.6051 |
| ogm-FastPD-LF1              | 174.04 sec   | 27509437.00 | -891985522.00 | 4.14 GB | 0    | 0   | 13.0723 |
| ogm-FastPD-LF2              | 7264.73 sec  | 27209149.50 | -891985522.00 | 4.15 GB | 0    | 0   | 12.6623 |
| ogm-ICM                     | 229.56 sec   | 32467486.00 | $-\infty$     | 0.47 GB | 0    | 0   | 27.8092 |
| ogm-LF-1                    | 75.71 sec    | 32467900.00 | $-\infty$     | 0.29 GB | 0    | 0   | 27.8123 |
| ogm-LF-2                    | 3600.18 sec  | 31188870.50 | $-\infty$     | 0.31 GB | 0    | 0   | 26.4747 |
| ogm-TRWS-LF1                | 679.63 sec   | 26464015.00 | 26462450.59   | 0.99 GB | 0    | 0   | 10.9921 |
| ogm-TRWS-LF2                | 2404.03 sec  | 26463829.00 | 26462450.59   | 1.01 GB | 1    | 0   | 10.9923 |
| mrf-LBP-TL                  | 573.91 sec   | 26597364.50 | $-\infty$     | 0.63 GB | 0    | 0   | 10.5954 |
| mrf-BPS-TL                  | 593.35 sec   | 26612532.50 | $-\infty$     | 0.82 GB | 0    | 0   | 12.0199 |
| ogm-BPS                     | 3677.40 sec  | 37224573.50 | $-\infty$     | 2.97 GB | 0    | 0   | 10.9955 |
| ogm-LBP-0.5                 | 3691.83 sec  | 29749485.50 | $-\infty$     | 2.97 GB | 0    | 0   | 10.5383 |
| ogm-LBP-0.95                | 3686.06 sec  | 32138459.50 | $-\infty$     | 2.97 GB | 0    | 0   | 10.9303 |
| ogm-TRBP-0.5                | 3642.96 sec  | 42541674.50 | $-\infty$     | 2.97 GB | 0    | 0   | 10.5823 |
| ogm-TRBP-0.95               | 3643.81 sec  | 33260789.50 | $-\infty$     | 2.97 GB | 0    | 0   | 10.9427 |
| ogm-TRBPS                   | 3711.16 sec  | 37905326.50 | $-\infty$     | 2.97 GB | 0    | 0   | 10.9502 |
| mrf-TRWS-TL                 | 563.15 sec   | 26464865.00 | 26462450.59   | 0.82 GB | 0    | 0   | 10.9940 |
| ogm-ADSAL                   | 3892.14 sec  | 26487768.50 | 26445564.61   | 2.46 GB | 0    | 0   | 10.9601 |
| ogm-BUNDLE-A                | 3611.21 sec  | 56032960.00 | 24444132.61   | 2.65 GB | 0    | 0   | 12.9335 |
| ogm-BUNDLE-H                | 3615.07 sec  | 56112406.50 | 24222273.40   | 2.65 GB | 0    | 0   | 12.2584 |
| ogm-SG-A                    | 3635.02 sec  | 47096469.50 | 24752599.88   | 0.79 GB | 0    | 0   | 11.4979 |
| ogm-BUNDLE-A+               | 3640.03 sec  | 56032960.00 | 24453012.25   | 2.65 GB | 0    | 0   | 12.9335 |
| ogm-SG-A+                   | 3643.39 sec  | 37714765.00 | 23395920.50   | 0.79 GB | 0    | 0   | 11.5747 |
| ogm-BUNDLE-A-               | 3635.32 sec  | 56032960.00 | 24383588.44   | 2.65 GB | 0    | 0   | 12.9335 |
| ogm-SG-A-                   | 3629.00 sec  | 56028769.00 | 24368624.32   | 0.79 GB | 0    | 0   | 12.9678 |
| ogm-CombiLP                 | 48723.23 sec | 26467926.00 | 26461874.39   | 2.09 GB | 1    | 0   | 10.9993 |

Table 23: mrf-photomontage (2 instances)

| algorithm                    | runtime     | value          | bound     | mem     | best | opt |
|------------------------------|-------------|----------------|-----------|---------|------|-----|
| mrf- $\alpha$ -Exp-TAB       | 7.37 sec    | 168457.00      | $-\infty$ | 0.89 GB | 2    | 0   |
| mrf- $\alpha\beta$ -Swap-TAB | 9.77 sec    | 170858.50      | $-\infty$ | 0.89 GB | 0    | 0   |
| ogm-ICM                      | 1.66 sec    | 18096281786.50 | $-\infty$ | 0.58 GB | 0    | 0   |
| ogm-LF-1                     | 0.84 sec    | 18096281786.50 | $-\infty$ | 0.72 GB | 0    | 0   |
| ogm-LF-2                     | 11.15 sec   | 18096106879.00 | $-\infty$ | 0.81 GB | 0    | 0   |
| ogm-TRWS-LF1                 | 341.22 sec  | 1239959.00     | 166827.12 | 1.37 GB | 0    | 0   |
| ogm-TRWS-LF2                 | 323.48 sec  | 735193.00      | 166827.12 | 1.46 GB | 0    | 0   |
| mrf-LBP-TAB                  | 458.73 sec  | 438611.00      | $-\infty$ | 0.93 GB | 0    | 0   |
| mrf-BPS-TAB                  | 188.37 sec  | 2217579.50     | $-\infty$ | 1.36 GB | 0    | 0   |
| ogm-BPS                      | 1244.90 sec | 1176748.50     | $-\infty$ | 2.35 GB | 0    | 0   |
| ogm-LBP-0.5                  | 1056.92 sec | 457885.50      | $-\infty$ | 2.35 GB | 0    | 0   |
| ogm-LBP-0.95                 | 717.99 sec  | 454990.50      | $-\infty$ | 2.35 GB | 0    | 0   |
| ogm-TRBP-0.5                 | 1795.88 sec | 457885.50      | $-\infty$ | 2.42 GB | 0    | 0   |
| ogm-TRBP-0.95                | 1508.58 sec | 454990.50      | $-\infty$ | 2.42 GB | 0    | 0   |
| ogm-TRBPS                    | 1483.45 sec | 1176748.50     | $-\infty$ | 2.42 GB | 0    | 0   |
| mrf-TRWS-TAB                 | 203.79 sec  | 1243144.00     | 166827.07 | 1.36 GB | 0    | 0   |
| ogm-ADSAL                    | 3605.24 sec | 185560.00      | 167274.34 | 1.09 GB | 0    | 0   |
| ogm-BUNDLE-A                 | 1641.14 sec | 676079.00      | 112183.66 | 1.78 GB | 0    | 0   |
| ogm-BUNDLE-H                 | 1588.13 sec | 599206.00      | 111100.35 | 1.79 GB | 0    | 0   |
| ogm-SG-A                     | 1616.05 sec | 3846787.00     | 26005.24  | 1.26 GB | 0    | 0   |
| ogm-BUNDLE-A+                | 1631.83 sec | 676079.00      | 112183.66 | 1.77 GB | 0    | 0   |
| ogm-SG-A+                    | 1624.53 sec | 5248670.00     | 23229.00  | 1.26 GB | 0    | 0   |
| ogm-BUNDLE-A-                | 1579.42 sec | 707036.50      | 112227.62 | 1.78 GB | 0    | 0   |
| ogm-SG-A-                    | 1499.23 sec | 3006179.50     | 96295.35  | 1.26 GB | 0    | 0   |
| ogm-CombiLP                  | 3707.70 sec | 214853.50      | 165504.72 | 7.63 GB | 0    | 0   |

Table 24: mrf-stereo (3 instances)

| algorithm                    | runtime     | value      | bound        | mem      | best | opt | PA2    | PA4    | HD     |
|------------------------------|-------------|------------|--------------|----------|------|-----|--------|--------|--------|
| FastPD                       | 3.11 sec    | 1614255.00 | 301059.33    | 2.55 GB  | 0    | 0   | 0.6828 | 0.8235 | 2.7121 |
| mrf- $\alpha$ -Exp-TAB       | 17.14 sec   | 1612676.33 | $-\infty$    | 9.75 GB  | 1    | 0   | 0.6836 | 0.8232 | 2.7130 |
| mrf- $\alpha$ -Exp-TL        | 11.36 sec   | 1615349.00 | $-\infty$    | 0.26 GB  | 0    | 0   | 0.6835 | 0.8235 | 2.7250 |
| mrf- $\alpha\beta$ -Swap-TAB | 16.15 sec   | 1662173.33 | $-\infty$    | 9.75 GB  | 0    | 0   | 0.6921 | 0.8294 | 2.6658 |
| mrf- $\alpha\beta$ -Swap-TL  | 13.12 sec   | 1927265.67 | $-\infty$    | 0.26 GB  | 0    | 0   | 0.6781 | 0.7921 | 2.7541 |
| ogm-FastPD-LF1               | 16.84 sec   | 1613252.00 | -33495282.00 | 2.66 GB  | 0    | 0   | 0.6828 | 0.8235 | 2.7121 |
| ogm-FastPD-LF2               | 156.87 sec  | 1611484.33 | -33495282.00 | 2.70 GB  | 0    | 0   | 0.6828 | 0.8234 | 2.7121 |
| ogm-ICM                      | 5.17 sec    | 8359199.00 | $-\infty$    | 0.22 GB  | 0    | 0   | 0.2354 | 0.3543 | 9.2714 |
| ogm-LF-1                     | 2.07 sec    | 8355303.67 | $-\infty$    | 0.27 GB  | 0    | 0   | 0.2345 | 0.3537 | 9.2853 |
| ogm-LF-2                     | 286.95 sec  | 7396373.00 | $-\infty$    | 0.30 GB  | 0    | 0   | 0.3491 | 0.4785 | 6.2699 |
| ogm-TRWS-LF1                 | 226.77 sec  | 1587732.67 | 1584746.53   | 0.56 GB  | 0    | 0   | 0.6803 | 0.8228 | 2.6928 |
| ogm-TRWS-LF2                 | 365.90 sec  | 1587043.67 | 1584746.53   | 0.59 GB  | 0    | 0   | 0.6803 | 0.8228 | 2.6930 |
| mrf-LBP-TAB                  | 1892.20 sec | 1633306.00 | $-\infty$    | 9.94 GB  | 0    | 0   | 0.6804 | 0.8261 | 2.7835 |
| mrf-LBP-TL                   | 242.10 sec  | 1633343.00 | $-\infty$    | 0.45 GB  | 0    | 0   | 0.6804 | 0.8261 | 2.7835 |
| mrf-BPS-TAB                  | 1473.43 sec | 1738697.00 | $-\infty$    | 19.62 GB | 0    | 0   | 0.7051 | 0.8282 | 2.7832 |
| mrf-BPS-TL                   | 224.42 sec  | 1738696.00 | $-\infty$    | 0.56 GB  | 0    | 0   | 0.7051 | 0.8282 | 2.7831 |
| ogm-BPS                      | 2387.10 sec | 2548773.00 | $-\infty$    | 2.22 GB  | 0    | 0   | 0.6726 | 0.8268 | 2.8554 |
| ogm-LBP-0.5                  | 2468.07 sec | 1650000.67 | $-\infty$    | 2.22 GB  | 0    | 0   | 0.6728 | 0.8258 | 2.8612 |
| ogm-LBP-0.95                 | 2369.21 sec | 1656829.67 | $-\infty$    | 2.22 GB  | 0    | 0   | 0.6735 | 0.8272 | 2.8465 |
| ogm-TRBP-0.5                 | 3019.34 sec | 1656850.00 | $-\infty$    | 2.24 GB  | 0    | 0   | 0.6712 | 0.8245 | 2.8775 |
| ogm-TRBP-0.95                | 3019.90 sec | 1656103.33 | $-\infty$    | 2.24 GB  | 0    | 0   | 0.6735 | 0.8272 | 2.8542 |
| ogm-TRBPS                    | 2959.70 sec | 2551872.67 | $-\infty$    | 2.24 GB  | 0    | 0   | 0.6726 | 0.8267 | 2.8532 |
| mrf-TRWS-TAB                 | 1518.01 sec | 1587932.00 | 1584745.90   | 19.62 GB | 0    | 0   | 0.6803 | 0.8229 | 2.6893 |
| mrf-TRWS-TL                  | 216.41 sec  | 1587928.67 | 1584746.53   | 0.56 GB  | 0    | 0   | 0.6803 | 0.8228 | 2.6928 |
| ogm-ADSAL                    | 3163.13 sec | 1589318.00 | 1584664.58   | 0.98 GB  | 1    | 1   | 0.6814 | 0.8232 | 2.6948 |
| ogm-BUNDLE-A                 | 2112.27 sec | 1649017.67 | 1584324.98   | 1.93 GB  | 1    | 1   | 0.6803 | 0.8224 | 2.7086 |
| ogm-BUNDLE-H                 | 2152.15 sec | 1645250.33 | 1584466.49   | 1.94 GB  | 1    | 0   | 0.6802 | 0.8226 | 2.6986 |
| ogm-SG-A                     | 2098.93 sec | 1789576.67 | 1576864.87   | 0.75 GB  | 1    | 0   | 0.6796 | 0.8221 | 2.7329 |
| ogm-BUNDLE-A+                | 2003.83 sec | 1648854.67 | 1584323.27   | 1.91 GB  | 1    | 1   | 0.6803 | 0.8225 | 2.7075 |
| ogm-SG-A+                    | 2110.78 sec | 2500828.67 | 1402271.83   | 0.75 GB  | 0    | 0   | 0.6763 | 0.8259 | 2.8594 |
| ogm-BUNDLE-A-                | 2150.49 sec | 1648723.33 | 1584316.40   | 1.93 GB  | 0    | 0   | 0.6803 | 0.8225 | 2.7082 |
| ogm-SG-A-                    | 2100.09 sec | 1762200.33 | 1579674.96   | 0.75 GB  | 0    | 0   | 0.6783 | 0.8207 | 2.7321 |
| ogm-CombiLP                  | 835.92 sec  | 1587560.67 | 1584724.04   | 7.17 GB  | 2    | 2   | 0.6809 | 0.8229 | 2.7141 |

Table 25: object-seg (5 instances)

| algorithm                   | runtime    | value    | bound     | mem     | best | opt |
|-----------------------------|------------|----------|-----------|---------|------|-----|
| $\alpha$ -Exp-pct           | 0.33 sec   | 31317.60 | $-\infty$ | 0.08 GB | 4    | 0   |
| FastPD                      | 0.11 sec   | 31317.60 | 29611.23  | 0.17 GB | 4    | 0   |
| FastPD-pct                  | 0.29 sec   | 31317.60 | $-\infty$ | 0.08 GB | 4    | 0   |
| mrf- $\alpha$ -Exp-TL       | 0.40 sec   | 31317.60 | $-\infty$ | 0.05 GB | 4    | 0   |
| mrf- $\alpha\beta$ -Swap-TL | 0.23 sec   | 31323.23 | $-\infty$ | 0.05 GB | 2    | 0   |
| ogm-FastPD-LF1              | 0.39 sec   | 31317.60 | 24148.97  | 0.22 GB | 4    | 0   |
| ogm-FastPD-LF2              | 1.73 sec   | 31317.60 | 24148.97  | 0.23 GB | 4    | 0   |
| ogm-ICM                     | 0.22 sec   | 65180.15 | $-\infty$ | 0.06 GB | 0    | 0   |
| ogm-LF-1                    | 0.11 sec   | 65180.15 | $-\infty$ | 0.08 GB | 0    | 0   |
| ogm-LF-2                    | 1.48 sec   | 64937.24 | $-\infty$ | 0.10 GB | 0    | 0   |
| ogm-TRWS-LF1                | 3.27 sec   | 31317.23 | 31317.23  | 0.09 GB | 5    | 5   |
| ogm-TRWS-LF2                | 3.29 sec   | 31317.23 | 31317.23  | 0.09 GB | 5    | 5   |
| mrf-LBP-TL                  | 29.43 sec  | 32400.01 | $-\infty$ | 0.05 GB | 0    | 0   |
| mrf-BPS-TL                  | 11.19 sec  | 35775.27 | $-\infty$ | 0.05 GB | 0    | 0   |
| ogm-BPS                     | 137.34 sec | 33564.55 | $-\infty$ | 0.31 GB | 0    | 0   |
| ogm-LBP-0.5                 | 110.84 sec | 32663.86 | $-\infty$ | 0.31 GB | 0    | 0   |
| ogm-LBP-0.95                | 61.38 sec  | 32673.75 | $-\infty$ | 0.31 GB | 0    | 0   |
| ogm-TRBP-0.5                | 227.29 sec | 32663.86 | $-\infty$ | 0.32 GB | 0    | 0   |
| ogm-TRBP-0.95               | 222.82 sec | 32668.92 | $-\infty$ | 0.32 GB | 0    | 0   |
| ogm-TRBPS                   | 180.23 sec | 33530.63 | $-\infty$ | 0.32 GB | 0    | 0   |
| MCR-TC-MTC                  | 421.26 sec | 32376.56 | 31317.23  | 0.86 GB | 4    | 4   |
| MCR-pct                     | 62.36 sec  | 31674.41 | 31317.23  | 0.20 GB | 2    | 2   |
| mrf-TRWS-TL                 | 2.21 sec   | 31317.23 | 31317.23  | 0.05 GB | 5    | 5   |
| ogm-ADSAL                   | 99.50 sec  | 31317.23 | 31317.23  | 0.14 GB | 5    | 5   |
| ogm-BUNDLE-A                | 128.82 sec | 31317.31 | 31316.93  | 0.24 GB | 4    | 1   |
| ogm-BUNDLE-H                | 193.92 sec | 31317.23 | 31316.88  | 0.27 GB | 5    | 0   |
| ogm-SG-A                    | 185.11 sec | 31432.95 | 31311.80  | 0.16 GB | 2    | 0   |
| ogm-BUNDLE-A+               | 196.92 sec | 31354.83 | 31312.49  | 0.24 GB | 1    | 0   |
| ogm-SG-A+                   | 187.17 sec | 39295.40 | 30371.94  | 0.16 GB | 0    | 0   |
| ogm-BUNDLE-A-               | 197.66 sec | 31321.98 | 31315.85  | 0.24 GB | 1    | 0   |
| ogm-SG-A-                   | 180.46 sec | 32137.62 | 31286.74  | 0.16 GB | 0    | 0   |
| TRWS-pct                    | 0.96 sec   | 31317.23 | 31317.23  | 0.08 GB | 5    | 5   |
| MCI-TC-MTC-TCI              | 428.32 sec | 31317.23 | 31317.23  | 1.81 GB | 5    | 5   |
| MCI-pct                     | 69.84 sec  | 31317.23 | 31317.23  | 0.39 GB | 5    | 5   |
| ogm-CombiLP                 | 32.61 sec  | 31317.23 | 31317.23  | 0.28 GB | 5    | 5   |

Table 26: protein-folding (21 instances)

| algorithm     | runtime        | value      | bound      | mem      | best | opt |
|---------------|----------------|------------|------------|----------|------|-----|
| ogm-ICM       | 0.09 sec       | -5446.55   | $-\infty$  | 0.25 GB  | 0    | 0   |
| ogm-LBP-LF1   | 102.71 sec     | -5892.69   | $-\infty$  | 0.25 GB  | 8    | 0   |
| ogm-LBP-LF2   | 130.17 sec     | -5923.01   | $-\infty$  | 0.25 GB  | 12   | 0   |
| ogm-LF-1      | 0.05 sec       | -5447.49   | $-\infty$  | 0.25 GB  | 0    | 0   |
| ogm-LF-2      | 54.54 sec      | -5747.56   | $-\infty$  | 0.25 GB  | 0    | 0   |
| ogm-LF-3      | 22422.07 sec   | -5780.11   | $-\infty$  | 0.25 GB  | 4    | 0   |
| ogm-TRWS-LF1  | 22.68 sec      | -5856.47   | -6041.38   | 0.25 GB  | 3    | 1   |
| ogm-TRWS-LF2  | 54.88 sec      | -5897.06   | -6041.38   | 0.25 GB  | 6    | 1   |
| BPS-TAB       | 24.48 sec      | -5917.15   | $-\infty$  | 0.25 GB  | 11   | 0   |
| ogm-BPS       | 119.08 sec     | -5638.92   | $-\infty$  | 0.25 GB  | 6    | 0   |
| ogm-LBP-0.5   | 106.99 sec     | -5846.70   | $-\infty$  | 0.25 GB  | 13   | 0   |
| ogm-LBP-0.95  | 102.34 sec     | -5812.01   | $-\infty$  | 0.25 GB  | 10   | 0   |
| ogm-TRBP-0.5  | 145.44 sec     | -5810.68   | $-\infty$  | 0.25 GB  | 9    | 0   |
| ogm-TRBP-0.95 | 140.09 sec     | -5644.96   | $-\infty$  | 0.25 GB  | 8    | 0   |
| ogm-TRBPS     | 135.42 sec     | -5396.82   | $-\infty$  | 0.25 GB  | 7    | 0   |
| ADDD          | 209.19 sec     | -4189.61   | -10364.42  | 0.85 GB  | 0    | 0   |
| MPLP          | 510.90 sec     | -5611.60   | -6033.98   | 0.62 GB  | 1    | 1   |
| MPLP-C        | 1639.52 sec    | -5765.28   | -5984.52   | 11.14 GB | 12   | 9   |
| ogm-ADSAL     | 1014.89 sec    | -5881.47   | -6128.90   | 0.51 GB  | 4    | 1   |
| ogm-BUNDLE-A  | 487.56 sec     | -5480.14   | -6477.78   | 0.25 GB  | 1    | 0   |
| ogm-BUNDLE-H  | 479.92 sec     | -5486.47   | -6413.00   | 0.25 GB  | 1    | 0   |
| ogm-SG-A      | 474.52 sec     | -5210.59   | -6706.79   | 0.25 GB  | 1    | 0   |
| ogm-BUNDLE-A+ | 483.77 sec     | -5448.23   | -6395.59   | 0.25 GB  | 2    | 1   |
| ogm-SG-A+     | 463.18 sec     | -5479.15   | -6494.35   | 0.25 GB  | 1    | 1   |
| ogm-BUNDLE-A- | 477.86 sec     | -5146.23   | -6996.84   | 0.25 GB  | 0    | 0   |
| ogm-SG-A-     | 473.93 sec     | -4806.12   | -7670.56   | 0.25 GB  | 0    | 0   |
| TRWS-TAB      | 22.18 sec      | -5771.50   | -6041.38   | 0.25 GB  | 2    | 1   |
| BRAOBB-1      | <i>NaN</i> sec | <i>NaN</i> | <i>NaN</i> | 1.10 GB  | 0    | 0   |
| BRAOBB-2      | <i>NaN</i> sec | <i>NaN</i> | <i>NaN</i> | 1.36 GB  | 1    | 0   |
| BRAOBB-3      | <i>NaN</i> sec | <i>NaN</i> | <i>NaN</i> | 2.68 GB  | 1    | 0   |
| ogm-CombiLP   | 700.10 sec     | -5955.77   | -5955.77   | 24.26 GB | 20   | 21  |

Table 27: protein-prediction (8 instances)

| algorithm     | runtime     | value     | bound     | mem     | best | opt |
|---------------|-------------|-----------|-----------|---------|------|-----|
| ogm-ICM       | 0.03 sec    | 60414.84  | $-\infty$ | 0.01 GB | 0    | 0   |
| ogm-LBP-LF2   | 25.03 sec   | 52942.95  | $-\infty$ | 0.06 GB | 1    | 0   |
| ogm-LF-1      | 0.03 sec    | 60427.60  | $-\infty$ | 0.02 GB | 0    | 0   |
| ogm-LF-2      | 0.70 sec    | 58682.74  | $-\infty$ | 0.02 GB | 0    | 0   |
| ogm-LF-3      | 19.08 sec   | 57944.06  | $-\infty$ | 0.09 GB | 0    | 0   |
| ogm-BPS       | 27.64 sec   | 75286.37  | $-\infty$ | 0.05 GB | 0    | 0   |
| ogm-LBP-0.5   | 24.79 sec   | 53798.89  | $-\infty$ | 0.05 GB | 0    | 0   |
| ogm-LBP-0.95  | 33.31 sec   | 110527.26 | $-\infty$ | 0.05 GB | 0    | 0   |
| ogm-TRBP-0.5  | 35.60 sec   | 61386.17  | $-\infty$ | 0.05 GB | 0    | 0   |
| ogm-TRBP-0.95 | 36.11 sec   | 70903.40  | $-\infty$ | 0.05 GB | 0    | 0   |
| ogm-TRBPS     | 28.38 sec   | 72110.77  | $-\infty$ | 0.05 GB | 0    | 0   |
| ADDD          | 10.70 sec   | 106216.86 | 41124.16  | 0.05 GB | 0    | 0   |
| MPLP          | 69.09 sec   | 101531.75 | 43123.68  | 0.05 GB | 0    | 0   |
| ogm-BUNDLE-A  | 1287.27 sec | 81035.49  | 44090.57  | 0.28 GB | 0    | 0   |
| ogm-BUNDLE-H  | 1301.94 sec | 81039.93  | 44092.42  | 0.28 GB | 0    | 0   |
| ogm-SG-A      | 1273.32 sec | 81587.06  | 42489.54  | 0.19 GB | 0    | 0   |
| ogm-BUNDLE-A+ | 1275.45 sec | 81993.85  | 43819.34  | 0.28 GB | 0    | 0   |
| ogm-SG-A+     | 1256.94 sec | 82606.85  | 34907.38  | 0.19 GB | 0    | 0   |
| ogm-BUNDLE-A- | 1300.65 sec | 81058.43  | 44090.07  | 0.26 GB | 0    | 0   |
| ogm-SG-A-     | 1312.12 sec | 81443.05  | 42676.66  | 0.18 GB | 0    | 0   |
| ogm-LP-LP     | 169.61 sec  | 102829.40 | 44347.16  | 0.24 GB | 0    | 0   |
| ogm-ILP       | 2263.46 sec | 57477.07  | 44674.02  | 0.65 GB | 7    | 3   |

Table 28: scene-decomposition (715 instances)

| algorithm        | runtime   | value   | bound     | mem     | best | opt | PA     |
|------------------|-----------|---------|-----------|---------|------|-----|--------|
| $\alpha$ -Fusion | 0.00 sec  | -866.85 | $-\infty$ | 0.01 GB | 587  | 0   | 0.7694 |
| ogm-ICM          | 0.00 sec  | -864.56 | $-\infty$ | 0.01 GB | 110  | 0   | 0.7649 |
| ogm-LBP-LF1      | 0.05 sec  | -866.76 | $-\infty$ | 0.01 GB | 576  | 0   | 0.7699 |
| ogm-LBP-LF2      | 0.06 sec  | -866.76 | $-\infty$ | 0.01 GB | 576  | 0   | 0.7699 |
| ogm-LF-1         | 0.00 sec  | -864.56 | $-\infty$ | 0.01 GB | 110  | 0   | 0.7649 |
| ogm-LF-2         | 0.02 sec  | -865.81 | $-\infty$ | 0.01 GB | 286  | 0   | 0.7685 |
| ogm-LF-3         | 0.45 sec  | -866.27 | $-\infty$ | 0.01 GB | 420  | 0   | 0.7699 |
| ogm-TRWS-LF1     | 0.00 sec  | -866.93 | -866.93   | 0.01 GB | 713  | 712 | 0.7693 |
| ogm-TRWS-LF2     | 0.00 sec  | -866.93 | -866.93   | 0.01 GB | 714  | 712 | 0.7693 |
| BPS-TAB          | 0.10 sec  | -866.73 | $-\infty$ | 0.01 GB | 566  | 0   | 0.7701 |
| ogm-BPS          | 0.02 sec  | -866.77 | $-\infty$ | 0.01 GB | 585  | 0   | 0.7694 |
| ogm-LBP-0.5      | 0.05 sec  | -866.76 | $-\infty$ | 0.01 GB | 576  | 0   | 0.7699 |
| ogm-LBP-0.95     | 0.02 sec  | -866.76 | $-\infty$ | 0.01 GB | 580  | 0   | 0.7696 |
| ogm-TRBP-0.5     | 0.24 sec  | -866.84 | $-\infty$ | 0.01 GB | 642  | 0   | 0.7704 |
| ogm-TRBP-0.95    | 0.11 sec  | -866.84 | $-\infty$ | 0.01 GB | 644  | 0   | 0.7708 |
| ogm-TRBPS        | 0.13 sec  | -866.79 | $-\infty$ | 0.01 GB | 644  | 0   | 0.7705 |
| ADDD             | 0.06 sec  | -866.92 | -866.93   | 0.01 GB | 701  | 697 | 0.7693 |
| MPLP             | 0.04 sec  | -866.91 | -866.93   | 0.01 GB | 700  | 561 | 0.7693 |
| MPLP-C           | 0.04 sec  | -866.92 | -866.93   | 0.01 GB | 710  | 567 | 0.7693 |
| ogm-ADSAL        | 0.04 sec  | -866.93 | -866.93   | 0.01 GB | 714  | 712 | 0.7693 |
| ogm-BUNDLE-A     | 0.94 sec  | -866.93 | -866.93   | 0.01 GB | 714  | 349 | 0.7693 |
| ogm-BUNDLE-H     | 0.26 sec  | -866.93 | -866.93   | 0.01 GB | 715  | 673 | 0.7693 |
| ogm-SG-A         | 0.96 sec  | -866.93 | -866.94   | 0.01 GB | 711  | 223 | 0.7692 |
| ogm-BUNDLE-A+    | 0.07 sec  | -866.93 | -866.93   | 0.01 GB | 715  | 712 | 0.7693 |
| ogm-SG-A+        | 0.07 sec  | -866.92 | -866.93   | 0.01 GB | 713  | 708 | 0.7693 |
| ogm-BUNDLE-A-    | 1.19 sec  | -866.92 | -867.13   | 0.01 GB | 683  | 0   | 0.7695 |
| ogm-SG-A-        | 1.08 sec  | -866.85 | -867.56   | 0.01 GB | 505  | 0   | 0.7700 |
| ogm-LP-LP        | 0.23 sec  | -866.92 | -866.93   | 0.05 GB | 712  | 712 | 0.7693 |
| TRWS-TAB         | 0.00 sec  | -866.93 | -866.93   | 0.01 GB | 714  | 712 | 0.7693 |
| BRAOBB-1         | 17.61 sec | -866.90 | $-\infty$ | 0.27 GB | 670  | 0   | 0.7688 |
| BRAOBB-2         | 23.40 sec | -866.86 | $-\infty$ | 0.27 GB | 625  | 0   | 0.7691 |
| BRAOBB-3         | 27.20 sec | -866.90 | $-\infty$ | 1.16 GB | 653  | 0   | 0.7693 |
| ADDD-BB          | 0.11 sec  | -866.93 | -866.93   | 0.01 GB | 715  | 715 | 0.7693 |
| ogm-CombiLP      | 0.02 sec  | -866.93 | -866.93   | 0.03 GB | 715  | 715 | 0.7693 |
| ogm-ILP          | 0.17 sec  | -866.93 | -866.93   | 0.09 GB | 715  | 715 | 0.7693 |

## 2 Anytime Evaluation per Model

Table 29: brain-3mm (4 instances)

| algorithm                | value       |             |             |             |             | time        |
|--------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                          | (1 sec)     | (10 sec)    | (60 sec)    | (600 sec)   | (end)       | (end)       |
| $\alpha$ -Exp-pct        | 92058776.50 | 92058776.50 | 25164182.75 | 25164182.75 | 25164182.75 | 9.57 sec    |
| $\alpha$ -Exp-VIEW       | 92058776.50 | 92058776.50 | 25178822.25 | 25164077.00 | 25164077.00 | 126.94 sec  |
| FastPD                   | $\infty$    | 25164518.25 | 25164518.25 | 25164518.25 | 25164518.25 | 2.02 sec    |
| FastPD-pct               | 92058776.50 | 25164326.50 | 25164326.50 | 25164326.50 | 25164326.50 | 7.56 sec    |
| ogm-FastPD-LF1           | $\infty$    | 25164518.25 | 25164518.25 | 25164518.25 | 25164518.25 | 10.51 sec   |
| ogm-FastPD-LF2           | $\infty$    | 25164518.25 | 25164518.25 | 25163683.50 | 25163683.50 | 61.27 sec   |
| ogm-ICM                  | 92058776.50 | 92058776.50 | 25425309.50 | 25425309.50 | 25425309.50 | 16.60 sec   |
| ogm-LF-1                 | 92058776.50 | 92058776.50 | 25425594.00 | 25425594.00 | 25425594.00 | 9.48 sec    |
| ogm-LF-2                 | 92058776.50 | 92058776.50 | 92058776.50 | 25238328.25 | 25238328.25 | 83.39 sec   |
| ogm-TRWS-LF1             | $\infty$    | $\infty$    | $\infty$    | $\infty$    | 25162580.25 | 285.81 sec  |
| ogm-TRWS-LF2             | $\infty$    | $\infty$    | $\infty$    | $\infty$    | 25162535.00 | 343.41 sec  |
| $\alpha\beta$ -Swap-VIEW | 92058776.50 | 92058776.50 | 25327586.25 | 25164671.25 | 25164671.25 | 136.43 sec  |
| BPS-TL                   | 25205321.50 | 25166078.25 | 25166078.25 | 25166078.25 | 25166078.25 | 644.82 sec  |
| ogm-BPS                  | 92058776.50 | 25875088.00 | 25206515.00 | 25176849.75 | 25176740.00 | 2707.16 sec |
| ogm-LBP-0.5              | 92058776.50 | 25955289.50 | 25298837.25 | 25167691.25 | 25166874.25 | 1930.36 sec |
| ogm-LBP-0.95             | 92058776.50 | 25872275.50 | 25207101.00 | 25166977.50 | 25166881.75 | 2718.32 sec |
| ogm-TRBP-0.5             | 92058776.50 | 92058776.50 | 25363893.00 | 25168686.75 | 25166817.00 | 2774.78 sec |
| ogm-TRBP-0.95            | 92058776.50 | 92058776.50 | 25246613.00 | 25166984.75 | 25166836.25 | 2723.08 sec |
| ogm-TRBPS                | 92058776.50 | 25874562.75 | 25222343.00 | 25176859.00 | 25176731.75 | 2710.81 sec |
| ogm-ADSAL                | $\infty$    | $\infty$    | $\infty$    | 25163371.25 | 25162633.25 | 2720.64 sec |
| ogm-BUNDLE-A             | $\infty$    | $\infty$    | 25813060.00 | 25224391.00 | 25166504.00 | 2708.90 sec |
| ogm-BUNDLE-H             | $\infty$    | $\infty$    | 25808456.75 | 25185603.00 | 25164211.25 | 2705.77 sec |
| ogm-SG-A                 | $\infty$    | $\infty$    | 25592601.25 | 25227633.75 | 25169609.25 | 2704.54 sec |
| ogm-BUNDLE-A+            | $\infty$    | $\infty$    | 25813060.00 | 25224817.00 | 25167143.75 | 2708.06 sec |
| ogm-SG-A+                | $\infty$    | $\infty$    | 25630657.25 | 25579004.00 | 25536961.50 | 2709.06 sec |
| ogm-BUNDLE-A-            | $\infty$    | $\infty$    | 25813060.00 | 25197483.00 | 25163766.00 | 2707.39 sec |
| ogm-SG-A-                | $\infty$    | $\infty$    | 25789387.75 | 25530009.75 | 25244347.75 | 2707.91 sec |
| TRWS-TL                  | $\infty$    | 25166520.50 | 25162889.25 | 25162670.75 | 25162670.75 | 282.73 sec  |
| TRWS-pct                 | 92058776.50 | 92058776.50 | 47605961.50 | 25162648.50 | 25162648.50 | 40.16 sec   |
| MCI-pct                  | 92058776.50 | 92058776.50 | 25162493.00 | 25162493.00 | 25162493.00 | 27.35 sec   |
| ogm-CombiLP              | $\infty$    | $\infty$    | $\infty$    | $\infty$    | 25162580.25 | 3874.65 sec |

Table 30: brain-5mm (4 instances)

| algorithm                | value       |             |             |             |             | time        |
|--------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                          | (1 sec)     | (10 sec)    | (60 sec)    | (600 sec)   | (end)       | (end)       |
| $\alpha$ -Exp-pct        | 74057213.75 | 19088999.75 | 19088999.75 | 19088999.75 | 19088999.75 | 7.02 sec    |
| $\alpha$ -Exp-VIEW       | 74057213.75 | 74057213.75 | 19089175.50 | 19089080.00 | 19089080.00 | 100.66 sec  |
| FastPD                   | $\infty$    | 19089484.75 | 19089484.75 | 19089484.75 | 19089484.75 | 1.32 sec    |
| FastPD-pct               | 74057213.75 | 19089255.25 | 19089255.25 | 19089255.25 | 19089255.25 | 4.66 sec    |
| ogm-FastPD-LF1           | $\infty$    | 19089484.75 | 19089484.75 | 19089484.75 | 19089484.75 | 7.19 sec    |
| ogm-FastPD-LF2           | $\infty$    | 19089484.75 | 19088812.00 | 19088812.00 | 19088812.00 | 48.51 sec   |
| ogm-ICM                  | 74057213.75 | 74057213.75 | 19272820.00 | 19272820.00 | 19272820.00 | 11.30 sec   |
| ogm-LF-1                 | 74057213.75 | 19273079.25 | 19273079.25 | 19273079.25 | 19273079.25 | 6.49 sec    |
| ogm-LF-2                 | 74057213.75 | 74057213.75 | 19140692.25 | 19140692.25 | 19140692.25 | 51.81 sec   |
| ogm-TRWS-LF1             | $\infty$    | $\infty$    | $\infty$    | 19087665.75 | 19087665.75 | 125.92 sec  |
| ogm-TRWS-LF2             | $\infty$    | $\infty$    | $\infty$    | 19087628.00 | 19087628.00 | 184.22 sec  |
| $\alpha\beta$ -Swap-VIEW | 74057213.75 | 74057213.75 | 19089841.25 | 19089768.00 | 19089768.00 | 91.84 sec   |
| BPS-TL                   | 19107324.50 | 19090723.25 | 19090723.25 | 19090723.25 | 19090723.25 | 450.43 sec  |
| ogm-BPS                  | 74057213.75 | 19357379.00 | 19105637.00 | 19098965.75 | 19098965.75 | 3601.41 sec |
| ogm-LBP-0.5              | 74057213.75 | 19702979.75 | 19140662.00 | 19091374.00 | 19091373.75 | 1116.27 sec |
| ogm-LBP-0.95             | 74057213.75 | 19559915.25 | 19100857.25 | 19091233.25 | 19091194.75 | 1574.35 sec |
| ogm-TRBP-0.5             | 74057213.75 | 19705227.50 | 19156870.25 | 19091423.50 | 19091287.25 | 2948.01 sec |
| ogm-TRBP-0.95            | 74057213.75 | 19630782.25 | 19106648.00 | 19091182.50 | 19091155.25 | 3463.22 sec |
| ogm-TRBPS                | 74057213.75 | 19633240.50 | 19108846.50 | 19098987.50 | 19098985.00 | 3602.69 sec |
| ogm-ADSAL                | $\infty$    | $\infty$    | $\infty$    | 19087896.25 | 19087679.25 | 3610.13 sec |
| ogm-BUNDLE-A             | $\infty$    | $\infty$    | 19424844.25 | 19108691.25 | 19088524.25 | 3605.55 sec |
| ogm-BUNDLE-H             | $\infty$    | $\infty$    | 19438281.50 | 19093912.75 | 19088322.25 | 3603.13 sec |
| ogm-SG-A                 | $\infty$    | 19639958.25 | 19372672.25 | 19117000.50 | 19090744.50 | 3602.62 sec |
| ogm-BUNDLE-A+            | $\infty$    | $\infty$    | 19424844.25 | 19110092.75 | 19089189.75 | 3602.10 sec |
| ogm-SG-A+                | $\infty$    | 19639958.25 | 19419230.50 | 19328785.25 | 19300226.00 | 3602.34 sec |
| ogm-BUNDLE-A-            | $\infty$    | $\infty$    | 19424844.25 | 19090981.25 | 19088258.25 | 3604.02 sec |
| ogm-SG-A-                | $\infty$    | 19639958.25 | 19592117.00 | 19324775.00 | 19125604.50 | 3603.66 sec |
| TRWS-TL                  | 19176098.25 | 19088592.50 | 19087807.75 | 19087727.75 | 19087727.75 | 120.29 sec  |
| TRWS-pct                 | 74057213.75 | 74057213.75 | 19087728.50 | 19087728.50 | 19087728.50 | 21.93 sec   |
| MCI-pct                  | 74057213.75 | 74057213.75 | 19087612.50 | 19087612.50 | 19087612.50 | 25.63 sec   |
| ogm-CombiLP              | $\infty$    | $\infty$    | $\infty$    | $\infty$    | 19087626.75 | 2022.87 sec |

Table 31: brain-9mm (4 instances)

| algorithm                | value       |             |            |            |            | time        |
|--------------------------|-------------|-------------|------------|------------|------------|-------------|
|                          | (1 sec)     | (10 sec)    | (60 sec)   | (600 sec)  | (end)      | (end)       |
| $\alpha$ -Exp-pct        | 32703082.50 | 9185804.25  | 9185804.25 | 9185804.25 | 9185804.25 | 3.51 sec    |
| $\alpha$ -Exp-VIEW       | 32703082.50 | 24922227.00 | 9185865.25 | 9185865.25 | 9185865.25 | 37.19 sec   |
| FastPD                   | 9186044.50  | 9186044.50  | 9186044.50 | 9186044.50 | 9186044.50 | 0.75 sec    |
| FastPD-pct               | 32703082.50 | 9185930.25  | 9185930.25 | 9185930.25 | 9185930.25 | 2.63 sec    |
| ogm-FastPD-LF1           | $\infty$    | 9186044.50  | 9186044.50 | 9186044.50 | 9186044.50 | 3.71 sec    |
| ogm-FastPD-LF2           | $\infty$    | 9186044.50  | 9185787.25 | 9185787.25 | 9185787.25 | 21.41 sec   |
| ogm-ICM                  | 32703082.50 | 9263471.50  | 9263471.50 | 9263471.50 | 9263471.50 | 5.78 sec    |
| ogm-LF-1                 | 32703082.50 | 9263538.00  | 9263538.00 | 9263538.00 | 9263538.00 | 3.24 sec    |
| ogm-LF-2                 | 32703082.50 | 32703082.50 | 9207084.00 | 9207084.00 | 9207084.00 | 29.25 sec   |
| ogm-TRWS-LF1             | $\infty$    | $\infty$    | $\infty$   | 9185326.75 | 9185326.75 | 80.76 sec   |
| ogm-TRWS-LF2             | $\infty$    | $\infty$    | $\infty$   | 9185298.75 | 9185298.75 | 93.10 sec   |
| $\alpha\beta$ -Swap-VIEW | 32703082.50 | 32703082.50 | 9186218.50 | 9186218.50 | 9186218.50 | 43.51 sec   |
| BPS-TL                   | 9186804.00  | 9186485.25  | 9186485.25 | 9186485.25 | 9186485.25 | 230.93 sec  |
| ogm-BPS                  | 32703082.50 | 9227185.00  | 9190336.75 | 9189963.75 | 9189963.75 | 1698.94 sec |
| ogm-LBP-0.5              | 32703082.50 | 9293083.25  | 9192691.00 | 9186565.25 | 9186565.25 | 408.82 sec  |
| ogm-LBP-0.95             | 32703082.50 | 9247096.75  | 9187037.25 | 9186506.00 | 9186506.00 | 346.08 sec  |
| ogm-TRBP-0.5             | 32703082.50 | 9325217.25  | 9195952.50 | 9186546.00 | 9186545.00 | 1166.47 sec |
| ogm-TRBP-0.95            | 32703082.50 | 9280426.25  | 9187769.25 | 9186479.25 | 9186479.25 | 1852.20 sec |
| ogm-TRBPS                | 32703082.50 | 9252194.75  | 9190400.00 | 9189701.00 | 9189701.00 | 2102.64 sec |
| ogm-ADSAL                | $\infty$    | $\infty$    | 9190630.00 | 9185384.25 | 9185310.25 | 2193.95 sec |
| ogm-BUNDLE-A             | $\infty$    | 9437061.75  | 9286393.50 | 9188242.50 | 9185566.25 | 2568.53 sec |
| ogm-BUNDLE-H             | $\infty$    | 9437059.00  | 9254424.25 | 9186255.75 | 9185559.25 | 2566.86 sec |
| ogm-SG-A                 | $\infty$    | 9399643.25  | 9265866.25 | 9189086.75 | 9186984.25 | 2292.96 sec |
| ogm-BUNDLE-A+            | $\infty$    | 9437061.75  | 9292726.50 | 9188765.00 | 9185798.75 | 2568.56 sec |
| ogm-SG-A+                | $\infty$    | 9410237.75  | 9321592.00 | 9277875.75 | 9266873.50 | 2468.36 sec |
| ogm-BUNDLE-A-            | $\infty$    | 9437061.75  | 9279901.00 | 9185980.25 | 9185626.75 | 2581.48 sec |
| ogm-SG-A-                | $\infty$    | 9435592.75  | 9400388.50 | 9252103.75 | 9193050.50 | 2353.54 sec |
| TRWS-TL                  | 9198264.00  | 9185487.50  | 9185371.50 | 9185363.25 | 9185363.25 | 68.40 sec   |
| TRWS-pct                 | 32703082.50 | 24760882.50 | 9185347.00 | 9185347.00 | 9185347.00 | 14.13 sec   |
| MCI-pct                  | 32703082.50 | 32703082.50 | 9185280.75 | 9185280.75 | 9185280.75 | 8.32 sec    |
| ogm-CombiLP              | $\infty$    | $\infty$    | $\infty$   | $\infty$   | 9185280.75 | 895.43 sec  |

Table 32: cell-tracking (1 instances)

| algorithm     | value            |                  |                  |               |               | time         |
|---------------|------------------|------------------|------------------|---------------|---------------|--------------|
|               | (1 sec)          | (10 sec)         | (60 sec)         | (600 sec)     | (end)         | (end)        |
| ogm-ICM       | 40359847.14      | 40359847.14      | 40359847.14      | 40359847.14   | 40359847.14   | 0.03 sec     |
| ogm-LBP-LF2   | $\infty$         | $\infty$         | $\infty$         | 7515575.61    | 7515575.61    | 62.12 sec    |
| ogm-LF-1      | 40359847.14      | 40359847.14      | 40359847.14      | 40359847.14   | 40359847.14   | 0.04 sec     |
| ogm-LF-2      | 14075743.46      | 14075743.46      | 14075743.46      | 14075743.46   | 14075743.46   | 0.43 sec     |
| ogm-LF-3      | 45364196.24      | 8461693.24       | 8461693.24       | 8461693.24    | 8461693.24    | 1.55 sec     |
| ogm-BPS       | 1207557577.48    | 107515639.76     | 107515319.56     | 107515319.56  | 107515319.56  | 60.07 sec    |
| ogm-LBP-0.5   | 8807557216.69    | 107522947.52     | 7520989.51       | 7520989.51    | 7520989.51    | 62.01 sec    |
| ogm-LBP-0.95  | 2507556975.16    | 7518377.63       | 7518377.63       | 7518377.63    | 7518377.63    | 61.97 sec    |
| ogm-TRBP-0.5  | 16207487147.14   | 1207526572.16    | 107518566.78     | 107518566.78  | 107518566.78  | 65.94 sec    |
| ogm-TRBP-0.95 | 6107511577.69    | 107519956.23     | 107517017.88     | 107517017.88  | 107517017.88  | 65.97 sec    |
| ogm-TRBPS     | 6607520261.74    | 407521662.88     | 7528407.72       | 7528407.72    | 7528407.72    | 62.92 sec    |
| ADDD          | 45364196.24      | 45364196.24      | 45364196.24      | 45364196.24   | 45364196.24   | 11.74 sec    |
| MPLP          | 1248810837307.02 | 1248810837307.02 | 1248810837307.02 | 107514359.61  | 107514359.61  | 459.93 sec   |
| ogm-BUNDLE-A  | $\infty$         | 4008557514.22    | 2007687774.66    | 7696631.53    | 7696631.53    | 532.38 sec   |
| ogm-BUNDLE-H  | $\infty$         | 4508746916.51    | 2207692125.44    | 7748583.42    | 7748583.42    | 522.32 sec   |
| ogm-SG-A      | 6215592373.05    | 1726802266.16    | 927583594.52     | 927583594.52  | 927583594.52  | 522.89 sec   |
| ogm-BUNDLE-A+ | $\infty$         | 4008557514.22    | 2007687774.66    | 7696631.53    | 7696631.53    | 527.17 sec   |
| ogm-SG-A+     | 6215592373.05    | 529475189.95     | 230126875.07     | 230126875.07  | 230126875.07  | 526.20 sec   |
| ogm-BUNDLE-A- | $\infty$         | 4008557514.22    | 2007687774.66    | 7696631.53    | 7696631.53    | 531.65 sec   |
| ogm-SG-A-     | 6215592373.05    | 3023370752.77    | 2019684226.99    | 1819216014.16 | 1819216014.16 | 518.55 sec   |
| ogm-LP-LP     | 45364196.24      | 7516359.61       | 7516359.61       | 7516359.61    | 7516359.61    | 4.21 sec     |
| ADDD-BB       | 45364196.24      | 45364196.24      | 45364196.24      | 45364196.24   | 7514421.21    | 38698.77 sec |
| ogm-ILP-pct   | 45364196.24      | 45364196.24      | 7514421.21       | 7514421.21    | 7514421.21    | 12.36 sec    |
| ogm-ILP       | 45364196.24      | 45364196.24      | 7514421.21       | 7514421.21    | 7514421.21    | 11.99 sec    |

Table 33: color-seg-n4 (9 instances)

| algorithm                   | value    |          |          |           |          | time       |
|-----------------------------|----------|----------|----------|-----------|----------|------------|
|                             | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)    | (end)      |
| $\alpha$ -Exp-pct           | 54757.02 | 20034.80 | 20034.80 | 20034.80  | 20034.80 | 2.98 sec   |
| FastPD                      | 20034.80 | 20034.80 | 20034.80 | 20034.80  | 20034.80 | 0.29 sec   |
| FastPD-pct                  | 54757.02 | 20034.85 | 20034.85 | 20034.85  | 20034.85 | 1.52 sec   |
| mrf- $\alpha$ -Exp-TL       | 20034.40 | 20033.56 | 20033.56 | 20033.56  | 20033.56 | 1.11 sec   |
| mrf- $\alpha\beta$ -Swap-TL | 20061.01 | 20060.91 | 20060.91 | 20060.91  | 20060.91 | 0.64 sec   |
| ogm-FastPD-LF1              | $\infty$ | 20034.80 | 20034.80 | 20034.80  | 20034.80 | 1.07 sec   |
| ogm-FastPD-LF2              | $\infty$ | 20033.21 | 20033.21 | 20033.21  | 20033.21 | 6.15 sec   |
| ogm-ICM                     | 59361.98 | 26329.45 | 26329.45 | 26329.45  | 26329.45 | 1.07 sec   |
| ogm-LF-1                    | 26330.66 | 26330.66 | 26330.66 | 26330.66  | 26330.66 | 0.50 sec   |
| ogm-LF-2                    | 63093.85 | 30316.54 | 23775.38 | 23775.38  | 23775.38 | 7.38 sec   |
| ogm-TRWS-LF1                | $\infty$ | $\infty$ | 20012.17 | 20012.17  | 20012.17 | 7.58 sec   |
| ogm-TRWS-LF2                | $\infty$ | $\infty$ | 20012.17 | 20012.17  | 20012.17 | 9.08 sec   |
| mrf-LBP-TL                  | 20054.06 | 20053.19 | 20053.19 | 20053.19  | 20053.19 | 39.87 sec  |
| mrf-BPS-TL                  | 20094.10 | 20094.03 | 20094.03 | 20094.03  | 20094.03 | 23.35 sec  |
| ogm-BPS                     | 21806.66 | 20410.00 | 20372.89 | 20372.87  | 20372.87 | 419.65 sec |
| ogm-LBP-0.5                 | 22042.86 | 20221.85 | 20055.62 | 20054.26  | 20054.26 | 335.15 sec |
| ogm-LBP-0.95                | 21789.03 | 20117.88 | 20058.43 | 20058.42  | 20058.42 | 267.45 sec |
| ogm-TRBP-0.5                | 22145.59 | 20303.28 | 20059.55 | 20054.07  | 20054.06 | 505.13 sec |
| ogm-TRBP-0.95               | 21817.84 | 20169.44 | 20058.02 | 20058.00  | 20058.00 | 371.35 sec |
| ogm-TRBPS                   | 21840.93 | 20433.84 | 20367.61 | 20367.42  | 20367.42 | 555.27 sec |
| MCR-TC-MTC                  | 27589.26 | 21972.40 | 21137.95 | 20621.79  | 20450.12 | 440.57 sec |
| MCR-pct                     | 62394.57 | 40144.27 | 32136.67 | 20944.63  | 20944.63 | 424.71 sec |
| mrf-TRWS-TL                 | 20016.77 | 20013.16 | 20012.17 | 20012.17  | 20012.17 | 23.52 sec  |
| ogm-ADSAL                   | $\infty$ | 20068.01 | 20015.00 | 20012.16  | 20012.15 | 311.96 sec |
| ogm-BUNDLE-A                | $\infty$ | 20792.64 | 20323.80 | 20024.78  | 20024.78 | 224.07 sec |
| ogm-BUNDLE-H                | $\infty$ | 20572.43 | 20064.02 | 20012.44  | 20012.44 | 431.91 sec |
| ogm-SG-A                    | 21494.76 | 20561.04 | 20089.97 | 20027.98  | 20027.98 | 391.10 sec |
| ogm-BUNDLE-A+               | $\infty$ | 21071.90 | 20450.02 | 20156.95  | 20156.95 | 404.59 sec |
| ogm-SG-A+                   | 21676.63 | 20795.95 | 20325.79 | 20198.96  | 20198.96 | 404.87 sec |
| ogm-BUNDLE-A-               | $\infty$ | 20649.30 | 20058.21 | 20012.19  | 20012.19 | 392.39 sec |
| ogm-SG-A-                   | 21778.78 | 21137.37 | 20530.48 | 20055.22  | 20055.22 | 425.47 sec |
| TRWS-pct                    | 57704.26 | 40144.16 | 20889.89 | 20012.17  | 20012.17 | 45.62 sec  |
| MCI-TC-MTC-TCI              | 27591.57 | 21996.35 | 21188.09 | 20621.78  | 20450.11 | 442.89 sec |
| MCI-pct                     | 62394.57 | 49974.51 | 32135.88 | 20889.89  | 20889.89 | 429.54 sec |
| ogm-CombiLP                 | $\infty$ | $\infty$ | $\infty$ | 20012.14  | 20012.14 | 36.68 sec  |

Table 34: color-seg-n8 (9 instances)

| algorithm                | value    |          |          |           |          | time        |
|--------------------------|----------|----------|----------|-----------|----------|-------------|
|                          | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)    | (end)       |
| $\alpha$ -Exp-pct        | 62394.68 | 20900.32 | 20011.24 | 20011.24  | 20011.24 | 5.56 sec    |
| $\alpha$ -Exp-VIEW       | 57709.51 | 20011.13 | 20011.13 | 20011.13  | 20011.13 | 7.79 sec    |
| FastPD                   | 20011.14 | 20011.14 | 20011.14 | 20011.14  | 20011.14 | 0.45 sec    |
| FastPD-pct               | 57707.85 | 20011.24 | 20011.24 | 20011.24  | 20011.24 | 2.79 sec    |
| ogm-FastPD-LF1           | $\infty$ | 20011.13 | 20011.13 | 20011.13  | 20011.13 | 2.07 sec    |
| ogm-FastPD-LF2           | $\infty$ | 20011.08 | 20010.28 | 20010.28  | 20010.28 | 21.51 sec   |
| ogm-ICM                  | 61942.65 | 25302.60 | 25302.60 | 25302.60  | 25302.60 | 1.98 sec    |
| ogm-LF-1                 | 56991.75 | 25306.40 | 25306.40 | 25306.40  | 25306.40 | 0.93 sec    |
| ogm-LF-2                 | 63093.85 | 58165.60 | 21129.07 | 21129.07  | 21129.07 | 25.21 sec   |
| ogm-TRWS-LF1             | $\infty$ | $\infty$ | $\infty$ | 19991.28  | 19991.28 | 25.26 sec   |
| ogm-TRWS-LF2             | $\infty$ | $\infty$ | $\infty$ | 19991.27  | 19991.27 | 43.80 sec   |
| $\alpha\beta$ -Swap-VIEW | 57715.97 | 20039.32 | 20038.26 | 20038.26  | 20038.26 | 9.75 sec    |
| BPS-TL                   | 20146.84 | 20120.78 | 20120.78 | 20120.78  | 20120.78 | 62.69 sec   |
| ogm-BPS                  | 54929.93 | 20179.38 | 20080.07 | 20079.74  | 20079.73 | 878.41 sec  |
| ogm-LBP-0.5              | 55301.83 | 20365.28 | 20102.32 | 20087.13  | 20087.10 | 850.32 sec  |
| ogm-LBP-0.95             | 54940.09 | 20198.34 | 20082.52 | 20081.48  | 20081.48 | 786.09 sec  |
| ogm-TRBP-0.5             | 55301.77 | 20436.86 | 20113.84 | 20086.99  | 20086.96 | 1157.33 sec |
| ogm-TRBP-0.95            | 55116.61 | 20248.91 | 20083.63 | 20081.13  | 20081.13 | 1078.86 sec |
| ogm-TRBPS                | 54930.14 | 20236.77 | 20080.74 | 20079.32  | 20079.31 | 1146.51 sec |
| MCR-TC-MTC               | 26448.30 | 23281.06 | 21413.64 | 20671.36  | 20671.36 | 539.88 sec  |
| MCR-pct                  | 62394.67 | 44933.76 | 32134.24 | 20988.67  | 20988.67 | 486.38 sec  |
| ogm-ADSAL                | $\infty$ | $\infty$ | 20000.42 | 19991.31  | 19991.25 | 1320.13 sec |
| ogm-BUNDLE-A             | $\infty$ | 21341.34 | 20572.29 | 20049.68  | 20020.56 | 935.94 sec  |
| ogm-BUNDLE-H             | $\infty$ | 20916.90 | 20218.04 | 19997.28  | 19991.94 | 978.44 sec  |
| ogm-SG-A                 | $\infty$ | 20922.02 | 20512.15 | 20376.65  | 20360.63 | 854.29 sec  |
| ogm-BUNDLE-A+            | $\infty$ | 21461.41 | 21109.98 | 20382.07  | 20248.63 | 973.20 sec  |
| ogm-SG-A+                | $\infty$ | 21207.63 | 21206.35 | 21203.83  | 21203.60 | 896.80 sec  |
| ogm-BUNDLE-A-            | $\infty$ | 21124.12 | 20282.12 | 19993.44  | 19991.51 | 986.69 sec  |
| ogm-SG-A-                | $\infty$ | 21200.82 | 20765.81 | 20219.82  | 20128.58 | 913.14 sec  |
| TRWS-TL                  | 20016.38 | 19992.10 | 19991.32 | 19991.31  | 19991.31 | 24.80 sec   |
| TRWS-pct                 | 62394.67 | 54762.34 | 40149.49 | 19991.33  | 19991.33 | 91.86 sec   |
| MCI-TC-MTC-TCI           | 26448.30 | 26198.68 | 21413.52 | 20671.29  | 20670.74 | 836.33 sec  |
| MCI-pct                  | 62394.67 | 59449.29 | 40149.46 | 30086.09  | 20893.77 | 534.88 sec  |
| ogm-CombiLP              | $\infty$ | $\infty$ | $\infty$ | 19991.21  | 19991.21 | 118.53 sec  |



Table 35: color-seg (3 instances)

| algorithm                | value        |              |              |              |              | time        |
|--------------------------|--------------|--------------|--------------|--------------|--------------|-------------|
|                          | (1 sec)      | (10 sec)     | (60 sec)     | (600 sec)    | (end)        | (end)       |
| $\alpha$ -Exp-pct        | 345351303.33 | 308472274.33 | 308472274.33 | 308472274.33 | 308472274.33 | 0.82 sec    |
| $\alpha$ -Exp-VIEW       | 345351304.67 | 308472275.67 | 308472275.67 | 308472275.67 | 308472275.67 | 5.98 sec    |
| FastPD                   | 308472275.00 | 308472275.00 | 308472275.00 | 308472275.00 | 308472275.00 | 0.31 sec    |
| FastPD-pct               | 345351303.67 | 308472274.67 | 308472274.67 | 308472274.67 | 308472274.67 | 0.80 sec    |
| ogm-FastPD-LF1           | $\infty$     | 308472275.00 | 308472275.00 | 308472275.00 | 308472275.00 | 1.25 sec    |
| ogm-FastPD-LF2           | $\infty$     | 308472275.00 | 308472275.00 | 308472275.00 | 308472275.00 | 8.58 sec    |
| ogm-ICM                  | 345549316.00 | 311333421.33 | 311333421.33 | 311333421.33 | 311333421.33 | 2.38 sec    |
| ogm-LF-1                 | 345549665.67 | 311333771.00 | 311333771.00 | 311333771.00 | 311333771.00 | 1.94 sec    |
| ogm-LF-2                 | 345719073.00 | 345399441.67 | 309850181.00 | 309850181.00 | 309850181.00 | 11.82 sec   |
| ogm-TRWS-LF1             | $\infty$     | $\infty$     | $\infty$     | 308472294.33 | 308472294.33 | 104.99 sec  |
| ogm-TRWS-LF2             | $\infty$     | $\infty$     | $\infty$     | 308472294.33 | 308472294.33 | 102.44 sec  |
| $\alpha\beta$ -Swap-VIEW | 345351321.33 | 308472292.33 | 308472292.33 | 308472292.33 | 308472292.33 | 6.25 sec    |
| BPS-TL                   | 308758625.67 | 308733349.67 | 308733349.67 | 308733349.67 | 308733349.67 | 68.24 sec   |
| ogm-BPS                  | 345352413.67 | 308604967.00 | 308495577.33 | 308494459.00 | 308494459.00 | 106.57 sec  |
| ogm-LBP-0.5              | 345353763.67 | 308651580.00 | 308507008.00 | 308492950.67 | 308492950.67 | 362.43 sec  |
| ogm-LBP-0.95             | 345352677.00 | 308613159.00 | 308497940.33 | 308494213.33 | 308494213.33 | 117.09 sec  |
| ogm-TRBP-0.5             | 345354014.67 | 308667578.33 | 308509416.00 | 308492909.67 | 308492909.67 | 1245.66 sec |
| ogm-TRBP-0.95            | 345353110.67 | 308647141.67 | 308500132.00 | 308494370.67 | 308494370.67 | 1237.92 sec |
| ogm-TRBPS                | 345352786.33 | 308628521.67 | 308497583.67 | 308494514.33 | 308494514.33 | 1009.29 sec |
| MCR-TC-MTC               | 328487770.67 | 309842256.00 | 308718278.00 | 308472274.33 | 308472274.33 | 89.46 sec   |
| MCR-pct                  | 328323156.00 | 308472274.33 | 308472274.33 | 308472274.33 | 308472274.33 | 0.82 sec    |
| ogm-ADSAL                | $\infty$     | $\infty$     | 308477694.67 | 308472407.67 | 308472289.00 | 2156.82 sec |
| ogm-BUNDLE-A             | $\infty$     | 308605794.33 | 308554883.33 | 308473557.67 | 308472329.33 | 1661.42 sec |
| ogm-BUNDLE-H             | $\infty$     | 308602001.00 | 308528112.67 | 308475047.67 | 308472407.67 | 1585.35 sec |
| ogm-SG-A                 | $\infty$     | 308720891.00 | 308717737.67 | 308568128.33 | 308485243.33 | 1399.36 sec |
| ogm-BUNDLE-A+            | $\infty$     | 308605613.33 | 308539575.67 | 308473286.67 | 308472378.67 | 1625.25 sec |
| ogm-SG-A+                | $\infty$     | 308736418.67 | 308736418.67 | 308736418.67 | 308736418.67 | 1325.76 sec |
| ogm-BUNDLE-A-            | $\infty$     | 308605662.67 | 308539580.00 | 308473417.33 | 308472520.67 | 1522.39 sec |
| ogm-SG-A-                | $\infty$     | 308591352.00 | 308529965.33 | 308495789.33 | 308481954.33 | 1389.03 sec |
| TRWS-TL                  | 308481039.00 | 308473445.67 | 308472322.67 | 308472294.33 | 308472294.33 | 90.76 sec   |
| TRWS-pct                 | 345351303.33 | 308472290.67 | 308472290.67 | 308472290.67 | 308472290.67 | 1.07 sec    |
| MCI-TC-MTC-TCI           | 320296118.67 | 309399034.00 | 308693040.33 | 308472274.33 | 308472274.33 | 80.19 sec   |
| MCI-pct                  | 345351303.33 | 308472274.33 | 308472274.33 | 308472274.33 | 308472274.33 | 0.98 sec    |
| ogm-CombiLP              | $\infty$     | $\infty$     | $\infty$     | $\infty$     | 308472274.33 | 483.64 sec  |

Table 36: correlation-clustering (715 instances)

| algorithm       | value   |          |          |           |         | time     |
|-----------------|---------|----------|----------|-----------|---------|----------|
|                 | (1 sec) | (10 sec) | (60 sec) | (600 sec) | (end)   | (end)    |
| ogm-ICM         | -579.85 | -585.60  | -585.60  | -585.60   | -585.60 | 1.28 sec |
| ogm-LF-1        | -584.76 | -585.60  | -585.60  | -585.60   | -585.60 | 0.63 sec |
| MCR-CC          | -626.90 | -626.91  | -626.91  | -626.91   | -626.91 | 0.12 sec |
| MCR-CCFDB       | -626.91 | -626.91  | -626.91  | -626.91   | -626.91 | 0.08 sec |
| MCR-CCFDB-OWC   | -626.93 | -626.93  | -626.93  | -626.93   | -626.93 | 0.08 sec |
| MCI-CCFDB-CCIFD | -627.91 | -628.12  | -628.16  | -628.16   | -628.16 | 0.78 sec |
| MCI-CCI         | -626.26 | -628.15  | -628.16  | -628.16   | -628.16 | 1.28 sec |
| MCI-CCIFD       | -626.05 | -628.15  | -628.16  | -628.16   | -628.16 | 1.25 sec |

Table 37: dtf-chinesechar (100 instances)

| algorithm     | value     |           |           |           |           | time           |
|---------------|-----------|-----------|-----------|-----------|-----------|----------------|
|               | (1 sec)   | (10 sec)  | (60 sec)  | (600 sec) | (end)     | (end)          |
| LSA-TR (euc.) | -49548.10 | -49548.10 | -49548.10 | -49548.10 | -49548.10 | 0.05 sec       |
| LSA-TR (ham.) | -49536.76 | -49536.76 | -49536.76 | -49536.76 | -49536.76 | 0.06 sec       |
| ogm-ICM       | -49515.98 | -49515.98 | -49515.98 | -49515.98 | -49515.98 | 0.21 sec       |
| ogm-LBP-LF2   | $\infty$  | $\infty$  | $\infty$  | $\infty$  | -49520.26 | 481.32 sec     |
| ogm-LF-1      | -49516.08 | -49516.08 | -49516.08 | -49516.08 | -49516.08 | 0.23 sec       |
| ogm-LF-2      | -45591.15 | -47666.27 | -49531.11 | -49531.11 | -49531.11 | 7.34 sec       |
| ogm-LF-3      | -45591.15 | -45602.09 | -45602.19 | -45992.19 | -49535.37 | 637.92 sec     |
| ogm-TRWS-LF1  | $\infty$  | $\infty$  | $\infty$  | -49505.84 | -49505.84 | 84.41 sec      |
| ogm-TRWS-LF2  | $\infty$  | $\infty$  | $\infty$  | -49519.42 | -49519.42 | 83.78 sec      |
| BPS-TAB       | -49527.72 | -49536.05 | -49537.77 | -49538.16 | -49538.16 | 62.69 sec      |
| ogm-BPS       | -49469.46 | -49503.03 | -49511.09 | -49516.46 | -49516.46 | 456.79 sec     |
| ogm-LBP-0.5   | -48870.75 | -49499.07 | -49510.35 | -49519.68 | -49519.71 | 482.00 sec     |
| ogm-LBP-0.95  | -48872.13 | -49503.49 | -49513.31 | -49520.37 | -49520.40 | 479.42 sec     |
| ogm-TRBP-0.5  | -48871.44 | -49498.15 | -49509.10 | -49529.27 | -49529.34 | 561.98 sec     |
| ogm-TRBP-0.95 | -48873.61 | -49500.01 | -49513.58 | -49531.36 | -49531.45 | 540.99 sec     |
| ogm-TRBPS     | -49471.72 | -49497.96 | -49512.83 | -49523.74 | -49523.79 | 466.66 sec     |
| ADDD          | -36020.24 | -46417.40 | -48565.34 | -48656.71 | -48656.71 | 9.74 sec       |
| MPLP          | -49434.39 | -49442.39 | -49445.12 | -49445.12 | -49445.12 | 516.36 sec     |
| ogm-ADSAL     | $\infty$  | -49516.11 | -49521.27 | -49524.17 | -49524.30 | 730.78 sec     |
| ogm-BUNDLE-A  | $\infty$  | -49454.71 | -49468.59 | -49471.84 | -49471.84 | 445.46 sec     |
| ogm-BUNDLE-H  | $\infty$  | -49486.94 | -49487.46 | -49489.73 | -49489.73 | 454.52 sec     |
| ogm-SG-A      | -49426.72 | -49476.22 | -49476.24 | -49476.28 | -49476.28 | 420.01 sec     |
| ogm-BUNDLE-A+ | $\infty$  | -49442.23 | -49446.44 | -49469.23 | -49469.63 | 443.99 sec     |
| ogm-SG-A+     | -49427.18 | -49453.12 | -49457.32 | -49457.71 | -49457.71 | 417.48 sec     |
| ogm-BUNDLE-A- | $\infty$  | -49462.49 | -49462.49 | -49468.24 | -49468.24 | 447.56 sec     |
| ogm-SG-A-     | $\infty$  | -49460.05 | -49477.18 | -49478.46 | -49478.46 | 416.33 sec     |
| ogm-LP-LP     | -36020.24 | -36020.24 | -36020.24 | -40843.15 | -48928.96 | 1335.58 sec    |
| QPBO          | -49501.95 | -49501.95 | -49501.95 | -49501.95 | -49501.95 | 0.17 sec       |
| TRWS-TAB      | -49499.99 | -49512.54 | -49514.06 | -49514.06 | -49514.06 | 78.84 sec      |
| TRWS-pct      | -39258.01 | -48384.86 | -49496.76 | -49496.76 | -49496.76 | 4.43 sec       |
| ogm-ILP-pct   | -36020.24 | -36020.24 | -36020.24 | -36020.24 | -49547.41 | 3553.71 sec    |
| MCBC-pct      | $\infty$  | $\infty$  | $\infty$  | $\infty$  | -49550.10 | 2053.89 sec    |
| ogm-ILP       | -36020.24 | -36020.24 | -36020.24 | -36020.24 | -49536.00 | 3569.52 sec    |
| SA            | $\infty$  | $\infty$  | $\infty$  | $\infty$  | -49533.02 | <i>NaN</i> sec |

Table 38: geo-surf-3 (300 instances)

| algorithm        | value    |          |          |           |        | time      |
|------------------|----------|----------|----------|-----------|--------|-----------|
|                  | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)  | (end)     |
| $\alpha$ -Fusion | 132.14   | 132.14   | 132.14   | 132.14    | 132.14 | 0.00 sec  |
| ogm-ICM          | 132.77   | 132.77   | 132.77   | 132.77    | 132.77 | 0.00 sec  |
| ogm-LBP-LF1      | 132.17   | 132.17   | 132.17   | 132.17    | 132.17 | 0.07 sec  |
| ogm-LBP-LF2      | 132.17   | 132.17   | 132.17   | 132.17    | 132.17 | 0.09 sec  |
| ogm-LF-1         | 132.77   | 132.77   | 132.77   | 132.77    | 132.77 | 0.00 sec  |
| ogm-LF-2         | 132.36   | 132.36   | 132.36   | 132.36    | 132.36 | 0.02 sec  |
| ogm-LF-3         | 132.21   | 132.21   | 132.21   | 132.21    | 132.21 | 0.30 sec  |
| ogm-BPS          | 132.17   | 132.17   | 132.17   | 132.17    | 132.17 | 0.02 sec  |
| ogm-LBP-0.5      | 132.17   | 132.17   | 132.17   | 132.17    | 132.17 | 0.07 sec  |
| ogm-LBP-0.95     | 132.17   | 132.17   | 132.17   | 132.17    | 132.17 | 0.03 sec  |
| ogm-TRBP-0.5     | 132.16   | 132.16   | 132.16   | 132.16    | 132.16 | 1.14 sec  |
| ogm-TRBP-0.95    | 132.16   | 132.16   | 132.16   | 132.16    | 132.16 | 0.60 sec  |
| ogm-TRBPS        | 132.16   | 132.16   | 132.16   | 132.16    | 132.16 | 0.49 sec  |
| ADDD             | 163.99   | 132.14   | 132.14   | 132.14    | 132.14 | 0.12 sec  |
| MPLP             | 132.14   | 132.14   | 132.14   | 132.14    | 132.14 | 0.15 sec  |
| MPLP-C           | 132.14   | 132.14   | 132.14   | 132.14    | 132.14 | 0.16 sec  |
| ogm-BUNDLE-A     | 132.28   | 132.14   | 132.14   | 132.14    | 132.14 | 21.88 sec |
| ogm-BUNDLE-H     | 132.35   | 132.14   | 132.14   | 132.14    | 132.14 | 9.57 sec  |
| ogm-SG-A         | 132.39   | 132.14   | 132.14   | 132.14    | 132.14 | 22.37 sec |
| ogm-BUNDLE-A+    | 132.67   | 132.15   | 132.14   | 132.14    | 132.14 | 2.78 sec  |
| ogm-SG-A+        | 132.16   | 132.14   | 132.14   | 132.14    | 132.14 | 1.80 sec  |
| ogm-BUNDLE-A-    | 132.49   | 132.15   | 132.14   | 132.14    | 132.14 | 31.74 sec |
| ogm-SG-A-        | 133.54   | 132.32   | 132.15   | 132.14    | 132.14 | 31.06 sec |
| ogm-LP-LP        | 549.17   | 132.14   | 132.14   | 132.14    | 132.14 | 0.55 sec  |
| BRAOBB-1         | $\infty$ | 132.14   | 132.14   | 132.14    | 132.14 | 0.66 sec  |
| BRAOBB-2         | $\infty$ | 132.14   | 132.14   | 132.14    | 132.14 | 6.67 sec  |
| BRAOBB-3         | $\infty$ | 132.14   | 132.14   | 132.14    | 132.14 | 17.99 sec |
| ogm-ILP          | 132.14   | 132.14   | 132.14   | 132.14    | 132.14 | 0.13 sec  |

Table 39: geo-surf-7 (300 instances)

| algorithm        | value    |          |          |           |        | time       |
|------------------|----------|----------|----------|-----------|--------|------------|
|                  | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)  | (end)      |
| $\alpha$ -Fusion | 477.83   | 477.83   | 477.83   | 477.83    | 477.83 | 0.02 sec   |
| ogm-ICM          | 839.28   | 839.28   | 839.28   | 839.28    | 839.28 | 0.01 sec   |
| ogm-LBP-LF1      | $\infty$ | $\infty$ | 498.45   | 498.45    | 498.45 | 0.60 sec   |
| ogm-LBP-LF2      | $\infty$ | $\infty$ | 498.44   | 498.44    | 498.44 | 0.65 sec   |
| ogm-LF-1         | 838.32   | 838.32   | 838.32   | 838.32    | 838.32 | 0.01 sec   |
| ogm-LF-2         | 692.68   | 692.68   | 692.68   | 692.68    | 692.68 | 0.12 sec   |
| ogm-LF-3         | 688.52   | 632.48   | 620.69   | 620.56    | 620.56 | 9.45 sec   |
| ogm-BPS          | 498.14   | 498.14   | 498.14   | 498.14    | 498.14 | 0.37 sec   |
| ogm-LBP-0.5      | 498.53   | 498.43   | 498.43   | 498.43    | 498.43 | 0.60 sec   |
| ogm-LBP-0.95     | 498.50   | 498.50   | 498.50   | 498.50    | 498.50 | 0.33 sec   |
| ogm-TRBP-0.5     | 488.56   | 485.87   | 485.86   | 485.86    | 485.86 | 8.07 sec   |
| ogm-TRBP-0.95    | 487.10   | 486.68   | 486.68   | 486.68    | 486.68 | 7.42 sec   |
| ogm-TRBPS        | 487.47   | 487.17   | 487.17   | 487.17    | 487.17 | 6.85 sec   |
| ADDD             | 675.32   | 476.95   | 476.95   | 476.95    | 476.95 | 0.55 sec   |
| MPLP             | 482.65   | 477.05   | 477.04   | 477.04    | 477.04 | 1.31 sec   |
| MPLP-C           | 482.65   | 477.03   | 477.02   | 477.02    | 477.02 | 1.43 sec   |
| ogm-BUNDLE-A     | 523.81   | 482.02   | 477.35   | 477.08    | 477.08 | 53.41 sec  |
| ogm-BUNDLE-H     | 517.94   | 483.34   | 477.04   | 476.95    | 476.95 | 41.45 sec  |
| ogm-SG-A         | 513.69   | 489.06   | 480.35   | 479.02    | 479.02 | 52.38 sec  |
| ogm-BUNDLE-A+    | 534.38   | 489.43   | 477.00   | 476.95    | 476.95 | 32.29 sec  |
| ogm-SG-A+        | 522.62   | 484.47   | 477.65   | 477.28    | 477.28 | 29.96 sec  |
| ogm-BUNDLE-A-    | 517.61   | 488.72   | 480.43   | 479.37    | 479.37 | 53.63 sec  |
| ogm-SG-A-        | 530.70   | 504.29   | 490.26   | 488.41    | 488.41 | 52.14 sec  |
| ogm-LP-LP        | 1338.89  | 545.33   | 476.95   | 476.95    | 476.95 | 2.74 sec   |
| BRAOBB-1         | $\infty$ | $\infty$ | $\infty$ | $\infty$  | 479.82 | 973.62 sec |
| BRAOBB-2         | $\infty$ | $\infty$ | $\infty$ | $\infty$  | 477.10 | 957.11 sec |
| BRAOBB-3         | $\infty$ | $\infty$ | $\infty$ | $\infty$  | 477.11 | 685.84 sec |
| ogm-ILP          | 1064.96  | 476.95   | 476.95   | 476.95    | 476.95 | 0.95 sec   |

Table 40: image-seg-3rdorder (100 instances)

| algorithm       | value   |          |          |           |         | time      |
|-----------------|---------|----------|----------|-----------|---------|-----------|
|                 | (1 sec) | (10 sec) | (60 sec) | (600 sec) | (end)   | (end)     |
| ogm-ICM         | 7025.90 | 6131.73  | 6030.49  | 6030.49   | 6030.49 | 6.30 sec  |
| ogm-LF-1        | 6508.75 | 6030.29  | 6030.29  | 6030.29   | 6030.29 | 2.23 sec  |
| MCR-CC          | 6467.09 | 5836.44  | 5812.73  | 5812.02   | 5812.02 | 32.56 sec |
| MCR-CCFDB       | 6408.53 | 5832.20  | 5815.58  | 5815.37   | 5815.37 | 20.85 sec |
| MCR-CCFDB-OWC   | 6476.91 | 5832.08  | 5816.05  | 5815.31   | 5815.31 | 21.63 sec |
| MCI-CCFDB-CCIFD | 6928.83 | 6268.50  | 5771.32  | 5627.52   | 5627.52 | 46.68 sec |
| MCI-CCI         | 6966.93 | 6068.11  | 5633.09  | 5628.20   | 5628.11 | 70.54 sec |
| MCI-CCIFD       | 6975.68 | 6063.25  | 5633.13  | 5628.18   | 5627.52 | 50.78 sec |

Table 41: image-seg (100 instances)

| algorithm       | value   |          |          |           |         | time     |
|-----------------|---------|----------|----------|-----------|---------|----------|
|                 | (1 sec) | (10 sec) | (60 sec) | (600 sec) | (end)   | (end)    |
| ogm-ICM         | 6093.27 | 4906.92  | 4705.07  | 4705.07   | 4705.07 | 3.98 sec |
| ogm-KL          | 4952.04 | 4608.52  | 4608.49  | 4608.49   | 4608.49 | 1.46 sec |
| ogm-LF-1        | 5044.70 | 4705.01  | 4705.01  | 4705.01   | 4705.01 | 1.35 sec |
| MCR-CC          | 5379.29 | 4652.68  | 4465.35  | 4446.69   | 4446.69 | 8.54 sec |
| MCR-CCFDB       | 5313.23 | 4575.19  | 4457.85  | 4446.57   | 4446.57 | 4.34 sec |
| MCR-CCFDB-OWC   | 5324.28 | 4575.14  | 4457.80  | 4446.52   | 4446.52 | 4.34 sec |
| MCI-CCFDB-CCIFD | 5319.44 | 4573.18  | 4453.69  | 4442.64   | 4442.64 | 4.89 sec |
| MCI-CCI         | 4518.35 | 4447.07  | 4442.64  | 4442.64   | 4442.64 | 2.43 sec |
| MCI-CCIFD       | 4922.30 | 4453.89  | 4442.64  | 4442.64   | 4442.64 | 2.24 sec |

Table 42: inclusion (10 instances)

| algorithm        | value    |          |          |           |            | time        |
|------------------|----------|----------|----------|-----------|------------|-------------|
|                  | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)      | (end)       |
| $\alpha$ -Fusion | 1587.13  | 1587.13  | 1587.13  | 1587.13   | 1587.13    | 0.04 sec    |
| ogm-ICM          | 1556.20  | 1556.20  | 1556.20  | 1556.20   | 1556.20    | 0.01 sec    |
| ogm-LBP-LF1      | $\infty$ | $\infty$ | 1400.66  | 1400.66   | 1400.66    | 19.33 sec   |
| ogm-LBP-LF2      | $\infty$ | $\infty$ | 1400.61  | 1400.61   | 1400.61    | 19.37 sec   |
| ogm-LF-1         | 1556.20  | 1556.20  | 1556.20  | 1556.20   | 1556.20    | 0.01 sec    |
| ogm-LF-2         | 1476.39  | 1476.39  | 1476.39  | 1476.39   | 1476.39    | 0.09 sec    |
| ogm-LF-3         | 1462.71  | 1461.23  | 1461.23  | 1461.23   | 1461.23    | 1.14 sec    |
| ogm-BPS          | 1400.66  | 1400.66  | 1400.66  | 1400.66   | 1400.66    | 21.42 sec   |
| ogm-LBP-0.5      | 1400.64  | 1400.59  | 1400.59  | 1400.59   | 1400.59    | 19.77 sec   |
| ogm-LBP-0.95     | 1400.62  | 1400.58  | 1400.58  | 1400.58   | 1400.58    | 19.82 sec   |
| ogm-TRBP-0.5     | 1400.63  | 1400.61  | 1400.61  | 1400.61   | 1400.61    | 21.42 sec   |
| ogm-TRBP-0.95    | 1400.62  | 1400.58  | 1400.58  | 1400.58   | 1400.58    | 20.65 sec   |
| ogm-TRBPS        | 1400.69  | 1400.64  | 1400.64  | 1400.64   | 1400.64    | 20.68 sec   |
| ADDD             | 1771.26  | 1694.72  | 1694.72  | 1694.72   | 1694.72    | 6.23 sec    |
| MPLP             | 3806.14  | 2201.00  | 1901.01  | 1901.01   | 1901.01    | 5.94 sec    |
| MPLP-C           | 3906.30  | 2201.00  | 1800.85  | 1600.79   | 1600.79    | 3579.25 sec |
| ogm-BUNDLE-A     | 22585.05 | 22585.05 | 7314.41  | 4007.73   | 4007.73    | 77.24 sec   |
| ogm-BUNDLE-H     | 15537.35 | 1601.27  | 1400.76  | 1400.76   | 1400.76    | 73.24 sec   |
| ogm-SG-A         | 26797.36 | 26797.36 | 26797.36 | 26797.36  | 26797.36   | 78.38 sec   |
| ogm-BUNDLE-A+    | 22585.05 | 22585.05 | 7514.88  | 3909.68   | 3909.68    | 77.44 sec   |
| ogm-SG-A+        | 26797.36 | 26797.36 | 26797.36 | 26797.36  | 26797.36   | 77.57 sec   |
| ogm-BUNDLE-A-    | 22585.05 | 20351.85 | 1400.71  | 1400.69   | 1400.69    | 76.92 sec   |
| ogm-SG-A-        | 26484.90 | 19836.87 | 13328.28 | 11920.10  | 11920.10   | 75.57 sec   |
| ogm-LP-LP        | 1771.26  | 1771.26  | 1663.97  | 1663.97   | 1663.97    | 18.27 sec   |
| MCI-TC-MTC-TCI   | 1771.26  | 1620.73  | 1483.16  | 1400.57   | 1400.57    | 61.46 sec   |
| BRAOBB-1         | $\infty$ | $\infty$ | $\infty$ | $\infty$  | <i>Inf</i> | 3600.01 sec |
| BRAOBB-2         | $\infty$ | $\infty$ | $\infty$ | $\infty$  | 1401.73    | 3600.01 sec |
| BRAOBB-3         | $\infty$ | $\infty$ | $\infty$ | $\infty$  | 1401.64    | 3600.01 sec |
| ogm-ILP          | 1771.26  | 1400.57  | 1400.57  | 1400.57   | 1400.57    | 6.21 sec    |

Table 43: inpainting-n4 (2 instances)

| algorithm                   | value     |           |           |           |           | time        |
|-----------------------------|-----------|-----------|-----------|-----------|-----------|-------------|
|                             | (1 sec)   | (10 sec)  | (60 sec)  | (600 sec) | (end)     | (end)       |
| $\alpha$ -Exp-pct           | 454.75    | 454.75    | 454.75    | 454.75    | 454.75    | 0.11 sec    |
| FastPD                      | 454.75    | 454.75    | 454.75    | 454.75    | 454.75    | 0.01 sec    |
| FastPD-pct                  | 454.75    | 454.75    | 454.75    | 454.75    | 454.75    | 0.07 sec    |
| mrf- $\alpha$ -Exp-TL       | 454.75    | 454.75    | 454.75    | 454.75    | 454.75    | 0.01 sec    |
| mrf- $\alpha\beta$ -Swap-TL | 454.35    | 454.35    | 454.35    | 454.35    | 454.35    | 0.01 sec    |
| ogm-FastPD-LF1              | 454.75    | 454.75    | 454.75    | 454.75    | 454.75    | 0.05 sec    |
| ogm-FastPD-LF2              | 454.75    | 454.75    | 454.75    | 454.75    | 454.75    | 0.20 sec    |
| ogm-ICM                     | 3453.29   | 3453.29   | 3453.29   | 3453.29   | 3453.29   | 0.06 sec    |
| ogm-LF-1                    | 3453.29   | 3453.29   | 3453.29   | 3453.29   | 3453.29   | 0.04 sec    |
| ogm-LF-2                    | 3453.29   | 3453.29   | 3453.29   | 3453.29   | 3453.29   | 0.18 sec    |
| ogm-TRWS-LF1                | $\infty$  | 489.70    | 489.70    | 489.70    | 489.70    | 1.32 sec    |
| ogm-TRWS-LF2                | $\infty$  | 489.30    | 489.30    | 489.30    | 489.30    | 1.45 sec    |
| mrf-LBP-TL                  | 477.91    | 475.56    | 475.56    | 475.56    | 475.56    | 4.25 sec    |
| mrf-BPS-TL                  | 454.35    | 454.35    | 454.35    | 454.35    | 454.35    | 1.69 sec    |
| ogm-BPS                     | 509.72    | 497.55    | 497.55    | 497.55    | 497.55    | 11.99 sec   |
| ogm-LBP-0.5                 | 510.51    | 480.66    | 479.09    | 479.09    | 479.09    | 17.10 sec   |
| ogm-LBP-0.95                | 508.94    | 482.63    | 479.49    | 479.49    | 479.49    | 14.70 sec   |
| ogm-TRBP-0.5                | 540.35    | 483.41    | 479.09    | 479.09    | 479.09    | 20.63 sec   |
| ogm-TRBP-0.95               | 508.55    | 482.63    | 479.49    | 479.49    | 479.49    | 17.29 sec   |
| ogm-TRBPS                   | 509.72    | 497.55    | 497.55    | 497.55    | 497.55    | 13.66 sec   |
| MCR-TC-MTC                  | 520.04    | 520.04    | 499.12    | 499.12    | 461.81    | 1386.81 sec |
| MCR-pct                     | 541514.00 | 541514.00 | 270523.38 | 270523.38 | 1179.00   | 1248.88 sec |
| mrf-TRWS-TL                 | 454.75    | 454.75    | 454.75    | 454.75    | 454.75    | 0.97 sec    |
| ogm-ADSAL                   | 484.98    | 454.75    | 454.75    | 454.75    | 454.75    | 59.91 sec   |
| ogm-BUNDLE-A                | 455.25    | 455.25    | 455.25    | 455.25    | 455.25    | 39.98 sec   |
| ogm-BUNDLE-H                | 455.25    | 455.25    | 455.25    | 455.25    | 455.25    | 19.13 sec   |
| ogm-SG-A                    | 455.25    | 455.25    | 455.25    | 455.25    | 455.25    | 21.38 sec   |
| ogm-BUNDLE-A+               | 455.25    | 455.25    | 455.25    | 455.25    | 455.25    | 19.49 sec   |
| ogm-SG-A+                   | 455.25    | 455.25    | 455.25    | 455.25    | 455.25    | 20.25 sec   |
| ogm-BUNDLE-A-               | 455.25    | 455.25    | 455.25    | 455.25    | 455.25    | 39.31 sec   |
| ogm-SG-A-                   | 455.25    | 454.35    | 454.35    | 454.35    | 454.35    | 39.48 sec   |
| TRWS-pct                    | 270479.80 | 489.30    | 489.30    | 489.30    | 489.30    | 2.77 sec    |
| MCI-TC-MTC-TCI              | 520.04    | 520.04    | 499.12    | 499.12    | 461.81    | 1812.16 sec |
| MCI-pct                     | 541514.00 | 541514.00 | 270479.80 | 270479.80 | 270479.80 | 1807.10 sec |
| ogm-CombiLP                 | $\infty$  | 461.81    | 461.81    | 461.81    | 461.81    | 129.04 sec  |

Table 44: inpainting-n8 (2 instances)

| algorithm                | value     |           |           |           |           | time        |
|--------------------------|-----------|-----------|-----------|-----------|-----------|-------------|
|                          | (1 sec)   | (10 sec)  | (60 sec)  | (600 sec) | (end)     | (end)       |
| $\alpha$ -Exp-pct        | 464.76    | 464.76    | 464.76    | 464.76    | 464.76    | 0.24 sec    |
| $\alpha$ -Exp-VIEW       | 465.02    | 465.02    | 465.02    | 465.02    | 465.02    | 0.43 sec    |
| FastPD                   | 465.02    | 465.02    | 465.02    | 465.02    | 465.02    | 0.09 sec    |
| FastPD-pct               | 464.76    | 464.76    | 464.76    | 464.76    | 464.76    | 0.17 sec    |
| ogm-FastPD-LF1           | 465.02    | 465.02    | 465.02    | 465.02    | 465.02    | 0.17 sec    |
| ogm-FastPD-LF2           | 465.02    | 465.02    | 465.02    | 465.02    | 465.02    | 0.67 sec    |
| ogm-ICM                  | 3451.36   | 3451.36   | 3451.36   | 3451.36   | 3451.36   | 0.09 sec    |
| ogm-LF-1                 | 3451.36   | 3451.36   | 3451.36   | 3451.36   | 3451.36   | 0.06 sec    |
| ogm-LF-2                 | 3451.36   | 3451.36   | 3451.36   | 3451.36   | 3451.36   | 0.56 sec    |
| ogm-TRWS-LF1             | $\infty$  | $\infty$  | 499.36    | 499.36    | 499.36    | 11.25 sec   |
| ogm-TRWS-LF2             | $\infty$  | 500.09    | 499.30    | 499.30    | 499.30    | 10.24 sec   |
| $\alpha\beta$ -Swap-VIEW | 465.02    | 465.02    | 465.02    | 465.02    | 465.02    | 0.38 sec    |
| BPS-TL                   | 468.21    | 468.21    | 468.21    | 468.21    | 468.21    | 7.76 sec    |
| ogm-BPS                  | 515.44    | 494.20    | 493.96    | 493.96    | 493.96    | 9.64 sec    |
| ogm-LBP-0.5              | 627.51    | 495.58    | 493.79    | 493.79    | 493.79    | 41.77 sec   |
| ogm-LBP-0.95             | 567.63    | 493.96    | 493.96    | 493.96    | 493.96    | 11.67 sec   |
| ogm-TRBP-0.5             | 668.25    | 495.58    | 493.79    | 493.79    | 493.79    | 49.98 sec   |
| ogm-TRBP-0.95            | 613.66    | 494.24    | 493.96    | 493.96    | 493.96    | 13.36 sec   |
| ogm-TRBPS                | 541.27    | 494.20    | 493.96    | 493.96    | 493.96    | 10.58 sec   |
| MCR-TC-MTC               | 522.04    | 522.04    | 522.04    | 503.13    | 497.23    | 2073.02 sec |
| MCR-pct                  | 541514.00 | 541514.00 | 541514.00 | 270610.25 | 270610.25 | 1871.11 sec |
| ogm-ADSAL                | 481.59    | 481.06    | 467.20    | 467.20    | 467.20    | 297.48 sec  |
| ogm-BUNDLE-A             | 473.48    | 465.40    | 465.26    | 465.26    | 465.26    | 96.60 sec   |
| ogm-BUNDLE-H             | 468.37    | 466.37    | 465.34    | 465.34    | 465.34    | 97.86 sec   |
| ogm-SG-A                 | 466.81    | 466.81    | 466.81    | 466.81    | 466.81    | 93.27 sec   |
| ogm-BUNDLE-A+            | 478.38    | 478.38    | 465.26    | 465.26    | 465.26    | 102.22 sec  |
| ogm-SG-A+                | 465.76    | 465.76    | 465.76    | 465.76    | 465.76    | 100.14 sec  |
| ogm-BUNDLE-A-            | 466.81    | 466.81    | 465.26    | 465.26    | 465.26    | 99.87 sec   |
| ogm-SG-A-                | 466.81    | 466.81    | 466.81    | 466.81    | 466.81    | 86.02 sec   |
| TRWS-TL                  | 468.55    | 466.80    | 466.80    | 466.80    | 466.80    | 10.15 sec   |
| TRWS-pct                 | 541514.00 | 270485.10 | 467.69    | 467.69    | 467.69    | 19.74 sec   |
| MCI-TC-MTC-TCI           | 522.04    | 522.04    | 522.04    | 503.13    | 497.01    | 2166.25 sec |
| MCI-pct                  | 541514.00 | 541514.00 | 541514.00 | 270485.10 | 270485.10 | 1879.20 sec |
| ogm-CombiLP              | $\infty$  | $\infty$  | 465.35    | 465.35    | 465.35    | 2117.77 sec |

Table 45: knott-3d-150 (8 instances)

| algorithm       | value    |          |          |           |          | time      |
|-----------------|----------|----------|----------|-----------|----------|-----------|
|                 | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)    | (end)     |
| ogm-ICM         | -3741.57 | -4179.30 | -4179.30 | -4179.30  | -4179.30 | 2.05 sec  |
| ogm-KL          | -4431.67 | -4431.67 | -4431.67 | -4431.67  | -4431.67 | 0.15 sec  |
| ogm-LF-1        | -4179.30 | -4179.30 | -4179.30 | -4179.30  | -4179.30 | 0.72 sec  |
| MCR-CC          | -2226.74 | -4561.50 | -4568.75 | -4568.90  | -4568.90 | 17.72 sec |
| MCR-CCFDB       | -4098.00 | -4568.90 | -4568.90 | -4568.90  | -4568.90 | 1.87 sec  |
| MCR-CCFDB-OWC   | -4099.54 | -4570.87 | -4570.87 | -4570.87  | -4570.87 | 2.00 sec  |
| MCI-CCFDB-CCIFD | -4100.35 | -4571.69 | -4571.69 | -4571.69  | -4571.69 | 1.96 sec  |
| MCI-CCI         | -4548.31 | -4571.69 | -4571.69 | -4571.69  | -4571.69 | 0.57 sec  |
| MCI-CCIFD       | -4571.02 | -4571.69 | -4571.69 | -4571.69  | -4571.69 | 0.58 sec  |

Table 46: knott-3d-300 (8 instances)

| algorithm       | value    |           |           |           |           | time        |
|-----------------|----------|-----------|-----------|-----------|-----------|-------------|
|                 | (1 sec)  | (10 sec)  | (60 sec)  | (600 sec) | (end)     | (end)       |
| ogm-ICM         | 0.00     | -632.20   | -2305.45  | -25196.51 | -25196.51 | 84.37 sec   |
| ogm-KL          | -1989.98 | -25547.53 | -25556.93 | -25556.93 | -25556.93 | 13.16 sec   |
| ogm-LF-1        | -2088.80 | -17634.45 | -25243.76 | -25243.76 | -25243.76 | 29.08 sec   |
| MCR-CC          | -1989.98 | -1989.98  | -1989.98  | -10891.16 | -26786.99 | 3423.65 sec |
| MCR-CCFDB       | -1989.98 | -1989.98  | -1989.98  | -26664.02 | -27276.38 | 1338.99 sec |
| MCR-CCFDB-OWC   | -1989.98 | -1989.98  | -1989.98  | -26492.60 | -27287.49 | 1367.03 sec |
| MCI-CCFDB-CCIFD | -1989.98 | -1989.98  | -1989.98  | -27077.63 | -27280.79 | 1261.99 sec |
| MCI-CCI         | -1989.98 | -24081.39 | -27253.30 | -27290.39 | -27302.78 | 220.30 sec  |
| MCI-CCIFD       | -1989.98 | -4253.71  | -27274.93 | -27302.78 | -27302.78 | 104.55 sec  |

Table 47: knott-3d-450 (8 instances)

| algorithm       | value    |          |           |           |           | time        |
|-----------------|----------|----------|-----------|-----------|-----------|-------------|
|                 | (1 sec)  | (10 sec) | (60 sec)  | (600 sec) | (end)     | (end)       |
| ogm-ICM         | 0.00     | 0.00     | 0.00      | 0.00      | -72464.54 | 883.63 sec  |
| ogm-KL          | -4892.36 | -4892.36 | -72145.23 | -73188.82 | -73188.82 | 186.89 sec  |
| ogm-LF-1        | 0.00     | 0.00     | 0.00      | -72479.60 | -72479.60 | 298.07 sec  |
| MCR-CC          | -4892.36 | -4892.36 | -4892.36  | -4892.36  | -4892.36  | 9814.45 sec |
| MCR-CCFDB       | -4892.36 | -4892.36 | -4892.36  | -4892.36  | -4892.36  | 6404.34 sec |
| MCR-CCFDB-OWC   | -4892.36 | -4892.36 | -4892.36  | -4892.36  | -4892.36  | 6455.21 sec |
| MCI-CCFDB-CCIFD | -4892.36 | -4892.36 | -4892.36  | -4892.36  | -4892.36  | 6404.14 sec |
| MCI-CCI         | -4892.36 | -4892.36 | -50146.66 | -78324.79 | -78378.83 | 1196.62 sec |
| MCI-CCIFD       | -4892.36 | -4892.36 | -4892.36  | -72165.50 | -78379.99 | 1379.90 sec |

Table 48: matching (4 instances)

| algorithm        | value            |                  |                  |                  |                  | time        |
|------------------|------------------|------------------|------------------|------------------|------------------|-------------|
|                  | (1 sec)          | (10 sec)         | (60 sec)         | (600 sec)        | (end)            | (end)       |
| $\alpha$ -Fusion | 1465000000000.00 | 1465000000000.00 | 1465000000000.00 | 1465000000000.00 | 1465000000000.00 | 0.00 sec    |
| ogm-ICM          | 95.73            | 95.73            | 95.73            | 95.73            | 95.73            | 0.00 sec    |
| ogm-LBP-LF1      | 85.87            | 85.87            | 85.87            | 85.87            | 85.87            | 0.00 sec    |
| ogm-LBP-LF2      | 38.07            | 38.07            | 38.07            | 38.07            | 38.07            | 0.19 sec    |
| ogm-LF-1         | 95.73            | 95.73            | 95.73            | 95.73            | 95.73            | 0.00 sec    |
| ogm-LF-2         | 40.79            | 40.79            | 40.79            | 40.79            | 40.79            | 0.29 sec    |
| ogm-LF-3         | 40.79            | 39.81            | 39.81            | 39.81            | 39.81            | 12.35 sec   |
| ogm-TRWS-LF1     | 61.46            | 61.46            | 61.46            | 61.46            | 61.46            | 0.03 sec    |
| ogm-TRWS-LF2     | 33.31            | 33.31            | 33.31            | 33.31            | 33.31            | 0.32 sec    |
| BPS-TAB          | 33.97            | 33.97            | 33.97            | 33.97            | 33.97            | 0.11 sec    |
| ogm-BPS          | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 0.00 sec    |
| ogm-LBP-0.5      | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 0.00 sec    |
| ogm-LBP-0.95     | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 0.00 sec    |
| ogm-TRBP-0.5     | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 0.00 sec    |
| ogm-TRBP-0.95    | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 0.00 sec    |
| ogm-TRBPS        | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 97500000064.52   | 0.00 sec    |
| ADDD             | 985000000024.74  | 105000000039.23  | 105000000039.23  | 105000000039.23  | 105000000039.23  | 1.38 sec    |
| MPLP             | 32500000039.98   | 32500000039.98   | 32500000039.98   | 32500000039.98   | 32500000039.98   | 0.26 sec    |
| MPLP-C           | 30000000043.26   | 21.22            | 21.22            | 21.22            | 21.22            | 3.51 sec    |
| ogm-ADSAL        | $\infty$         | 35.70            | 34.08            | 32.47            | 32.47            | 1380.55 sec |
| ogm-BUNDLE-A     | 10000000044.34   | 10000000041.82   | 10000000041.82   | 10000000041.82   | 10000000041.82   | 1.56 sec    |
| ogm-BUNDLE-H     | 7500000055.96    | 7500000053.86    | 7500000053.86    | 7500000053.86    | 7500000053.86    | 1.57 sec    |
| ogm-SG-A         | 7500000099.44    | 7500000099.44    | 7500000099.44    | 7500000099.44    | 7500000099.44    | 1.38 sec    |
| ogm-BUNDLE-A+    | 10000000051.24   | 10000000048.72   | 10000000048.72   | 10000000048.72   | 10000000048.72   | 1.56 sec    |
| ogm-SG-A+        | 7500000101.08    | 7500000101.08    | 7500000101.08    | 7500000101.08    | 7500000101.08    | 1.37 sec    |
| ogm-BUNDLE-A-    | 10000000051.24   | 10000000048.72   | 10000000048.72   | 10000000048.72   | 10000000048.72   | 1.57 sec    |
| ogm-SG-A-        | 7500000096.42    | 7500000096.42    | 7500000096.42    | 7500000096.42    | 7500000096.42    | 1.32 sec    |
| ogm-LP-LP        | 1465000000000.00 | 1220000000010.31 | 102500000036.76  | 102500000036.76  | 102500000036.76  | 21.38 sec   |
| TRWS-TAB         | 43.38            | 43.38            | 43.38            | 43.38            | 43.38            | 0.03 sec    |
| BRAOBB-1         | $\infty$         | 21.22            | 21.22            | 21.22            | 21.22            | 2.05 sec    |
| BRAOBB-2         | $\infty$         | 21.22            | 21.22            | 21.22            | 21.22            | 8.09 sec    |
| BRAOBB-3         | $\infty$         | 21.22            | 21.22            | 21.22            | 21.22            | 41.69 sec   |
| ADDD-BB          | 1465000000000.00 | 1465000000000.00 | 1465000000000.00 | 1182500000004.84 | 37500000029.63   | 1192.02 sec |
| ogm-ASTAR        | $\infty$         | 21.22            | 21.22            | 21.22            | 21.22            | 0.80 sec    |
| ogm-CombiLP      | 34.66            | 34.66            | 34.66            | 21.22            | 21.22            | 314.52 sec  |
| ogm-ILP          | 1465000000000.00 | 1465000000000.00 | 1465000000000.00 | 282500000016.38  | 21.22            | 402.09 sec  |

Table 49: modularity-clustering (6 instances)

| algorithm       | value   |          |          |           |         | time        |
|-----------------|---------|----------|----------|-----------|---------|-------------|
|                 | (1 sec) | (10 sec) | (60 sec) | (600 sec) | (end)   | (end)       |
| ogm-ICM         | 0.0000  | 0.0000   | 0.0000   | 0.0000    | 0.0000  | 0.09 sec    |
| ogm-KL          | -0.4860 | -0.4860  | -0.4860  | -0.4860   | -0.4860 | 0.01 sec    |
| ogm-LF-1        | 0.0000  | 0.0000   | 0.0000   | 0.0000    | 0.0000  | 0.03 sec    |
| MCR-CC          | -0.1618 | -0.2505  | -0.2926  | -0.4619   | -0.4619 | 100.37 sec  |
| MCR-CCFDB       | -0.3486 | -0.4643  | -0.4643  | -0.4643   | -0.4643 | 2.15 sec    |
| MCR-CCFDB-OWC   | -0.3486 | -0.4682  | -0.4682  | -0.4682   | -0.4682 | 602.75 sec  |
| MCI-CCFDB-CCIFD | -0.4204 | -0.4613  | -0.4613  | -0.4613   | -0.4613 | 601.38 sec  |
| MCI-CCI         | -0.1518 | -0.3994  | -0.4279  | -0.4279   | -0.4312 | 1207.07 sec |
| MCI-CCIFD       | -0.1633 | -0.3519  | -0.3521  | -0.3685   | -0.4399 | 1204.03 sec |

Table 50: mrf-inpainting (2 instances)

| algorithm                   | value        |              |              |              |             | time         |
|-----------------------------|--------------|--------------|--------------|--------------|-------------|--------------|
|                             | (1 sec)      | (10 sec)     | (60 sec)     | (600 sec)    | (end)       | (end)        |
| FastPD                      | $\infty$     | $\infty$     | 32939430.00  | 32939430.00  | 32939430.00 | 7.72 sec     |
| mrf- $\alpha$ -Exp-TL       | 682736616.00 | 27343867.00  | 27266183.50  | 27266168.50  | 27266168.50 | 42.07 sec    |
| mrf- $\alpha\beta$ -Swap-TL | 621974336.50 | 28340738.50  | 27066162.00  | 27055552.00  | 27055552.00 | 92.06 sec    |
| ogm-FastPD-LF1              | $\infty$     | $\infty$     | $\infty$     | 27509437.00  | 27509437.00 | 174.04 sec   |
| ogm-FastPD-LF2              | $\infty$     | $\infty$     | $\infty$     | 32939430.00  | 27209149.50 | 7264.73 sec  |
| ogm-ICM                     | 682736616.00 | 682736616.00 | 682736616.00 | 32467486.00  | 32467486.00 | 229.56 sec   |
| ogm-LF-1                    | 682736616.00 | 682736616.00 | 570317775.00 | 32467900.00  | 32467900.00 | 75.71 sec    |
| ogm-LF-2                    | 682736616.00 | 682736616.00 | 682736616.00 | 682736616.00 | 31188870.50 | 3600.18 sec  |
| ogm-TRWS-LF1                | $\infty$     | $\infty$     | $\infty$     | $\infty$     | 26464015.00 | 679.63 sec   |
| ogm-TRWS-LF2                | $\infty$     | $\infty$     | $\infty$     | $\infty$     | 26463829.00 | 2404.03 sec  |
| mrf-LBP-TL                  | 30426622.00  | 26712208.50  | 26604675.00  | 26597003.00  | 26596968.00 | 573.91 sec   |
| mrf-BPS-TL                  | 28879216.00  | 26660810.00  | 26612532.50  | 26612532.50  | 26612532.50 | 593.35 sec   |
| ogm-BPS                     | 682736616.00 | 682736616.00 | 606058330.00 | 50395190.00  | 37224573.50 | 3677.40 sec  |
| ogm-LBP-0.5                 | 682736616.00 | 682736616.00 | 611827775.00 | 52416721.50  | 29746129.50 | 3691.83 sec  |
| ogm-LBP-0.95                | 682736616.00 | 682736616.00 | 605463651.00 | 46984317.50  | 32136405.50 | 3686.06 sec  |
| ogm-TRBP-0.5                | 682736616.00 | 682736616.00 | 611865767.50 | 61666463.50  | 33445889.00 | 3642.96 sec  |
| ogm-TRBP-0.95               | 682736616.00 | 682736616.00 | 605483118.00 | 51207279.00  | 33260789.50 | 3643.81 sec  |
| ogm-TRBPS                   | 682736616.00 | 682736616.00 | 606058058.50 | 56086207.00  | 37903121.00 | 3711.16 sec  |
| mrf-TRWS-TL                 | 26961852.00  | 26479316.00  | 26465442.50  | 26464763.00  | 26464759.00 | 563.15 sec   |
| ogm-ADSAL                   | $\infty$     | $\infty$     | $\infty$     | $\infty$     | 26487768.50 | 3892.14 sec  |
| ogm-BUNDLE-A                | $\infty$     | $\infty$     | $\infty$     | 58328853.50  | 56032960.00 | 3611.21 sec  |
| ogm-BUNDLE-H                | $\infty$     | $\infty$     | $\infty$     | 58328853.50  | 56112406.50 | 3615.07 sec  |
| ogm-SG-A                    | $\infty$     | $\infty$     | $\infty$     | 51398361.50  | 47096469.50 | 3635.02 sec  |
| ogm-BUNDLE-A+               | $\infty$     | $\infty$     | $\infty$     | 58328853.50  | 56032960.00 | 3640.03 sec  |
| ogm-SG-A+                   | $\infty$     | $\infty$     | $\infty$     | 41504500.50  | 37714765.00 | 3643.39 sec  |
| ogm-BUNDLE-A-               | $\infty$     | $\infty$     | $\infty$     | 58328853.50  | 56032960.00 | 3635.32 sec  |
| ogm-SG-A-                   | $\infty$     | $\infty$     | $\infty$     | 58328853.50  | 56028769.00 | 3629.00 sec  |
| ogm-CombiLP                 | $\infty$     | $\infty$     | $\infty$     | $\infty$     | 26467926.00 | 48723.23 sec |

Table 51: mrf-photomontage (2 instances)

| algorithm                    | value          |                |                |                |                | time        |
|------------------------------|----------------|----------------|----------------|----------------|----------------|-------------|
|                              | (1 sec)        | (10 sec)       | (60 sec)       | (600 sec)      | (end)          | (end)       |
| mrf- $\alpha$ -Exp-TAB       | 18886350000.00 | 168457.00      | 168457.00      | 168457.00      | 168457.00      | 7.37 sec    |
| mrf- $\alpha\beta$ -Swap-TAB | 15885856221.50 | 171984.00      | 170858.50      | 170858.50      | 170858.50      | 9.77 sec    |
| ogm-ICM                      | 18886350000.00 | 18096281786.50 | 18096281786.50 | 18096281786.50 | 18096281786.50 | 1.66 sec    |
| ogm-LF-1                     | 18096281786.50 | 18096281786.50 | 18096281786.50 | 18096281786.50 | 18096281786.50 | 0.84 sec    |
| ogm-LF-2                     | 18886350000.00 | 18096106879.00 | 18096106879.00 | 18096106879.00 | 18096106879.00 | 11.15 sec   |
| ogm-TRWS-LF1                 | $\infty$       | $\infty$       | $\infty$       | 1239959.00     | 1239959.00     | 341.22 sec  |
| ogm-TRWS-LF2                 | $\infty$       | $\infty$       | $\infty$       | 735193.00      | 735193.00      | 323.48 sec  |
| mrf-LBP-TAB                  | 1487005.50     | 444353.00      | 438611.00      | 438611.00      | 438611.00      | 458.73 sec  |
| mrf-BPS-TAB                  | 2960515.50     | 2217579.50     | 2217579.50     | 2217579.50     | 2217579.50     | 188.37 sec  |
| ogm-BPS                      | 18099911217.50 | 9241560.00     | 7342037.50     | 1146880.50     | 1146880.50     | 1244.90 sec |
| ogm-LBP-0.5                  | 18099911217.50 | 9285927.50     | 7338664.50     | 460337.50      | 457883.00      | 1056.92 sec |
| ogm-LBP-0.95                 | 18099911217.50 | 9285927.50     | 7005868.00     | 454989.00      | 454989.00      | 717.99 sec  |
| ogm-TRBP-0.5                 | 18886350000.00 | 9285927.50     | 8299253.50     | 470758.00      | 457883.00      | 1795.88 sec |
| ogm-TRBP-0.95                | 18886350000.00 | 9285927.50     | 7376060.50     | 454989.00      | 454989.00      | 1508.58 sec |
| ogm-TRBPS                    | 18886350000.00 | 9241560.00     | 7342037.50     | 1146880.50     | 1146880.50     | 1483.45 sec |
| mrf-TRWS-TAB                 | 3753569.00     | 2766318.50     | 237464.50      | 234150.50      | 234150.50      | 203.79 sec  |
| ogm-ADSAL                    | $\infty$       | $\infty$       | $\infty$       | 227982.50      | 185560.00      | 3605.24 sec |
| ogm-BUNDLE-A                 | $\infty$       | 4834214.00     | 4155468.00     | 3031855.50     | 676079.00      | 1641.14 sec |
| ogm-BUNDLE-H                 | $\infty$       | 4741151.00     | 3939884.00     | 2785483.00     | 599206.00      | 1588.13 sec |
| ogm-SG-A                     | $\infty$       | 5248670.00     | 5248670.00     | 4350224.00     | 3846787.00     | 1616.05 sec |
| ogm-BUNDLE-A+                | $\infty$       | 4834214.00     | 4155468.00     | 3031855.50     | 676079.00      | 1631.83 sec |
| ogm-SG-A+                    | $\infty$       | 5248670.00     | 5248670.00     | 5248670.00     | 5248670.00     | 1624.53 sec |
| ogm-BUNDLE-A-                | $\infty$       | 4834214.00     | 4155468.00     | 3009022.50     | 707036.50      | 1579.42 sec |
| ogm-SG-A-                    | $\infty$       | 5248670.00     | 4223916.50     | 3980452.50     | 3006179.50     | 1499.23 sec |
| ogm-CombiLP                  | $\infty$       | $\infty$       | $\infty$       | $\infty$       | 214853.50      | 3707.70 sec |

Table 52: mrf-stereo (3 instances)

| algorithm                    | value       |             |            |            |            | time        |
|------------------------------|-------------|-------------|------------|------------|------------|-------------|
|                              | (1 sec)     | (10 sec)    | (60 sec)   | (600 sec)  | (end)      | (end)       |
| FastPD                       | $\infty$    | 1614255.00  | 1614255.00 | 1614255.00 | 1614255.00 | 3.11 sec    |
| mrf- $\alpha$ -Exp-TAB       | 10027534.33 | 1613079.00  | 1612676.33 | 1612676.33 | 1612676.33 | 17.14 sec   |
| mrf- $\alpha$ -Exp-TL        | 10027534.33 | 1615762.33  | 1615349.00 | 1615349.00 | 1615349.00 | 11.36 sec   |
| mrf- $\alpha\beta$ -Swap-TAB | 5012532.00  | 1696662.00  | 1662173.33 | 1662173.33 | 1662173.33 | 16.15 sec   |
| mrf- $\alpha\beta$ -Swap-TL  | 4803266.00  | 1929772.33  | 1927265.67 | 1927265.67 | 1927265.67 | 13.12 sec   |
| ogm-FastPD-LF1               | $\infty$    | $\infty$    | 1613252.00 | 1613252.00 | 1613252.00 | 16.84 sec   |
| ogm-FastPD-LF2               | $\infty$    | $\infty$    | 1611510.33 | 1611484.33 | 1611484.33 | 156.87 sec  |
| ogm-ICM                      | 10027534.33 | 9141586.33  | 8359199.00 | 8359199.00 | 8359199.00 | 5.17 sec    |
| ogm-LF-1                     | 9133605.33  | 8355303.67  | 8355303.67 | 8355303.67 | 8355303.67 | 2.07 sec    |
| ogm-LF-2                     | 10027534.33 | 10027534.33 | 8726116.00 | 8726116.00 | 7396373.00 | 286.95 sec  |
| ogm-TRWS-LF1                 | $\infty$    | $\infty$    | $\infty$   | 1587732.67 | 1587732.67 | 226.77 sec  |
| ogm-TRWS-LF2                 | $\infty$    | $\infty$    | $\infty$   | 1587913.33 | 1587043.67 | 365.90 sec  |
| mrf-LBP-TAB                  | 8559473.33  | 3215104.00  | 1656965.67 | 1634055.00 | 1633281.67 | 1892.20 sec |
| mrf-LBP-TL                   | 1839357.67  | 1639193.67  | 1633282.00 | 1633281.67 | 1633281.67 | 242.10 sec  |
| mrf-BPS-TAB                  | 3575274.00  | 1775825.67  | 1738964.00 | 1738695.67 | 1738695.67 | 1473.43 sec |
| mrf-BPS-TL                   | 1968150.67  | 1738695.67  | 1738695.67 | 1738695.67 | 1738695.67 | 224.42 sec  |
| ogm-BPS                      | 10027534.33 | 7338618.00  | 3643065.00 | 2587689.00 | 2546538.00 | 2387.10 sec |
| ogm-LBP-0.5                  | 10027534.33 | 8802924.33  | 3191849.33 | 1728523.33 | 1649939.33 | 2468.07 sec |
| ogm-LBP-0.95                 | 10027534.33 | 8452225.00  | 2673491.00 | 1681252.00 | 1656538.67 | 2369.21 sec |
| ogm-TRBP-0.5                 | 10027534.33 | 9009951.00  | 3809355.33 | 1819825.00 | 1656827.67 | 3019.34 sec |
| ogm-TRBP-0.95                | 10027534.33 | 8870275.67  | 3160349.67 | 1733102.33 | 1655865.00 | 3019.90 sec |
| ogm-TRBPS                    | 10027534.33 | 8889829.33  | 4154319.00 | 2607587.33 | 2547961.67 | 2959.70 sec |
| mrf-TRWS-TAB                 | 3167309.33  | 1634740.00  | 1607148.33 | 1589762.00 | 1587269.67 | 1518.01 sec |
| mrf-TRWS-TL                  | 1656885.67  | 1601896.67  | 1589708.00 | 1587269.67 | 1587269.67 | 216.41 sec  |
| ogm-ADSAL                    | $\infty$    | $\infty$    | $\infty$   | 1606706.67 | 1589318.00 | 3163.13 sec |
| ogm-BUNDLE-A                 | $\infty$    | $\infty$    | 2370221.33 | 1929417.67 | 1649017.67 | 2112.27 sec |
| ogm-BUNDLE-H                 | $\infty$    | $\infty$    | 2377809.67 | 1901431.00 | 1645250.33 | 2152.15 sec |
| ogm-SG-A                     | $\infty$    | $\infty$    | 2379223.33 | 1917664.33 | 1789576.67 | 2098.93 sec |
| ogm-BUNDLE-A+                | $\infty$    | $\infty$    | 2370221.33 | 1929417.67 | 1648854.67 | 2003.83 sec |
| ogm-SG-A+                    | $\infty$    | $\infty$    | 3246798.33 | 2716228.33 | 2500828.67 | 2110.78 sec |
| ogm-BUNDLE-A-                | $\infty$    | $\infty$    | 2370221.33 | 1930528.33 | 1648723.33 | 2150.49 sec |
| ogm-SG-A-                    | $\infty$    | $\infty$    | 2587393.67 | 1983317.33 | 1762200.33 | 2100.09 sec |
| ogm-CombiLP                  | $\infty$    | $\infty$    | $\infty$   | $\infty$   | 1587560.67 | 835.92 sec  |

Table 53: object-seg (5 instances)

| algorithm                   | value    |          |          |           |          | time       |
|-----------------------------|----------|----------|----------|-----------|----------|------------|
|                             | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)    | (end)      |
| $\alpha$ -Exp-pct           | 31317.60 | 31317.60 | 31317.60 | 31317.60  | 31317.60 | 0.33 sec   |
| FastPD                      | 31317.60 | 31317.60 | 31317.60 | 31317.60  | 31317.60 | 0.11 sec   |
| FastPD-pct                  | 31317.60 | 31317.60 | 31317.60 | 31317.60  | 31317.60 | 0.29 sec   |
| mrf- $\alpha$ -Exp-TL       | 31317.60 | 31317.60 | 31317.60 | 31317.60  | 31317.60 | 0.40 sec   |
| mrf- $\alpha\beta$ -Swap-TL | 31323.23 | 31323.23 | 31323.23 | 31323.23  | 31323.23 | 0.23 sec   |
| ogm-FastPD-LF1              | 31317.60 | 31317.60 | 31317.60 | 31317.60  | 31317.60 | 0.39 sec   |
| ogm-FastPD-LF2              | 31317.60 | 31317.60 | 31317.60 | 31317.60  | 31317.60 | 1.73 sec   |
| ogm-ICM                     | 65180.15 | 65180.15 | 65180.15 | 65180.15  | 65180.15 | 0.22 sec   |
| ogm-LF-1                    | 65180.15 | 65180.15 | 65180.15 | 65180.15  | 65180.15 | 0.11 sec   |
| ogm-LF-2                    | 65054.89 | 64937.24 | 64937.24 | 64937.24  | 64937.24 | 1.48 sec   |
| ogm-TRWS-LF1                | $\infty$ | 31317.23 | 31317.23 | 31317.23  | 31317.23 | 3.27 sec   |
| ogm-TRWS-LF2                | $\infty$ | 31317.23 | 31317.23 | 31317.23  | 31317.23 | 3.29 sec   |
| mrf-LBP-TL                  | 32430.93 | 32400.01 | 32400.01 | 32400.01  | 32400.01 | 29.43 sec  |
| mrf-BPS-TL                  | 35775.27 | 35775.27 | 35775.27 | 35775.27  | 35775.27 | 11.19 sec  |
| ogm-BPS                     | 36372.00 | 33547.88 | 33418.04 | 33418.04  | 33418.04 | 137.34 sec |
| ogm-LBP-0.5                 | 37914.09 | 33486.16 | 32705.04 | 32663.86  | 32663.86 | 110.84 sec |
| ogm-LBP-0.95                | 36835.67 | 32901.96 | 32673.75 | 32673.75  | 32673.75 | 61.38 sec  |
| ogm-TRBP-0.5                | 38125.97 | 33823.16 | 32727.55 | 32663.86  | 32663.86 | 227.29 sec |
| ogm-TRBP-0.95               | 37230.17 | 33031.33 | 32676.56 | 32668.92  | 32668.92 | 222.82 sec |
| ogm-TRBPS                   | 36945.22 | 33641.96 | 33383.95 | 33383.95  | 33383.95 | 180.23 sec |
| MCR-TC-MTC                  | 38506.33 | 35236.46 | 32633.57 | 32043.04  | 31375.53 | 421.26 sec |
| MCR-pct                     | 47817.10 | 33956.55 | 33956.55 | 31674.41  | 31674.41 | 62.36 sec  |
| mrf-TRWS-TL                 | 31349.87 | 31317.23 | 31317.23 | 31317.23  | 31317.23 | 2.21 sec   |
| ogm-ADSAL                   | $\infty$ | 31751.11 | 31336.57 | 31317.23  | 31317.23 | 99.50 sec  |
| ogm-BUNDLE-A                | 40445.37 | 39555.82 | 33282.30 | 31317.31  | 31317.31 | 128.82 sec |
| ogm-BUNDLE-H                | 40226.39 | 36897.80 | 31656.08 | 31317.23  | 31317.23 | 193.92 sec |
| ogm-SG-A                    | 40512.69 | 38586.74 | 32026.22 | 31432.95  | 31432.95 | 185.11 sec |
| ogm-BUNDLE-A+               | 40445.37 | 39854.50 | 34493.35 | 31354.83  | 31354.83 | 196.92 sec |
| ogm-SG-A+                   | 40512.69 | 40512.69 | 40177.39 | 39295.40  | 39295.40 | 187.17 sec |
| ogm-BUNDLE-A-               | 40445.37 | 36816.37 | 31428.68 | 31321.98  | 31321.98 | 197.66 sec |
| ogm-SG-A-                   | 40250.58 | 36756.37 | 33018.94 | 32137.62  | 32137.62 | 180.46 sec |
| TRWS-pct                    | 33877.41 | 31317.23 | 31317.23 | 31317.23  | 31317.23 | 0.96 sec   |
| MCI-TC-MTC-TCI              | 38506.33 | 35305.03 | 32633.57 | 32000.46  | 31317.23 | 428.32 sec |
| MCI-pct                     | 47817.10 | 33877.41 | 33877.41 | 31317.23  | 31317.23 | 69.84 sec  |
| ogm-CombiLP                 | $\infty$ | $\infty$ | 31317.23 | 31317.23  | 31317.23 | 32.61 sec  |



Table 54: protein-folding (21 instances)

| algorithm     | value    |          |          |           |            | time           |
|---------------|----------|----------|----------|-----------|------------|----------------|
|               | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)      | (end)          |
| ogm-ICM       | -5446.55 | -5446.55 | -5446.55 | -5446.55  | -5446.55   | 0.09 sec       |
| ogm-LBP-LF1   | $\infty$ | $\infty$ | $\infty$ | -5892.69  | -5892.69   | 102.71 sec     |
| ogm-LBP-LF2   | $\infty$ | $\infty$ | $\infty$ | -5923.01  | -5923.01   | 130.17 sec     |
| ogm-LF-1      | -5447.49 | -5447.49 | -5447.49 | -5447.49  | -5447.49   | 0.05 sec       |
| ogm-LF-2      | -4165.04 | -5604.62 | -5733.35 | -5747.56  | -5747.56   | 54.54 sec      |
| ogm-LF-3      | -4165.00 | -4211.34 | -4347.96 | -4358.51  | -5780.11   | 22422.07 sec   |
| ogm-TRWS-LF1  | $\infty$ | $\infty$ | $\infty$ | -5856.47  | -5856.47   | 22.68 sec      |
| ogm-TRWS-LF2  | $\infty$ | $\infty$ | -5785.19 | -5897.06  | -5897.06   | 54.88 sec      |
| BPS-TAB       | -5957.94 | -5958.72 | -5958.72 | -5958.72  | -5958.72   | 24.48 sec      |
| ogm-BPS       | -4957.71 | -5824.23 | -5843.39 | -5872.91  | -5872.91   | 119.08 sec     |
| ogm-LBP-0.5   | -5108.73 | -5923.18 | -5941.37 | -5941.81  | -5941.81   | 106.99 sec     |
| ogm-LBP-0.95  | -5184.43 | -5939.74 | -5946.62 | -5949.16  | -5949.16   | 102.34 sec     |
| ogm-TRBP-0.5  | -5119.55 | -5866.04 | -5948.86 | -5949.53  | -5949.53   | 145.44 sec     |
| ogm-TRBP-0.95 | -5194.44 | -5908.87 | -5931.57 | -5941.09  | -5941.09   | 140.09 sec     |
| ogm-TRBPS     | -5064.32 | -5759.06 | -5834.45 | -5881.71  | -5881.71   | 135.42 sec     |
| ADDD          | 5512.48  | 929.27   | -1403.59 | -3859.53  | -4189.61   | 209.19 sec     |
| MPLP          | -1007.21 | -1165.88 | -1315.73 | -2593.92  | -5626.81   | 510.90 sec     |
| MPLP-C        | -3079.78 | -4471.06 | -5634.82 | -5836.19  | -5933.70   | 1639.52 sec    |
| ogm-ADSAL     | $\infty$ | $\infty$ | $\infty$ | $\infty$  | -5881.47   | 1014.89 sec    |
| ogm-BUNDLE-A  | $\infty$ | -3277.03 | -4344.59 | -5385.67  | -5480.14   | 487.56 sec     |
| ogm-BUNDLE-H  | $\infty$ | -2720.04 | -3795.50 | -5389.67  | -5486.47   | 479.92 sec     |
| ogm-SG-A      | $\infty$ | -3398.87 | -4494.51 | -5150.18  | -5210.59   | 474.52 sec     |
| ogm-BUNDLE-A+ | $\infty$ | -3294.66 | -4352.23 | -5331.72  | -5448.23   | 483.77 sec     |
| ogm-SG-A+     | $\infty$ | -3856.25 | -4702.67 | -5325.17  | -5479.15   | 463.18 sec     |
| ogm-BUNDLE-A- | $\infty$ | -2887.23 | -4142.93 | -5053.26  | -5146.23   | 477.86 sec     |
| ogm-SG-A-     | $\infty$ | -2584.64 | -3324.25 | -4708.51  | -4806.12   | 473.93 sec     |
| TRWS-TAB      | -5704.63 | -5797.52 | -5846.86 | -5846.86  | -5846.86   | 22.18 sec      |
| BRAOBB-1      | $\infty$ | $\infty$ | $\infty$ | $\infty$  | <i>NaN</i> | <i>NaN</i> sec |
| BRAOBB-2      | $\infty$ | $\infty$ | $\infty$ | $\infty$  | <i>NaN</i> | <i>NaN</i> sec |
| BRAOBB-3      | $\infty$ | $\infty$ | $\infty$ | $\infty$  | <i>NaN</i> | <i>NaN</i> sec |
| ogm-CombiLP   | $\infty$ | $\infty$ | -5807.58 | -5896.86  | -5955.77   | 700.10 sec     |

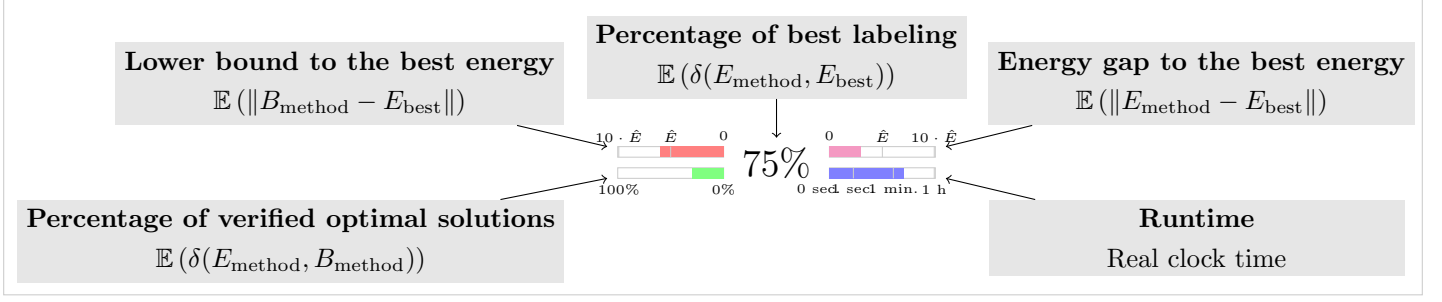
Table 55: protein-prediction (8 instances)

| algorithm     | value     |           |           |           |           | time        |
|---------------|-----------|-----------|-----------|-----------|-----------|-------------|
|               | (1 sec)   | (10 sec)  | (60 sec)  | (600 sec) | (end)     | (end)       |
| ogm-ICM       | 60414.84  | 60414.84  | 60414.84  | 60414.84  | 60414.84  | 0.03 sec    |
| ogm-LBP-LF2   | $\infty$  | $\infty$  | 52942.95  | 52942.95  | 52942.95  | 25.03 sec   |
| ogm-LF-1      | 60427.60  | 60427.60  | 60427.60  | 60427.60  | 60427.60  | 0.03 sec    |
| ogm-LF-2      | 58682.74  | 58682.74  | 58682.74  | 58682.74  | 58682.74  | 0.70 sec    |
| ogm-LF-3      | 127198.56 | 127173.36 | 57944.06  | 57944.06  | 57944.06  | 19.08 sec   |
| ogm-BPS       | 85804.13  | 74727.16  | 74673.63  | 74673.63  | 74673.63  | 27.64 sec   |
| ogm-LBP-0.5   | 54819.12  | 52985.35  | 52974.98  | 52974.98  | 52974.98  | 24.79 sec   |
| ogm-LBP-0.95  | 90346.43  | 71152.50  | 71137.01  | 71137.01  | 71137.01  | 33.31 sec   |
| ogm-TRBP-0.5  | 63572.76  | 60781.84  | 60268.84  | 60268.84  | 60268.84  | 35.60 sec   |
| ogm-TRBP-0.95 | 80412.07  | 68818.53  | 68284.15  | 68284.15  | 68284.15  | 36.11 sec   |
| ogm-TRBPS     | 81367.55  | 71136.97  | 71105.65  | 71105.65  | 71105.65  | 28.38 sec   |
| ADDD          | 128274.77 | 126562.66 | 106216.86 | 106216.86 | 106216.86 | 10.70 sec   |
| MPLP          | 97985.80  | 96835.43  | 96806.53  | 96806.53  | 96806.53  | 69.09 sec   |
| ogm-BUNDLE-A  | $\infty$  | 82473.51  | 82431.53  | 81283.83  | 81035.49  | 1287.27 sec |
| ogm-BUNDLE-H  | $\infty$  | 82413.75  | 81927.45  | 81118.39  | 81039.93  | 1301.94 sec |
| ogm-SG-A      | $\infty$  | 82451.55  | 81976.53  | 81636.80  | 81587.06  | 1273.32 sec |
| ogm-BUNDLE-A+ | $\infty$  | 82473.51  | 82469.04  | 82217.65  | 81993.85  | 1275.45 sec |
| ogm-SG-A+     | $\infty$  | 82606.85  | 82606.85  | 82606.85  | 82606.85  | 1256.94 sec |
| ogm-BUNDLE-A- | $\infty$  | 82473.51  | 81828.24  | 81107.41  | 81058.43  | 1300.65 sec |
| ogm-SG-A-     | $\infty$  | 82567.18  | 82391.94  | 81646.26  | 81443.05  | 1312.12 sec |
| ogm-LP-LP     | 128274.77 | 128274.77 | 128274.77 | 102829.40 | 102829.40 | 169.61 sec  |
| ogm-ILP       | 128274.77 | 128274.77 | 126560.76 | 126560.76 | 57477.07  | 2263.46 sec |

Table 56: scene-decomposition (715 instances)

| algorithm        | value    |          |          |           |         | time      |
|------------------|----------|----------|----------|-----------|---------|-----------|
|                  | (1 sec)  | (10 sec) | (60 sec) | (600 sec) | (end)   | (end)     |
| $\alpha$ -Fusion | -866.85  | -866.85  | -866.85  | -866.85   | -866.85 | 0.00 sec  |
| ogm-ICM          | -864.56  | -864.56  | -864.56  | -864.56   | -864.56 | 0.00 sec  |
| ogm-LBP-LF1      | -866.76  | -866.76  | -866.76  | -866.76   | -866.76 | 0.05 sec  |
| ogm-LBP-LF2      | -866.76  | -866.76  | -866.76  | -866.76   | -866.76 | 0.06 sec  |
| ogm-LF-1         | -864.56  | -864.56  | -864.56  | -864.56   | -864.56 | 0.00 sec  |
| ogm-LF-2         | -865.81  | -865.81  | -865.81  | -865.81   | -865.81 | 0.02 sec  |
| ogm-LF-3         | -866.27  | -866.27  | -866.27  | -866.27   | -866.27 | 0.45 sec  |
| ogm-TRWS-LF1     | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.00 sec  |
| ogm-TRWS-LF2     | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.00 sec  |
| BPS-TAB          | -866.73  | -866.73  | -866.73  | -866.73   | -866.73 | 0.10 sec  |
| ogm-BPS          | -866.77  | -866.77  | -866.77  | -866.77   | -866.77 | 0.02 sec  |
| ogm-LBP-0.5      | -866.76  | -866.76  | -866.76  | -866.76   | -866.76 | 0.05 sec  |
| ogm-LBP-0.95     | -866.76  | -866.76  | -866.76  | -866.76   | -866.76 | 0.02 sec  |
| ogm-TRBP-0.5     | -866.85  | -866.85  | -866.85  | -866.85   | -866.85 | 0.24 sec  |
| ogm-TRBP-0.95    | -866.85  | -866.85  | -866.85  | -866.85   | -866.85 | 0.11 sec  |
| ogm-TRBPS        | -866.85  | -866.85  | -866.85  | -866.85   | -866.85 | 0.13 sec  |
| ADDD             | -866.92  | -866.92  | -866.92  | -866.92   | -866.92 | 0.06 sec  |
| MPLP             | -866.92  | -866.92  | -866.92  | -866.92   | -866.92 | 0.04 sec  |
| MPLP-C           | -866.92  | -866.92  | -866.92  | -866.92   | -866.92 | 0.04 sec  |
| ogm-ADSAL        | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.04 sec  |
| ogm-BUNDLE-A     | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.94 sec  |
| ogm-BUNDLE-H     | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.26 sec  |
| ogm-SG-A         | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.96 sec  |
| ogm-BUNDLE-A+    | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.07 sec  |
| ogm-SG-A+        | -866.92  | -866.92  | -866.92  | -866.92   | -866.92 | 0.07 sec  |
| ogm-BUNDLE-A-    | -866.92  | -866.92  | -866.92  | -866.92   | -866.92 | 1.19 sec  |
| ogm-SG-A-        | -866.83  | -866.85  | -866.85  | -866.85   | -866.85 | 1.08 sec  |
| ogm-LP-LP        | -866.92  | -866.92  | -866.92  | -866.92   | -866.92 | 0.23 sec  |
| TRWS-TAB         | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.00 sec  |
| BRAOBB-1         | $\infty$ | $\infty$ | $\infty$ | $\infty$  | -866.90 | 17.61 sec |
| BRAOBB-2         | $\infty$ | $\infty$ | $\infty$ | $\infty$  | -866.86 | 23.40 sec |
| BRAOBB-3         | $\infty$ | $\infty$ | $\infty$ | -866.90   | -866.90 | 27.20 sec |
| ADDD-BB          | -848.17  | -865.23  | -866.93  | -866.93   | -866.93 | 0.11 sec  |
| ogm-CombiLP      | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.02 sec  |
| ogm-ILP          | -866.93  | -866.93  | -866.93  | -866.93   | -866.93 | 0.17 sec  |

### 3 Boxplots



|                 | mrf-stereo |     | mrf-photomontage |      | mrf-inpainting |     |
|-----------------|------------|-----|------------------|------|----------------|-----|
| ogm-ICM         |            | 0%  |                  | 0%   |                | 0%  |
| ogm-LF-1        |            | 0%  |                  | 0%   |                | 0%  |
| ogm-LF-2        |            | 0%  |                  | 0%   |                | 0%  |
| FastPD          |            | 0%  |                  | 0%   |                | 0%  |
| mrf-a-Exp-TL    |            | 0%  |                  | 0%   |                | 0%  |
| mrf-a-Exp-TAB   |            | 33% |                  | 100% |                | 0%  |
| mrf-ab-Swap-TL  |            | 0%  |                  | 0%   |                | 0%  |
| mrf-ab-Swap-TAB |            | 0%  |                  | 0%   |                | 0%  |
| ogm-FastPD-LF1  |            | 0%  |                  | 0%   |                | 0%  |
| ogm-FastPD-LF2  |            | 0%  |                  | 0%   |                | 0%  |
| ogm-TRWS-LF1    |            | 0%  |                  | 0%   |                | 0%  |
| ogm-TRWS-LF2    |            | 0%  |                  | 0%   |                | 50% |
| mrf-BPS-TL      |            | 0%  |                  | 0%   |                | 0%  |
| mrf-BPS-TAB     |            | 0%  |                  | 0%   |                | 0%  |
| mrf-LBP-TL      |            | 0%  |                  | 0%   |                | 0%  |
| mrf-LBP-TAB     |            | 0%  |                  | 0%   |                | 0%  |
| ogm-LBP-0.5     |            | 0%  |                  | 0%   |                | 0%  |
| ogm-LBP-0.95    |            | 0%  |                  | 0%   |                | 0%  |
| ogm-BPS         |            | 0%  |                  | 0%   |                | 0%  |
| ogm-TRBP-0.5    |            | 0%  |                  | 0%   |                | 0%  |
| ogm-TRBP-0.95   |            | 0%  |                  | 0%   |                | 0%  |
| ogm-TRBPS       |            | 0%  |                  | 0%   |                | 0%  |
| mrf-TRWS-TL     |            | 0%  |                  | 0%   |                | 0%  |
| mrf-TRWS-TAB    |            | 0%  |                  | 0%   |                | 0%  |
| ogm-BUNDLE-H    |            | 33% |                  | 0%   |                | 0%  |
| ogm-BUNDLE-A-   |            | 0%  |                  | 0%   |                | 0%  |
| ogm-BUNDLE-A    |            | 33% |                  | 0%   |                | 0%  |
| ogm-BUNDLE-A+   |            | 33% |                  | 0%   |                | 0%  |
| ogm-SG-A-       |            | 0%  |                  | 0%   |                | 0%  |
| ogm-SG-A        |            | 33% |                  | 0%   |                | 0%  |
| ogm-SG-A+       |            | 0%  |                  | 0%   |                | 0%  |
| ogm-ADSAL       |            | 33% |                  | 0%   |                | 0%  |
| ogm-CombiLP     |            | 66% |                  | 0%   |                | 50% |

Figure 2: **Results for grid structured models with four-pixel neighborhood-system and truncated convex regularization.** See Fig. 1 for the legend. For all models local polytope relaxations give good lower bounds. However, extracting a integer solution from the fractional one can be very difficult, especially for the photomontage instances, where soft constraints renders linear programming relaxation harder. FastPD and  $\alpha$ -expansion are the first choice if fast optimization is required. For stereo- and inpainting problems LP-relaxations as obtained by TRWS have given best results – especially when followed by lazy flipping post processing. For some instances we were able to solve them to optimality by CombiLP but often need more than 1 hour.

|                | color-seg-n4 |      |  | object-seg |      |  | inpainting-n4 |      |  |
|----------------|--------------|------|--|------------|------|--|---------------|------|--|
| ogm-ICM        |              | 0%   |  |            | 0%   |  |               | 0%   |  |
| ogm-LF-1       |              | 0%   |  |            | 0%   |  |               | 0%   |  |
| ogm-LF-2       |              | 0%   |  |            | 0%   |  |               | 0%   |  |
| FastPD         |              | 0%   |  |            | 80%  |  |               | 50%  |  |
| FastPD-pct     |              | 0%   |  |            | 80%  |  |               | 50%  |  |
| mrf-a-Exp-TL   |              | 0%   |  |            | 80%  |  |               | 50%  |  |
| a-Exp-pct      |              | 0%   |  |            | 80%  |  |               | 50%  |  |
| mrf-ab-Swap-TL |              | 0%   |  |            | 39%  |  |               | 100% |  |
| ogm-FastPD-LF1 |              | 0%   |  |            | 80%  |  |               | 50%  |  |
| ogm-FastPD-LF2 |              | 0%   |  |            | 80%  |  |               | 50%  |  |
| ogm-TRWS-LF1   |              | 77%  |  |            | 100% |  |               | 50%  |  |
| ogm-TRWS-LF2   |              | 77%  |  |            | 100% |  |               | 50%  |  |
| mrf-BPS-TL     |              | 0%   |  |            | 0%   |  |               | 100% |  |
| mrf-LBP-TL     |              | 0%   |  |            | 0%   |  |               | 50%  |  |
| ogm-LBP-0.5    |              | 0%   |  |            | 0%   |  |               | 50%  |  |
| ogm-LBP-0.95   |              | 0%   |  |            | 0%   |  |               | 50%  |  |
| ogm-BPS        |              | 0%   |  |            | 0%   |  |               | 50%  |  |
| ogm-TRBP-0.5   |              | 0%   |  |            | 0%   |  |               | 50%  |  |
| ogm-TRBP-0.95  |              | 0%   |  |            | 0%   |  |               | 50%  |  |
| ogm-TRBPS      |              | 0%   |  |            | 0%   |  |               | 50%  |  |
| mrf-TRWS-TL    |              | 88%  |  |            | 100% |  |               | 50%  |  |
| TRWS-pct       |              | 77%  |  |            | 100% |  |               | 50%  |  |
| ogm-BUNDLE-H   |              | 66%  |  |            | 100% |  |               | 50%  |  |
| ogm-BUNDLE-A-  |              | 77%  |  |            | 19%  |  |               | 50%  |  |
| ogm-BUNDLE-A   |              | 77%  |  |            | 80%  |  |               | 50%  |  |
| ogm-BUNDLE-A+  |              | 33%  |  |            | 19%  |  |               | 50%  |  |
| ogm-SG-A-      |              | 0%   |  |            | 0%   |  |               | 100% |  |
| ogm-SG-A       |              | 66%  |  |            | 39%  |  |               | 50%  |  |
| ogm-SG-A+      |              | 11%  |  |            | 0%   |  |               | 50%  |  |
| ogm-ADSAL      |              | 88%  |  |            | 100% |  |               | 50%  |  |
| MCR-TC-MTC     |              | 77%  |  |            | 80%  |  |               | 50%  |  |
| MCR-pct        |              | 33%  |  |            | 39%  |  |               | 0%   |  |
| MCI-TC-MTC-TCI |              | 88%  |  |            | 100% |  |               | 50%  |  |
| MCI-pct        |              | 88%  |  |            | 100% |  |               | 50%  |  |
| ogm-CombiLP    |              | 100% |  |            | 100% |  |               | 50%  |  |

Figure 3: **Results for grid structured models with four-pixel neighborhood-system and Potts regularization.** See Fig. 1 for the legend. For Potts models relaxations over the local polytope are often very tight and multicut relaxations works quite good and efficient. FastPD is the first choice if fast optimization is required. When ever persistence is available to reduce the problem size it helps a lot. In the color-seg-n4 dataset the instance pfau is the hardest one and rises the average runtime. Also the lower bound on the pfau instances produced by MCI within one hour are very weak. The fastest exact solver is here CombiLP. In the inpainting-n4 dataset the inverse instance is designed to be hard and make LP-solvers struggling.

|                | color-seg-n8                                                                        |      |                                                                                     | color-seg                                                                           |      |                                                                                     | inpainting-n8                                                                       |      |                                                                                       |
|----------------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------|
| ogm-ICM        |    | 0%   |    |    | 0%   |    |    | 0%   |    |
| ogm-LF-1       |    | 0%   |    |    | 0%   |    |    | 0%   |    |
| ogm-LF-2       |    | 0%   |    |    | 0%   |    |    | 0%   |    |
| FastPD         |    | 0%   |    |    | 66%  |    |    | 50%  |    |
| FastPD-pct     |    | 0%   |    |    | 66%  |    |    | 100% |    |
| a-Exp-VIEW     |    | 0%   |    |    | 66%  |    |    | 50%  |    |
| a-Exp-pct      |    | 0%   |    |    | 100% |    |    | 100% |    |
| ab-Swap-VIEW   |    | 0%   |    |    | 66%  |    |    | 50%  |    |
| ogm-FastPD-LF1 |    | 0%   |    |    | 66%  |    |    | 50%  |    |
| ogm-FastPD-LF2 |    | 0%   |    |    | 66%  |    |    | 50%  |    |
| ogm-TRWS-LF1   |    | 33%  |    |    | 66%  |    |    | 50%  |    |
| ogm-TRWS-LF2   |    | 44%  |    |    | 66%  |    |    | 50%  |    |
| BPS-TL         |    | 0%   |    |    | 0%   |    |    | 50%  |    |
| ogm-LBP-0.5    |    | 0%   |    |    | 0%   |    |    | 50%  |    |
| ogm-LBP-0.95   |    | 0%   |    |    | 0%   |    |    | 50%  |    |
| ogm-BPS        |    | 0%   |    |    | 0%   |    |    | 50%  |    |
| ogm-TRBP-0.5   |  | 0%   |  |  | 0%   |  |  | 50%  |  |
| ogm-TRBP-0.95  |  | 0%   |  |  | 0%   |  |  | 50%  |  |
| ogm-TRBPS      |  | 0%   |  |  | 0%   |  |  | 50%  |  |
| TRWS-TL        |  | 22%  |  |  | 66%  |  |  | 50%  |  |
| TRWS-pct       |  | 11%  |  |  | 66%  |  |  | 50%  |  |
| ogm-BUNDLE-H   |  | 33%  |  |  | 33%  |  |  | 0%   |  |
| ogm-BUNDLE-A-  |  | 11%  |  |  | 0%   |  |  | 50%  |  |
| ogm-BUNDLE-A   |  | 44%  |  |  | 33%  |  |  | 50%  |  |
| ogm-BUNDLE-A+  |  | 11%  |  |  | 33%  |  |  | 50%  |  |
| ogm-SG-A-      |  | 0%   |  |  | 0%   |  |  | 0%   |  |
| ogm-SG-A       |  | 11%  |  |  | 0%   |  |  | 0%   |  |
| ogm-SG-A+      |  | 0%   |  |  | 0%   |  |  | 0%   |  |
| ogm-ADSAL      |  | 55%  |  |  | 66%  |  |  | 50%  |  |
| MCR-TC-MTC     |  | 22%  |  |  | 100% |  |  | 0%   |  |
| MCR-pct        |  | 33%  |  |  | 100% |  |  | 0%   |  |
| ogm-CombiLP    |  | 100% |  |  | 100% |  |  | 50%  |  |
| MCI-TC-MTC-TCI |  | 88%  |  |  | 100% |  |  | 50%  |  |
| MCI-TC-TCI     |                                                                                     |      |                                                                                     |                                                                                     |      |                                                                                     |                                                                                     |      |                                                                                       |
| MCI-pct        |  | 77%  |  |  | 100% |  |  | 50%  |  |

Figure 4: **Results for grid structured models with 8-pixel neighborhood-system and Potts regularization.** See Fig. 1 for the legend. For Potts models relaxations over the local polytope are often very tight and multicut relaxations works quite good and efficient. Compare to the same models with a 4-pixel neighborhood-system, cf. Fig. 3, the local polytope relaxations becomes weaker. FastPD is the first choice if fast optimization is required. When ever persistence is available to reduce the problem size it helps a lot. In the color-seg-n8 dataset the instance pfau is the hardest one and rises the average runtime. Also the lower bound on the pfau instances produced by MCA within 6 hours are very weak. In the inpainting-n8 dataset the inverse instance is designed to be hard and make LP-solvers struggling.



Figure 5: **Results for grid structured models with 3D 6-voxel neighborhood-system and Potts regularization.** See Fig. 1 for the legend. The slice thick is given in millimeters. Thinner slices give larger models. FastPD is the fastest and TRWS the best approximative method. Local polytope relaxations give very good bounds for this models. Exact results are feasible by MCA-pct with runtimes comparable to approximative methods. Without preprocessing (pct) MCA is much slower and requires more than 10 GB of memory for the smallest instance.

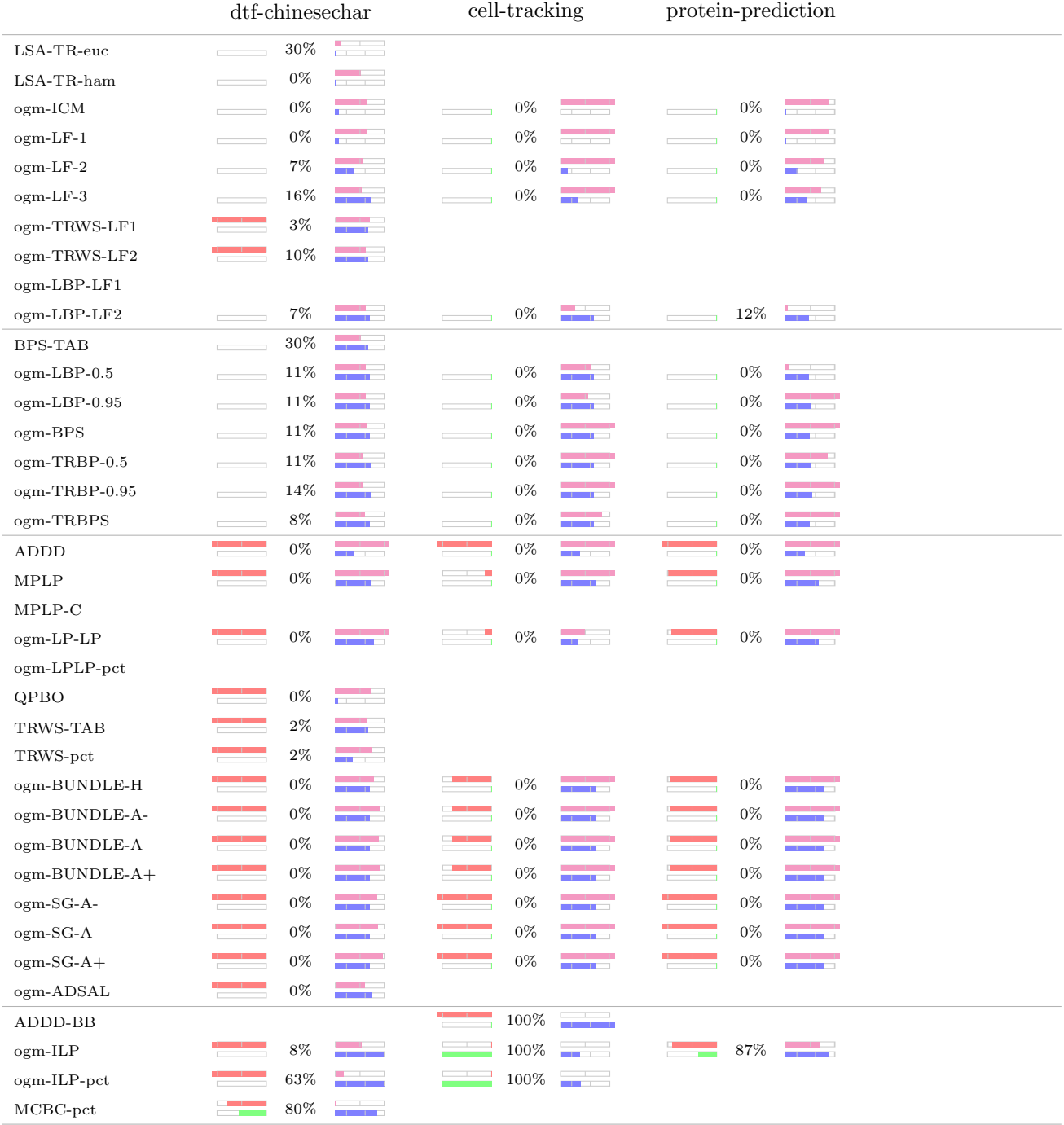


Figure 6: **Results for binary models.** See Fig. 1 for the legend. The Chinese character instances include Potts terms with negative coupling strength, that means the models are non-submodular. That why only a small subset of solvers is applicable. The large degree of the structure makes local polytope relaxations weak and the remaining rounding problem hard. BPS give good results, but we obtain best results by first using persistence to reduce the problem size and then tightening the relaxation. The cell-tracking and protein-prediction dataset include higher order terms. This render them more challenging and many standard methods are no longer applicable. For cell-tracking ogm-ILP performs best, many other methods violate soft-constraints which causes high objective values. For the protein-protein dataset ogm-ILP performs best except of 1 instance. This seemed to be caused by numerical problems. For all protein-protein instances LBP, optionally followed by lazy flipping give very good results in reasonable time.



|                | inclusion |      | geo-surf-3 |      | geo-surf-7 |      |
|----------------|-----------|------|------------|------|------------|------|
| A-Fusion       |           | 0%   |            | 97%  |            | 85%  |
| ogm-ICM        |           | 0%   |            | 44%  |            | 2%   |
| ogm-LF-1       |           | 0%   |            | 44%  |            | 2%   |
| ogm-LF-2       |           | 0%   |            | 69%  |            | 4%   |
| ogm-LF-3       |           | 0%   |            | 87%  |            | 8%   |
| ogm-LBP-LF1    |           | 39%  |            | 90%  |            | 22%  |
| ogm-LBP-LF2    |           | 69%  |            | 91%  |            | 22%  |
| ogm-LBP-0.5    |           | 69%  |            | 90%  |            | 22%  |
| ogm-LBP-0.95   |           | 80%  |            | 90%  |            | 22%  |
| ogm-BPS        |           | 60%  |            | 90%  |            | 22%  |
| ogm-TRBP-0.5   |           | 50%  |            | 94%  |            | 42%  |
| ogm-TRBP-0.95  |           | 80%  |            | 94%  |            | 39%  |
| ogm-TRBPS      |           | 69%  |            | 95%  |            | 38%  |
| ADDD           |           | 10%  |            | 97%  |            | 98%  |
| MPLP           |           | 19%  |            | 97%  |            | 92%  |
| MPLP-C         |           | 19%  |            | 98%  |            | 94%  |
| ogm-LP-LP      |           | 10%  |            | 100% |            | 99%  |
| ogm-BUNDLE-H   |           | 30%  |            | 100% |            | 99%  |
| ogm-BUNDLE-A-  |           | 30%  |            | 98%  |            | 37%  |
| ogm-BUNDLE-A   |           | 19%  |            | 100% |            | 86%  |
| ogm-BUNDLE-A+  |           | 0%   |            | 100% |            | 99%  |
| ogm-SG-A-      |           | 19%  |            | 89%  |            | 17%  |
| ogm-SG-A       |           | 0%   |            | 100% |            | 54%  |
| ogm-SG-A+      |           | 0%   |            | 100% |            | 97%  |
| MCI-TC-MTC-TCI |           | 100% |            | 100% |            | 100% |
| ogm-ILP        |           | 100% |            | 100% |            | 100% |
| BRAOBB-1       |           | 0%   |            | 99%  |            | 71%  |
| BRAOBB-2       |           | 0%   |            | 99%  |            | 88%  |
| BRAOBB-3       |           | 0%   |            | 100% |            | 89%  |

Figure 7: **Results for higher order models.** See Fig. 1 for the legend. For the inclusion instances ogm-ILP give best results, similar to those of LBP. LP relaxations of the local polytope are relatively tight but rounding is not trivial and often violates the inclusion soft-constraints. Adding additional cycle constraints does improve the results only marginally. The geo-surf instances are based on superpixels and therefor much smaller. Fastest and optimal results are produced by ogm-ILP and AD3-BB. Later is non-commercial available under the LGPL.

|                 | correlation-clustering |      | image-seg |      | image-seg-3rdorder |      | modularity-clustering |     |
|-----------------|------------------------|------|-----------|------|--------------------|------|-----------------------|-----|
| ogm-KL          |                        | 0%   |           | 0%   |                    | 0%   |                       | 50% |
| ogm-ICM         |                        | 0%   |           | 0%   |                    | 0%   |                       | 0%  |
| ogm-LF-1        |                        | 0%   |           | 0%   |                    | 0%   |                       | 0%  |
| MCR-CC          |                        | 23%  |           | 35%  |                    | 0%   |                       | 33% |
| MCR-CCFDB       |                        | 22%  |           | 35%  |                    | 0%   |                       | 16% |
| MCR-CCFDB-OWC   |                        | 23%  |           | 35%  |                    | 0%   |                       | 83% |
| MCI-CCI         |                        | 100% |           | 100% |                    | 99%  |                       | 66% |
| MCI-CCIFD       |                        | 100% |           | 100% |                    | 99%  |                       | 66% |
| MCI-CCFDB-CCIFD |                        | 100% |           | 100% |                    | 100% |                       | 83% |

Figure 8: **Results for unsupervised partition models.** See Fig. 1 for the legend. Over all MCI-CCFDB-CCIFD performs best. The KL-method used outside computer vision does not perform well on sparse computer vision problems and cannot be used for higher order models. Linear programming relaxations give worse results and are not necessary faster than ILP-based methods. The reason for this is, that the separation procedures for non-integer solutions are more complex and time consuming.

|                 | knott-3d-150                                                                             |                                                                                     | knott-3d-300                                                                             |                                                                                     | knott-3d-450                                                                            |                                                                                       |
|-----------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| ogm-KL          |  0%     |    |  0%     |    |  0%    |    |
| ogm-ICM         |  0%     |    |  0%     |    |  0%    |    |
| ogm-LF-1        |  0%    |   |  0%    |   |  0%   |   |
| MCR-CC          |  37%  |  |  12%  |  |  0%  |  |
| MCR-CCFDB       |  37%  |  |  12%  |  |  0%  |  |
| MCR-CCFDB-OWC   |  87%  |  |  75%  |  |  0%  |  |
| MCI-CCI         |  100% |  |  100% |  |  87% |  |
| MCI-CCIFD       |  100% |  |  100% |  |  87% |  |
| MCI-CCFDB-CCIFD |  100% |  |  75%  |  |  0%  |  |

Figure 9: **Evaluation of the 3D neuron segmentation datasets.** See Fig. 1 for the legend. With increasing problem size relaxations (MCR) get worse. Also integer variants (MCI) suffers and separating violated constraints becomes the most time consuming part. Consequently, for large models it is preferable to start directly with integer constraints and not to start with an LP-relaxation first, as done within MCI-CCFDB-CCIFD, because the separation procedure is than to slow.

|               | scene-decomposition                                                                 |      |                                                                                     | matching                                                                            |      |                                                                                     | protein-folding                                                                     |     |                                                                                       |
|---------------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------|
| A-Fusion      |    | 82%  |    |    | 0%   |    |                                                                                     |     |                                                                                       |
| ogm-ICM       |    | 15%  |    |    | 0%   |    |    | 0%  |    |
| ogm-LF-1      |    | 15%  |    |    | 0%   |    |    | 0%  |    |
| ogm-LF-2      |    | 39%  |    |    | 0%   |    |    | 0%  |    |
| ogm-LF-3      |    | 58%  |    |    | 0%   |    |    | 19% |    |
| ogm-TRWS-LF1  |    | 99%  |    |    | 0%   |    |    | 14% |    |
| ogm-TRWS-LF2  |    | 99%  |    |    | 0%   |    |    | 28% |    |
| ogm-LBP-LF1   |    | 80%  |    |    | 0%   |    |    | 38% |    |
| ogm-LBP-LF2   |    | 80%  |    |    | 25%  |    |    | 57% |    |
| BPS-TAB       |    | 79%  |    |    | 25%  |    |    | 52% |    |
| ogm-LBP-0.5   |    | 80%  |    |    | 0%   |    |    | 61% |    |
| ogm-LBP-0.95  |    | 81%  |    |    | 0%   |    |    | 47% |    |
| ogm-BPS       |    | 81%  |    |    | 0%   |    |    | 28% |    |
| ogm-TRBP-0.5  |    | 89%  |    |    | 0%   |    |    | 42% |    |
| ogm-TRBP-0.95 |    | 90%  |    |    | 0%   |    |    | 38% |    |
| ogm-TRBPS     |    | 90%  |    |    | 0%   |    |    | 33% |    |
| ADDD          |    | 98%  |    |    | 0%   |    |    | 0%  |    |
| MPLP          |    | 97%  |    |    | 0%   |    |    | 4%  |    |
| MPLP-C        |    | 99%  |    |    | 100% |    |    | 57% |    |
| ogm-LP-LP     |   | 99%  |   |   | 0%   |   |                                                                                     |     |                                                                                       |
| TRWS-TAB      |  | 99%  |  |  | 0%   |  |  | 9%  |  |
| ogm-BUNDLE-H  |  | 100% |  |  | 0%   |  |  | 4%  |  |
| ogm-BUNDLE-A- |  | 95%  |  |  | 0%   |  |  | 0%  |  |
| ogm-BUNDLE-A  |  | 99%  |  |  | 0%   |  |  | 4%  |  |
| ogm-BUNDLE-A+ |  | 100% |  |  | 0%   |  |  | 9%  |  |
| ogm-SG-A-     |  | 70%  |  |  | 0%   |  |  | 0%  |  |
| ogm-SG-A      |  | 99%  |  |  | 0%   |  |  | 4%  |  |
| ogm-SG-A+     |  | 99%  |  |  | 0%   |  |  | 4%  |  |
| ogm-ADSAL     |  | 99%  |  |  | 0%   |  |  | 19% |  |
| ogm-ILP       |  | 100% |  |  | 100% |  |                                                                                     |     |                                                                                       |
| ogm-ATSAR     |                                                                                     |      |                                                                                     |  | 100% |  |                                                                                     |     |                                                                                       |
| ogm-CombiLP   |  | 100% |  |  | 100% |  |  | 95% |  |
| ADDD-BB       |  | 100% |  |  | 75%  |  |                                                                                     |     |                                                                                       |
| BRAOBB-1      |  | 93%  |  |  | 100% |  |                                                                                     |     |                                                                                       |
| BRAOBB-2      |  | 87%  |  |  | 100% |  |                                                                                     |     |                                                                                       |
| BRAOBB-3      |  | 91%  |  |  | 100% |  |                                                                                     |     |                                                                                       |

Figure 10: **Evaluation of the second order models with no truncated convex regularizers.** See Fig. 1 for the legend. The scene decomposition instances are based on superpixels, such models are small and combinatorial methods like ogm-ILP or AD3-BB are fast and optimal. Contrary to scene-decomposition for the matching instances the local polytope relaxation is not tight. One can either tighten the relaxation MPLP-C or use alternative methods to obtain bounds ogm-AStar for obtaining fast optimal results. For protein-folding relaxations are weak too and the huge label-space renders the problem hard for many solvers, e.g. ogm-ILP. We obtain the best results by BPS.